



US012398891B2

(12) **United States Patent**
Karamanos et al.

(10) **Patent No.:** **US 12,398,891 B2**
(45) **Date of Patent:** ***Aug. 26, 2025**

(54) **AIR HANDLER DEVICES WITH U-BEND DESIGN**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/516,068**

(22) Filed: **Nov. 21, 2023**

(65) **Prior Publication Data**
US 2024/0085032 A1 Mar. 14, 2024

Related U.S. Application Data

(63) Continuation of application No. 18/314,420, filed on May 9, 2023, now Pat. No. 11,846,434, which is a (Continued)

(51) **Int. Cl.**
F24F 1/0029 (2019.01)
F24F 13/02 (2006.01)
F24F 13/08 (2006.01)
F24F 13/20 (2006.01)
F24F 13/24 (2006.01)

(52) **U.S. Cl.**
CPC **F24F 1/0029** (2013.01); **F24F 13/0236** (2013.01); **F24F 13/0245** (2013.01); **F24F 13/08** (2013.01); **F24F 13/20** (2013.01); **F24F 13/24** (2013.01); **F24F 2013/242** (2013.01)

(58) **Field of Classification Search**
CPC .. **F24F 1/0029**; **F24F 13/0236**; **F24F 13/0245**; **F24F 13/08**; **F24F 13/20**; **F24F 13/24**; **F24F 2013/242**
See application file for complete search history.

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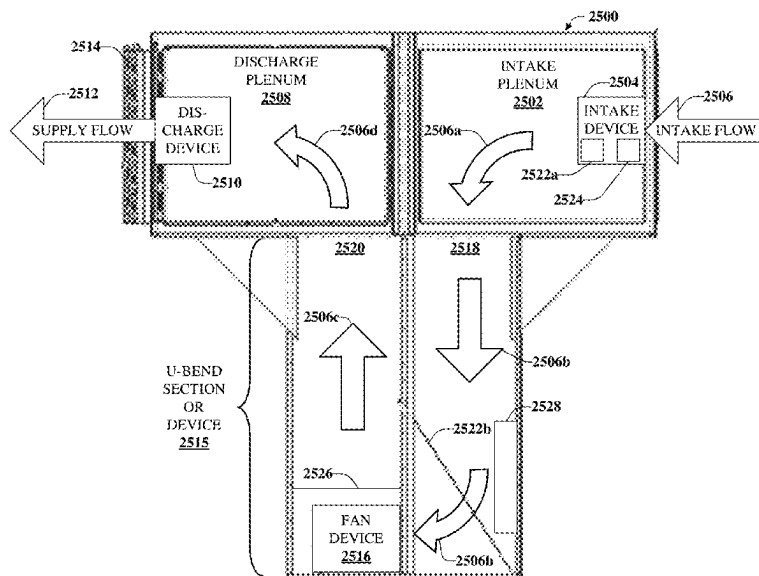
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(57) **ABSTRACT**

Architectures and techniques are presented that can facilitate improved design and function of certain air handler devices. Architectures directed to an improved air handler device can be designed to improve temperature control demands such as, e.g., concurrently heat and cool air and reducing device dimensions (e.g., size, weight) that can reduce costs and mitigate shipping and installation difficulties. Architectures directed to U-bend structures can further reduce footprint on leasable space and provide improved acoustics, service access, and reduced energy consumption and infrastructure costs.

29 Claims, 28 Drawing Sheets



Related U.S. Application Data

continuation-in-part of application No. 17/219,531, filed on Mar. 31, 2021, now Pat. No. 11,674,696, which is a continuation of application No. 16/930,635, filed on Jul. 16, 2020, now Pat. No. 11,346,564.

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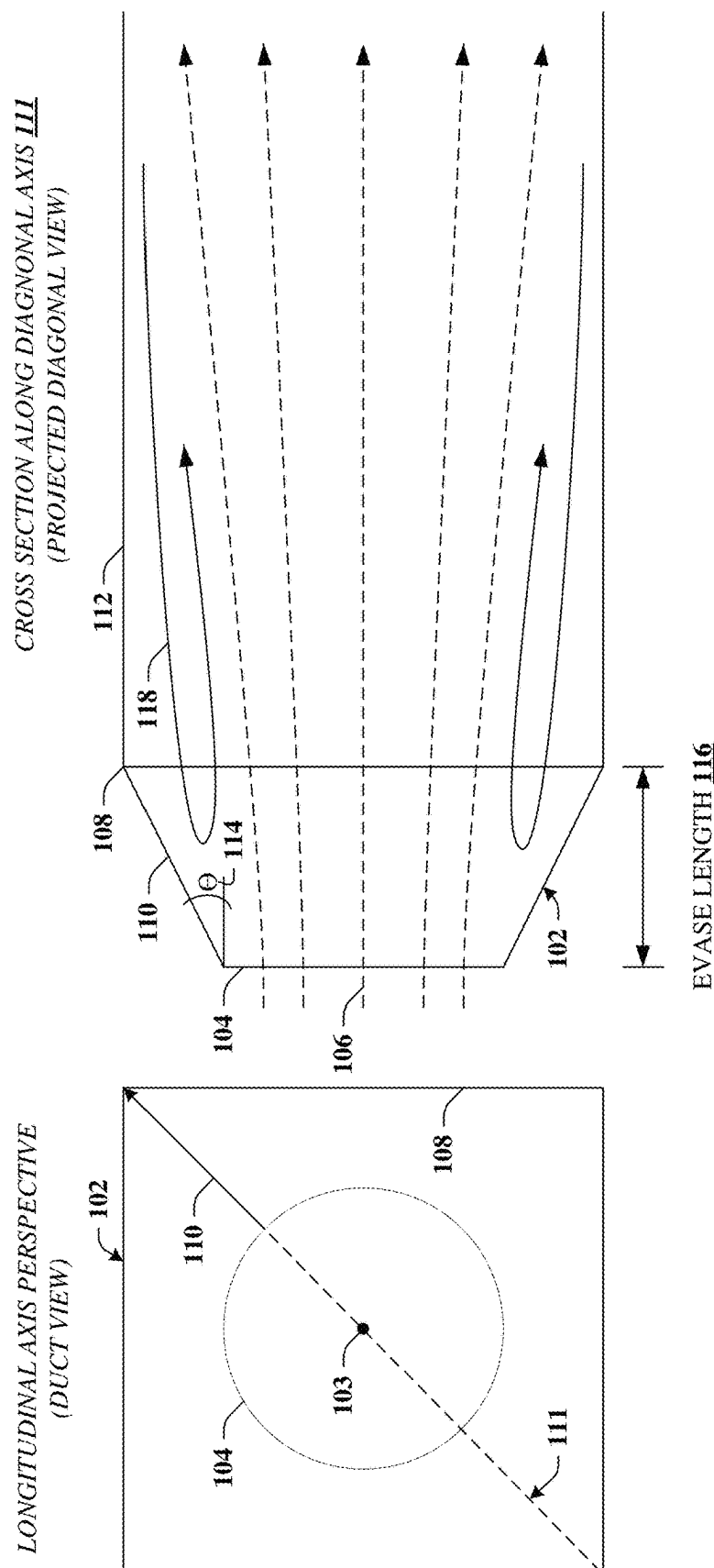


FIG. 1

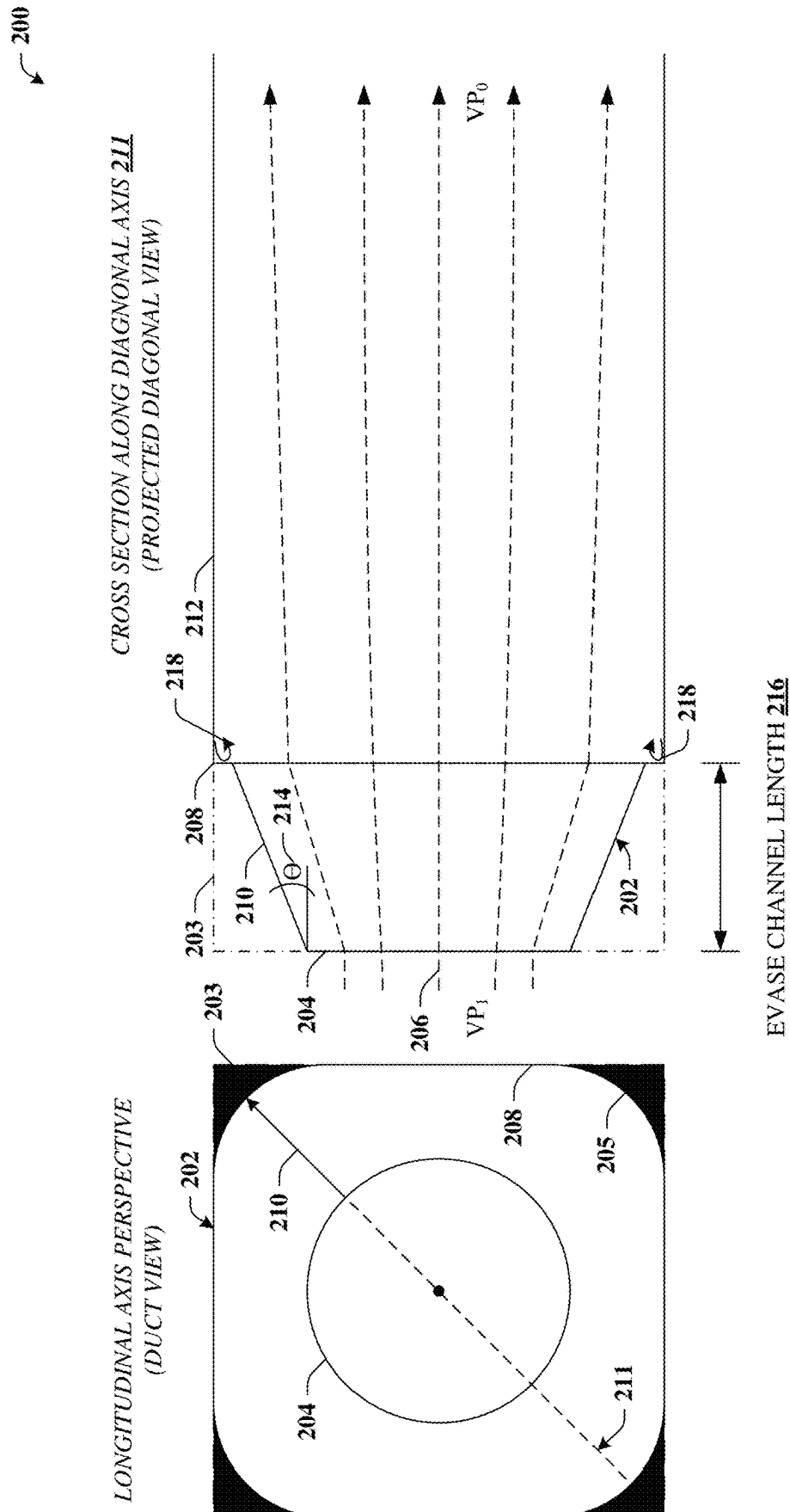


FIG. 2

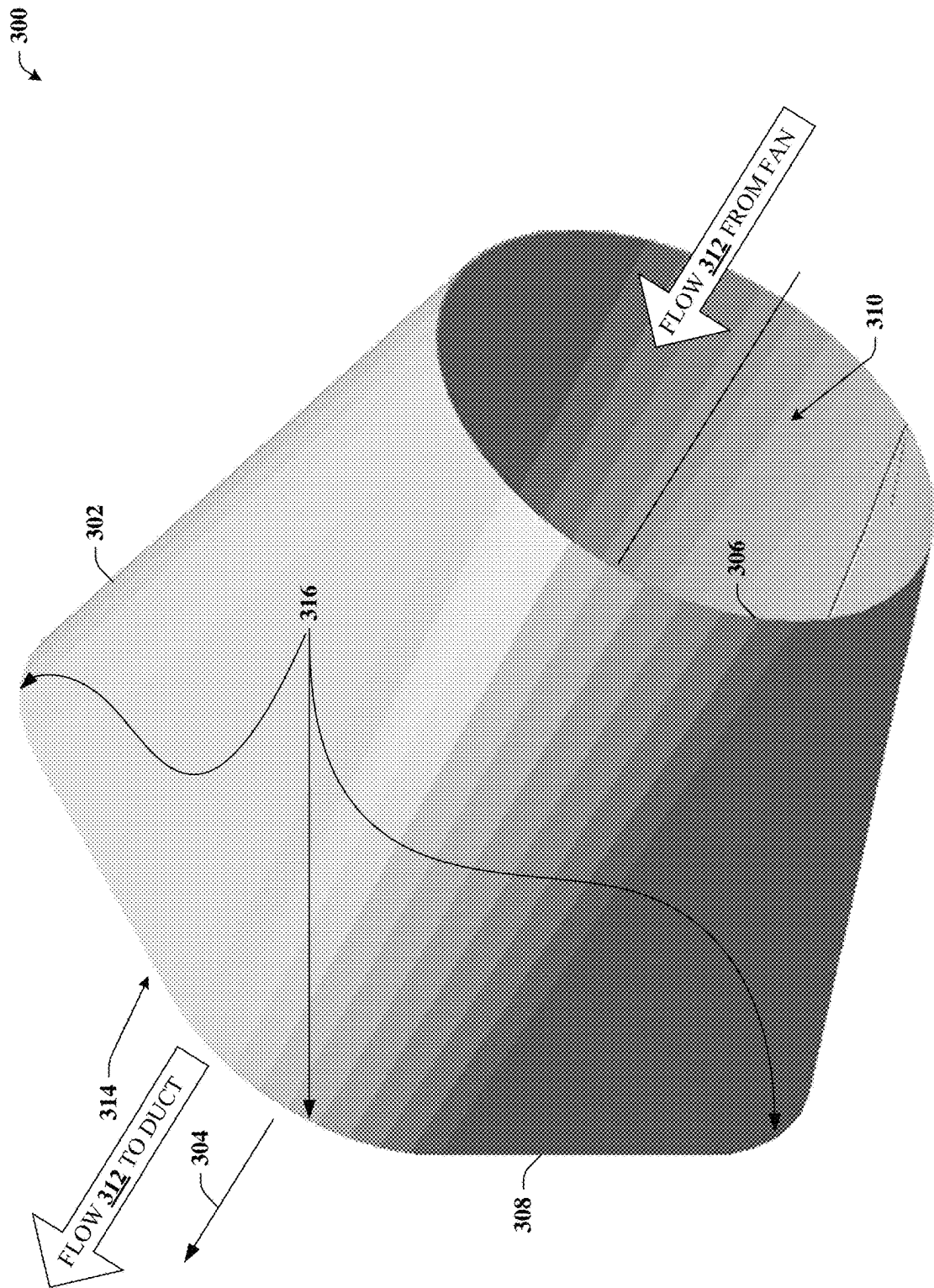


FIG. 3

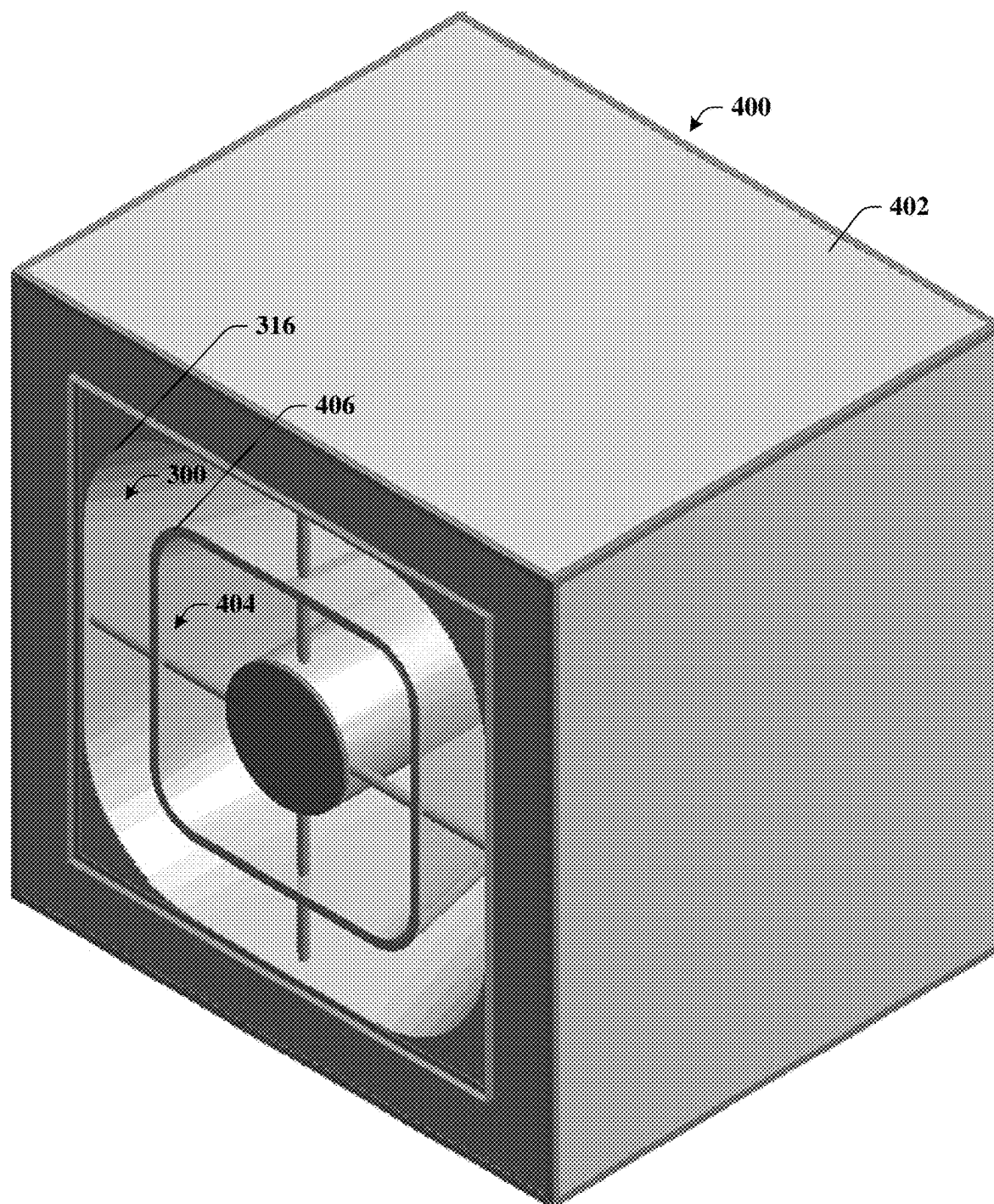


FIG. 4

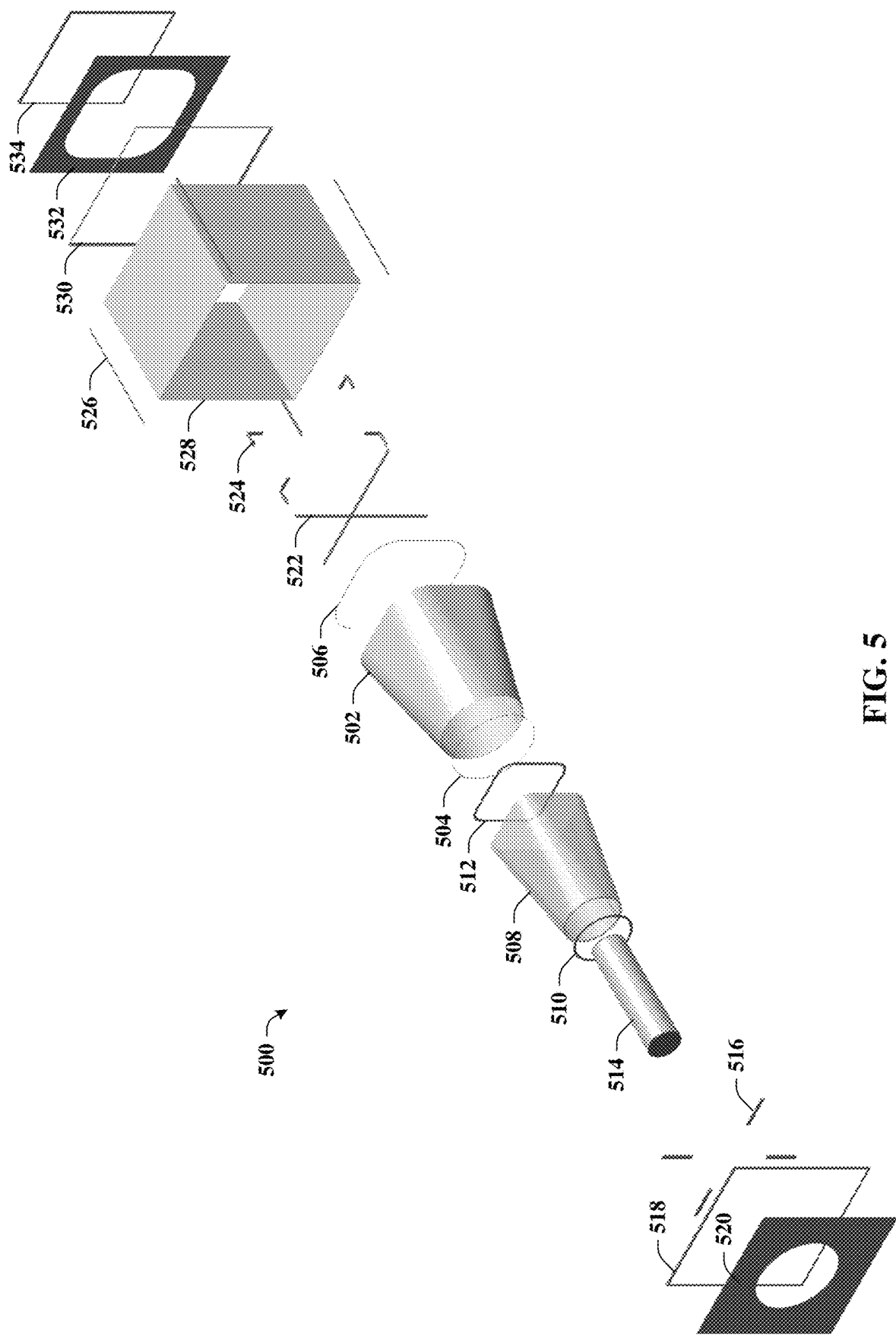


FIG. 5

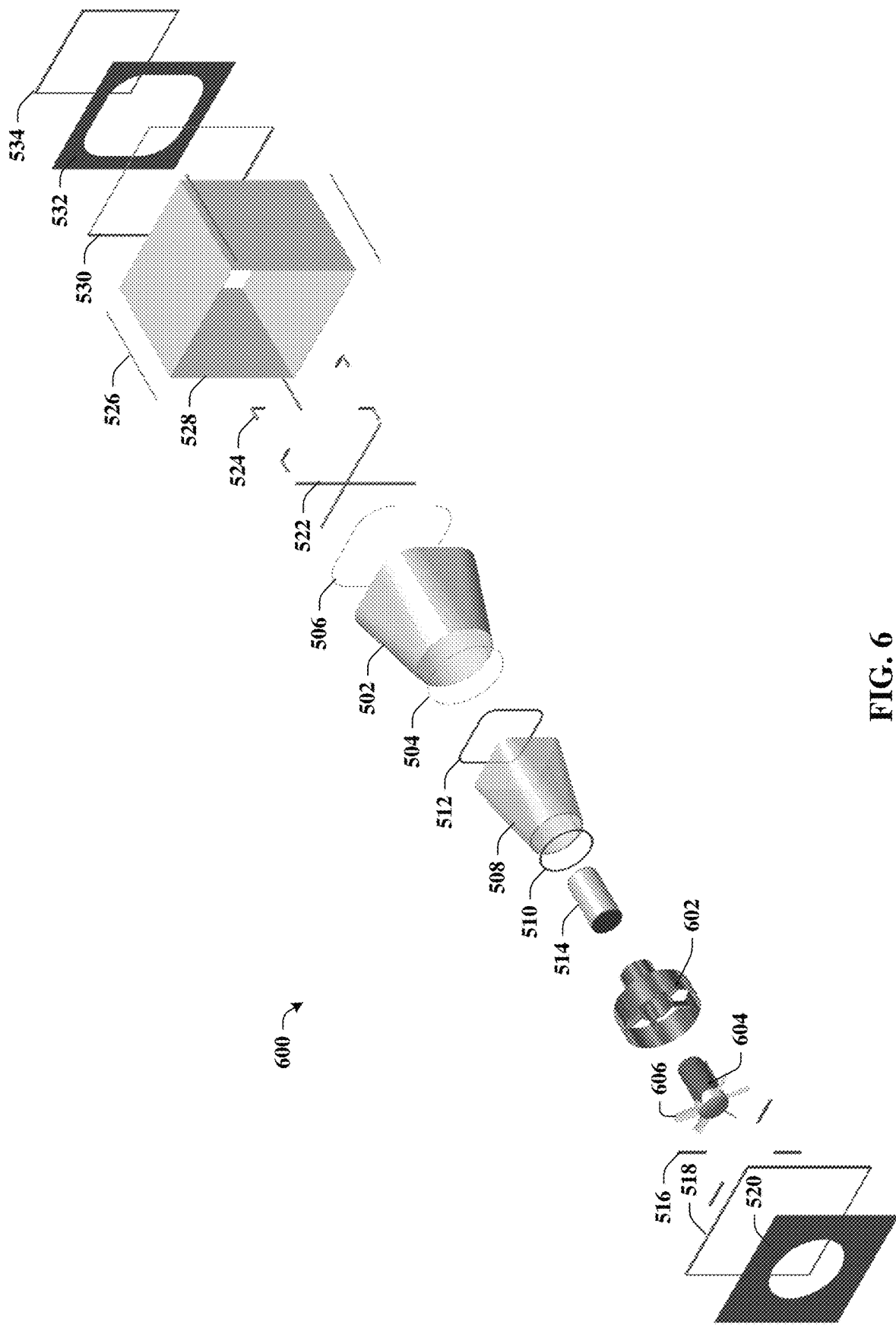
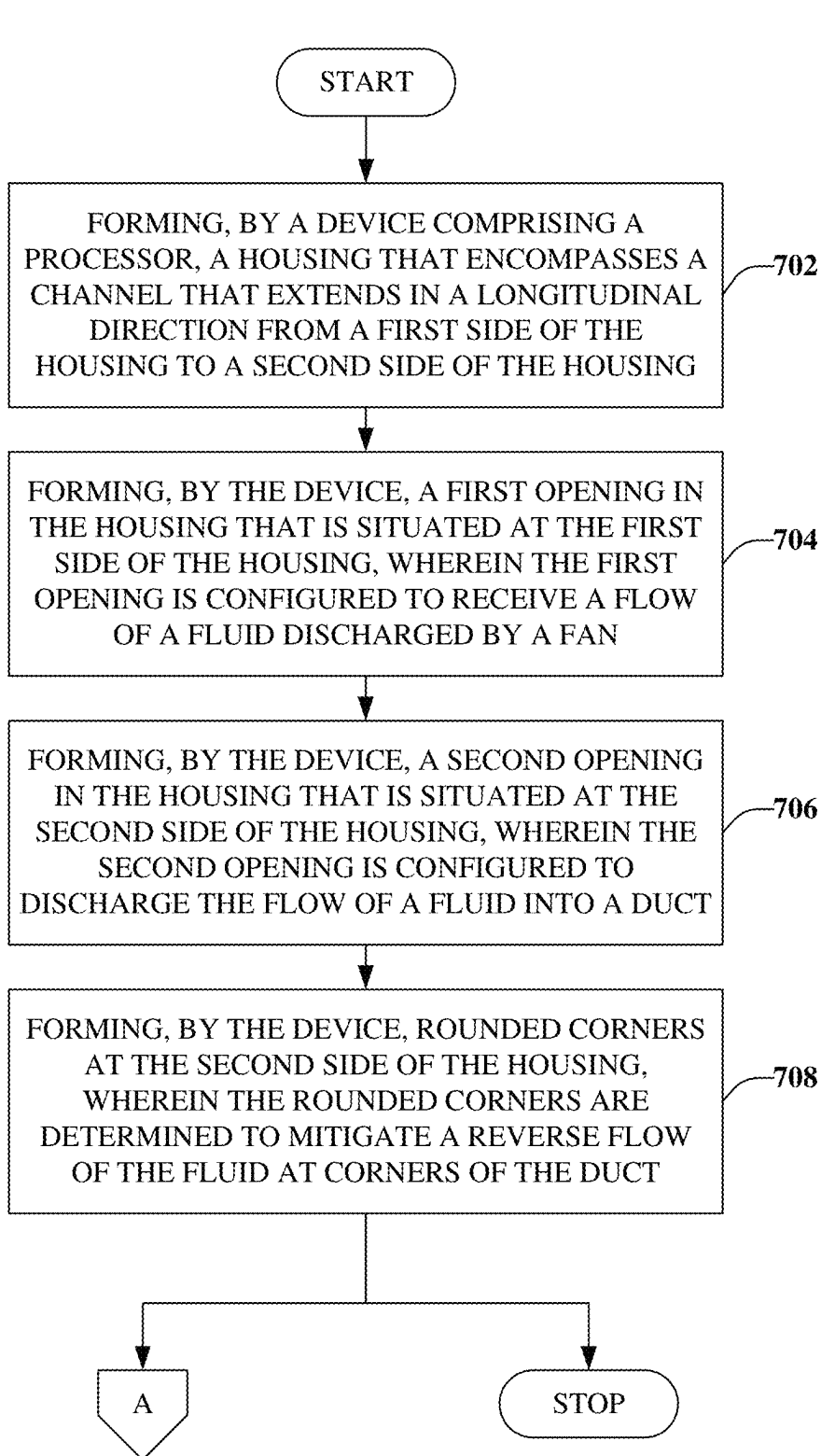
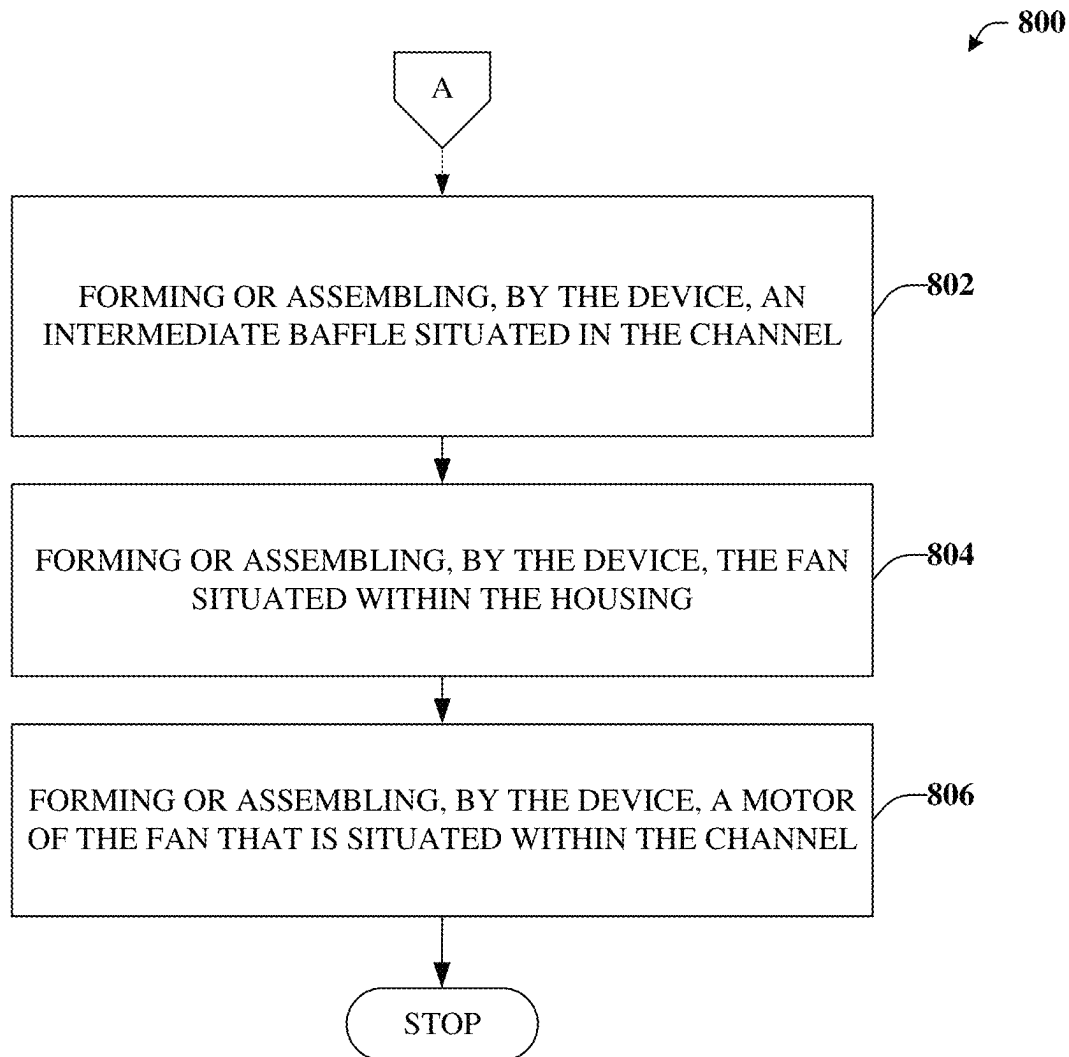


FIG. 6

**FIG. 7**

**FIG. 8**

900

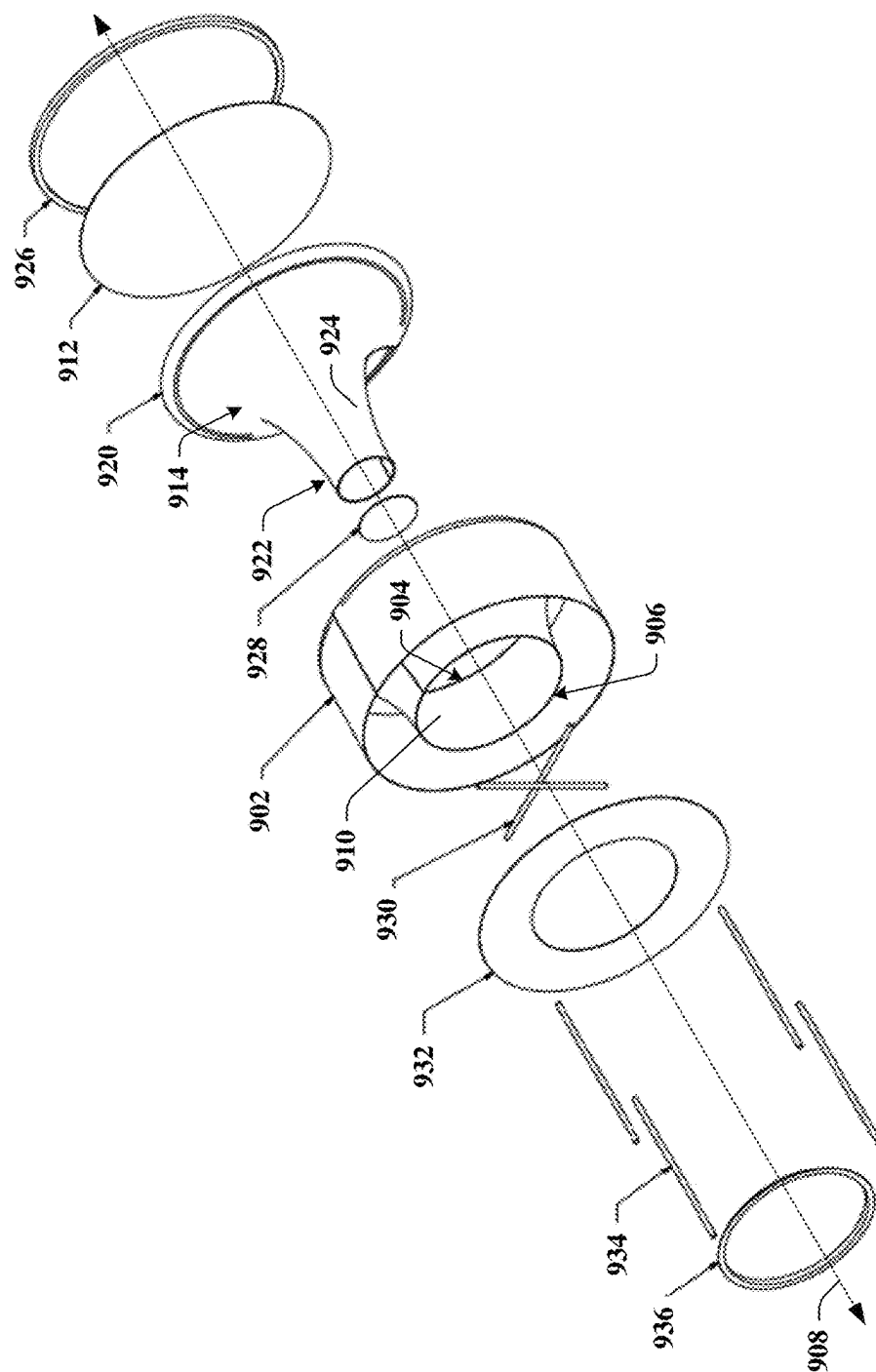


FIG. 9

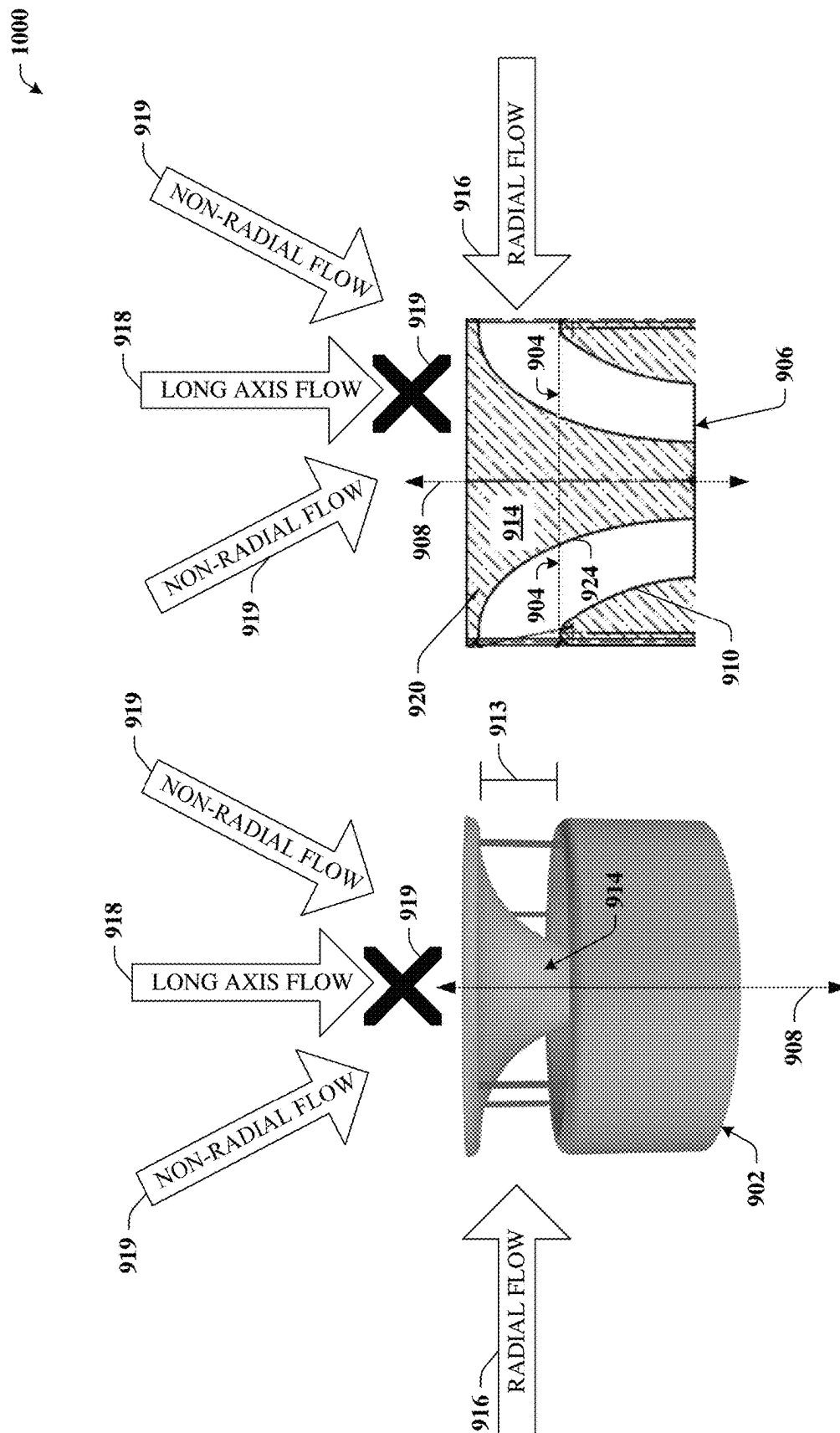


FIG. 10

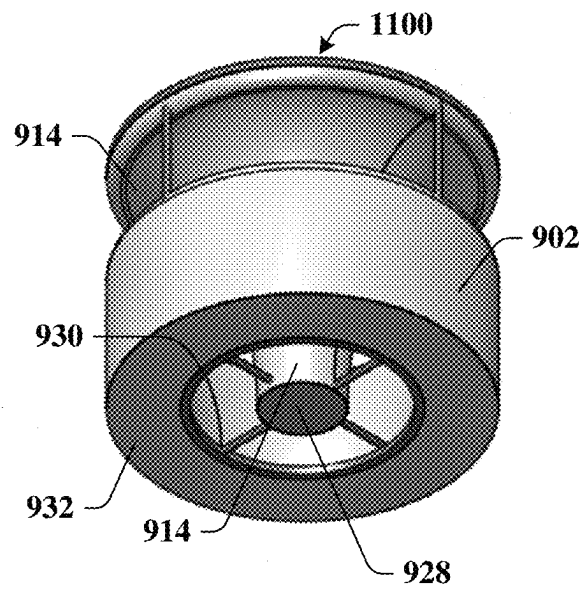


FIG. 11

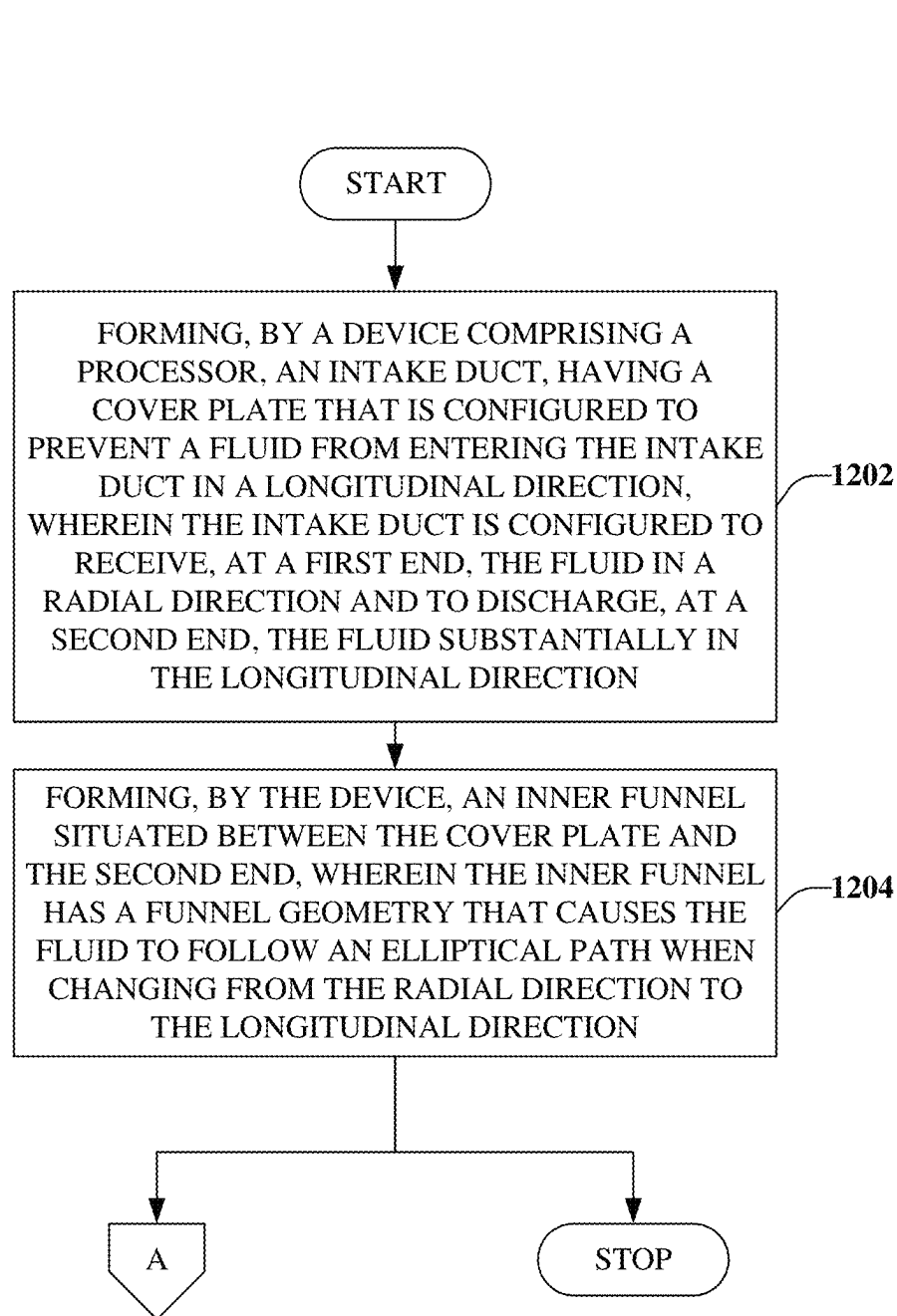
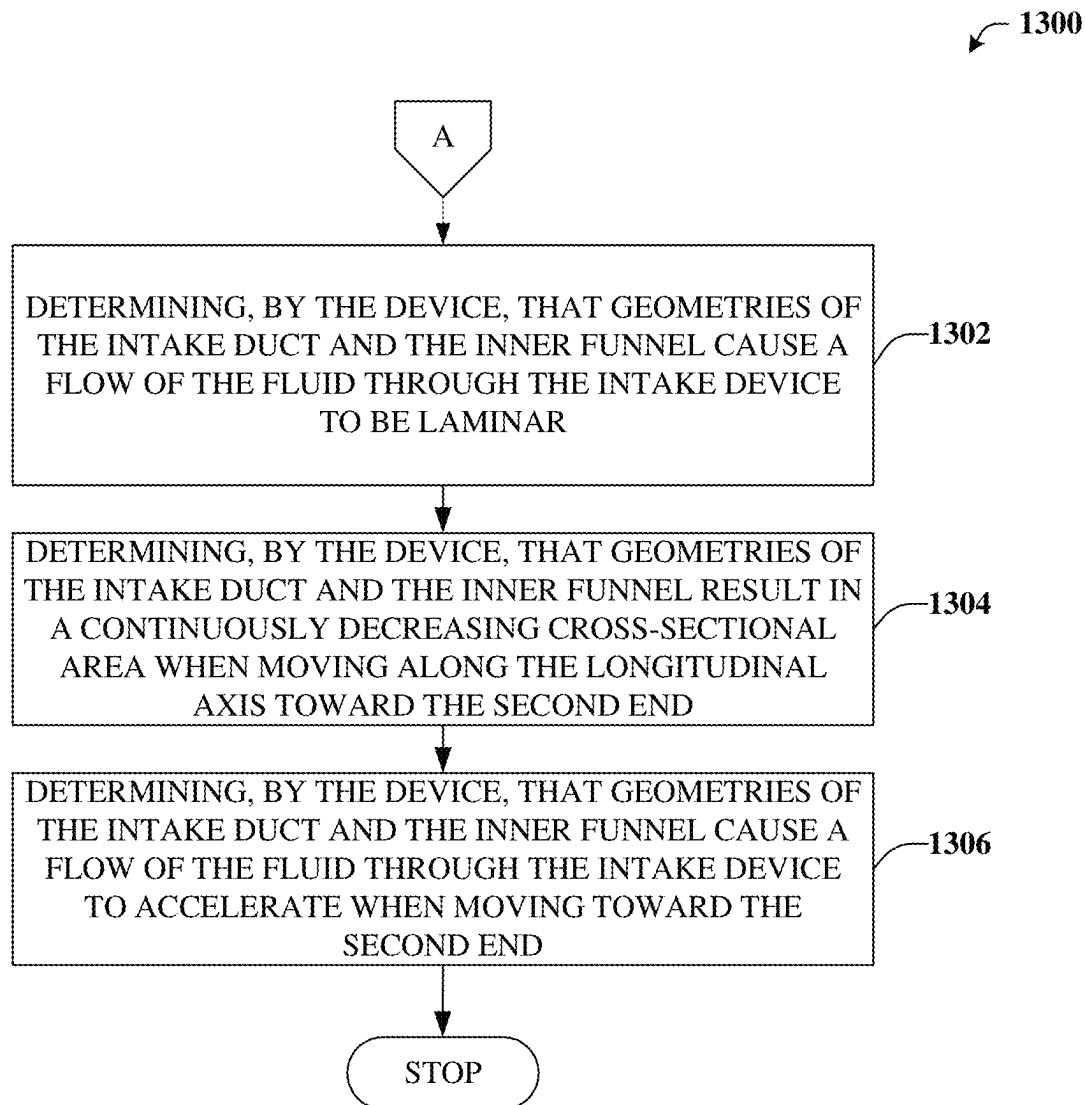


FIG. 12

**FIG. 13**

1400

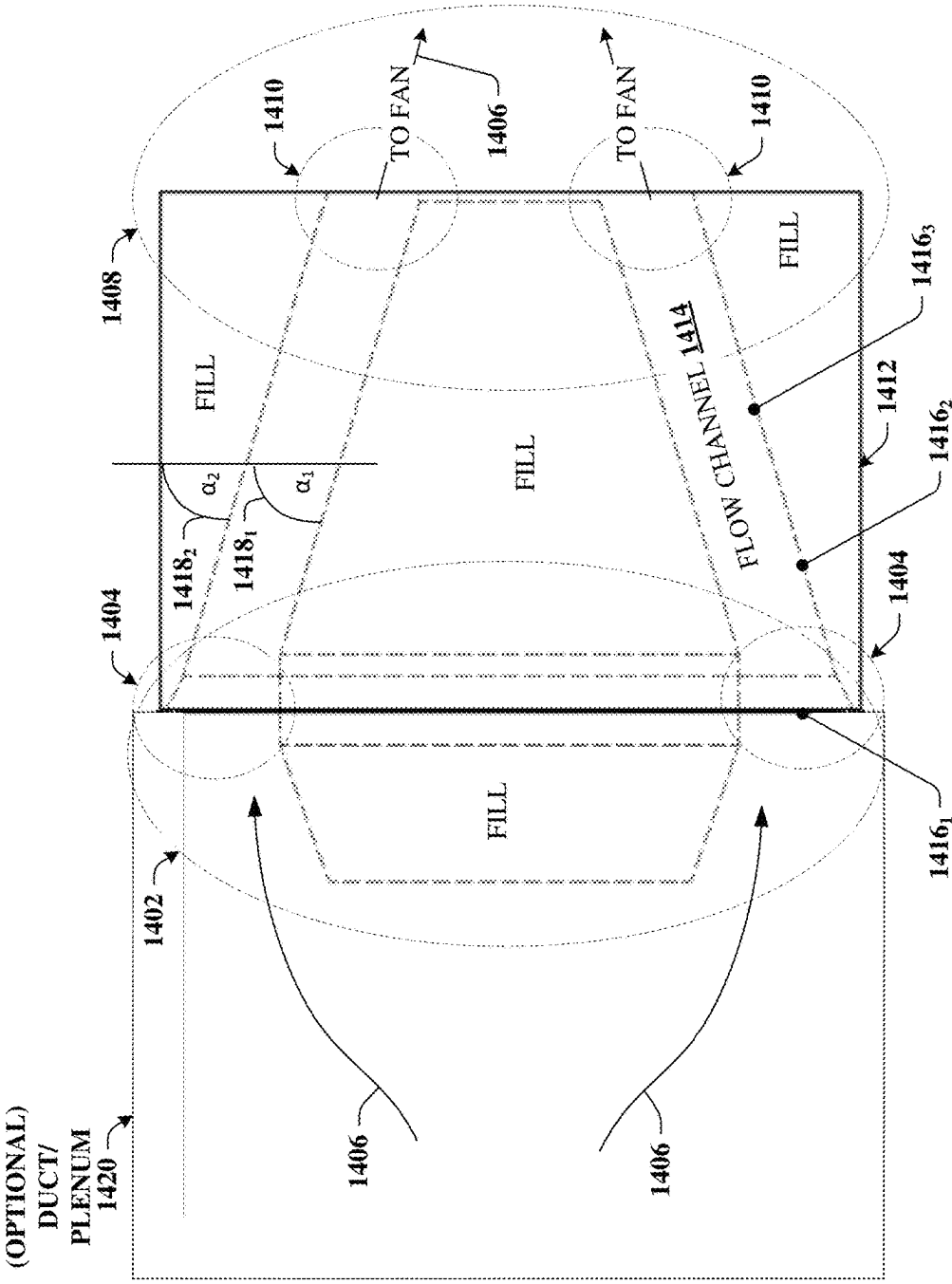


FIG. 14

1500

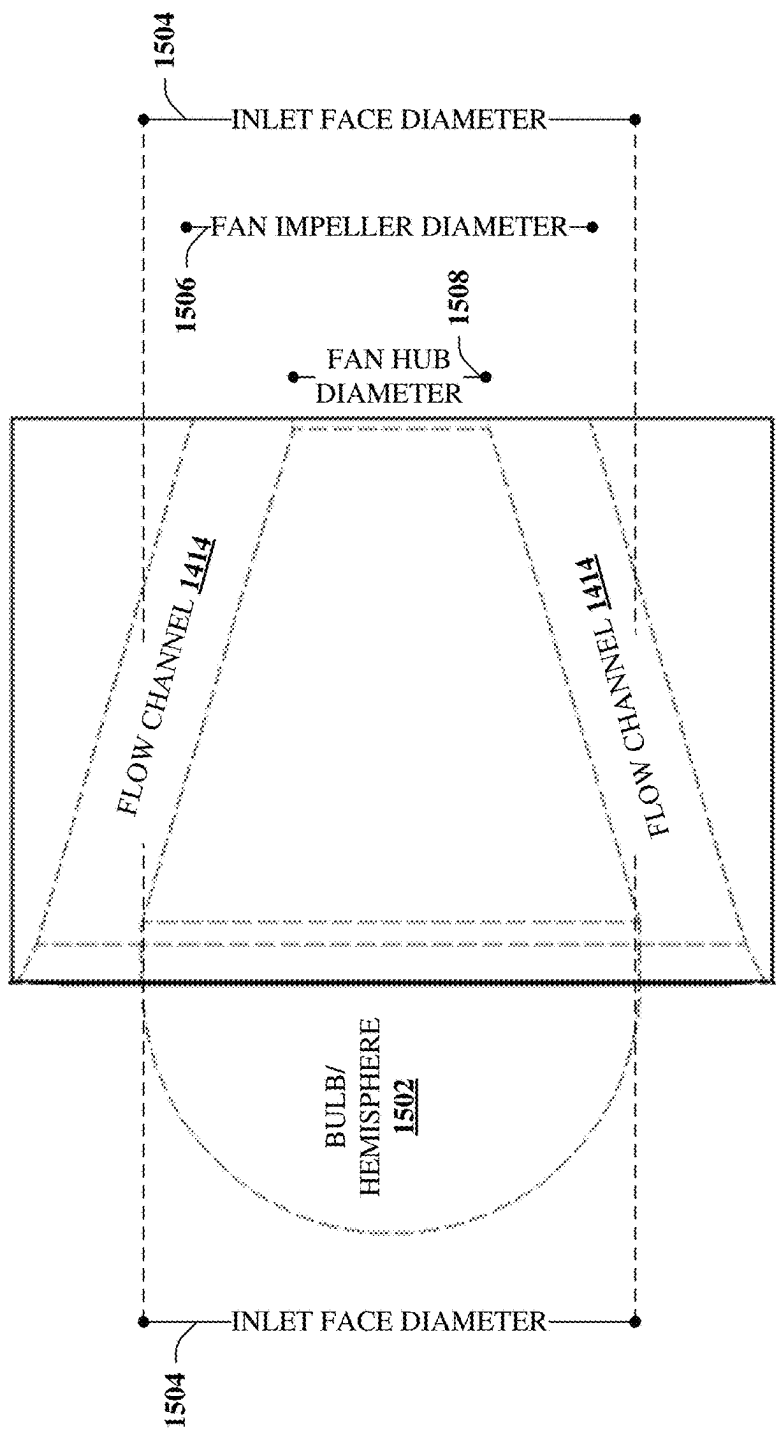
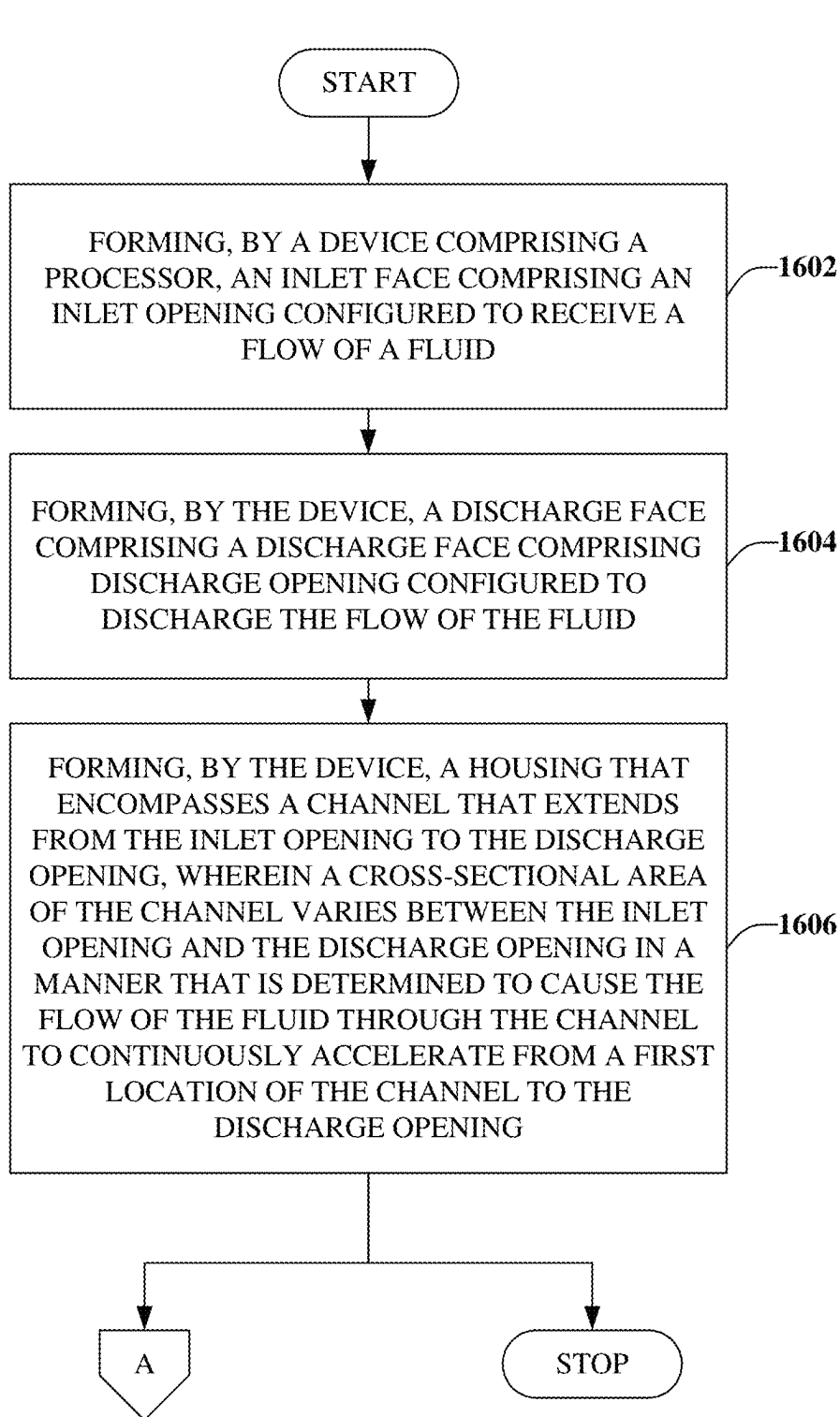
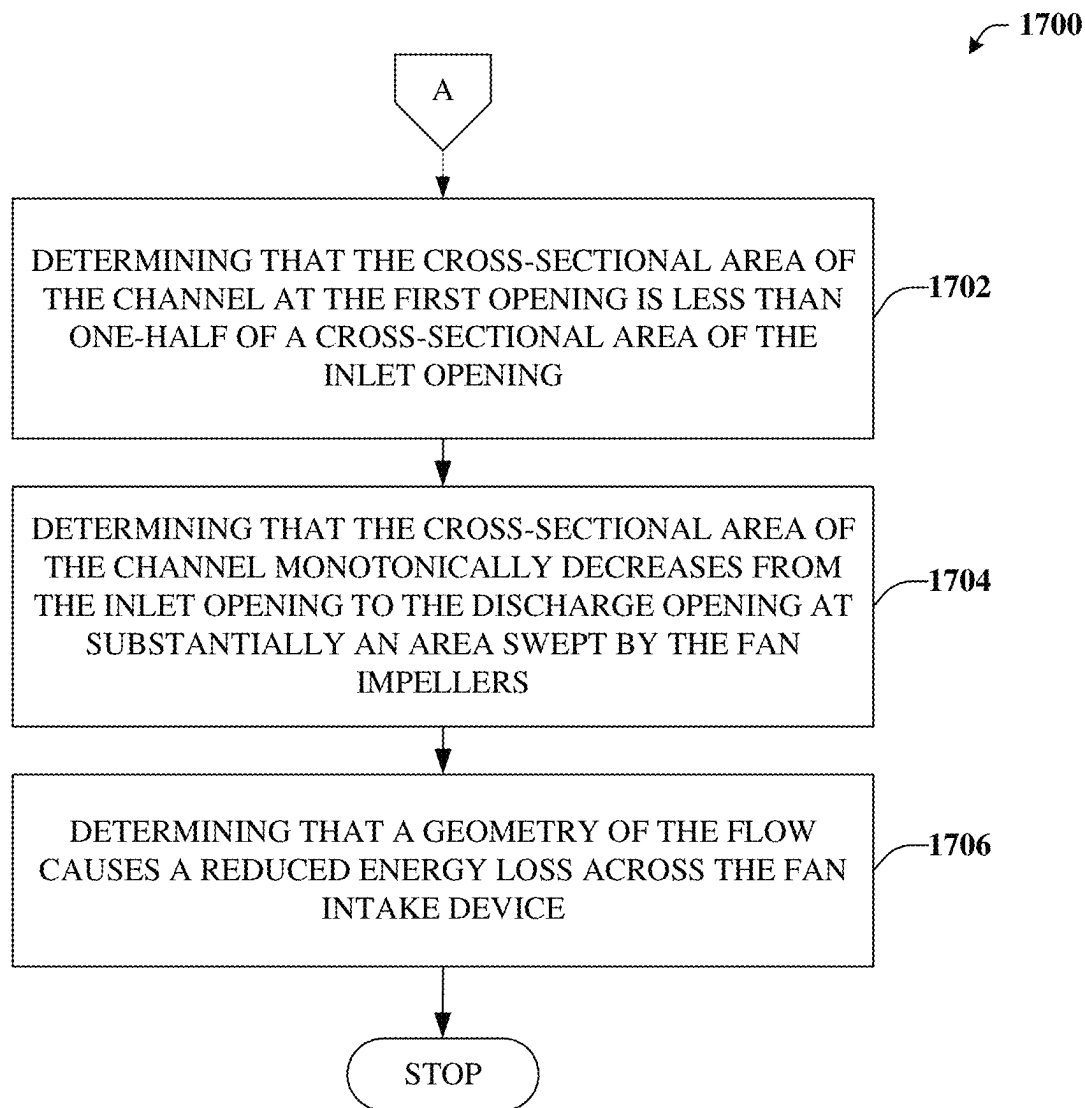


FIG. 15

**FIG. 16**

**FIG. 17**

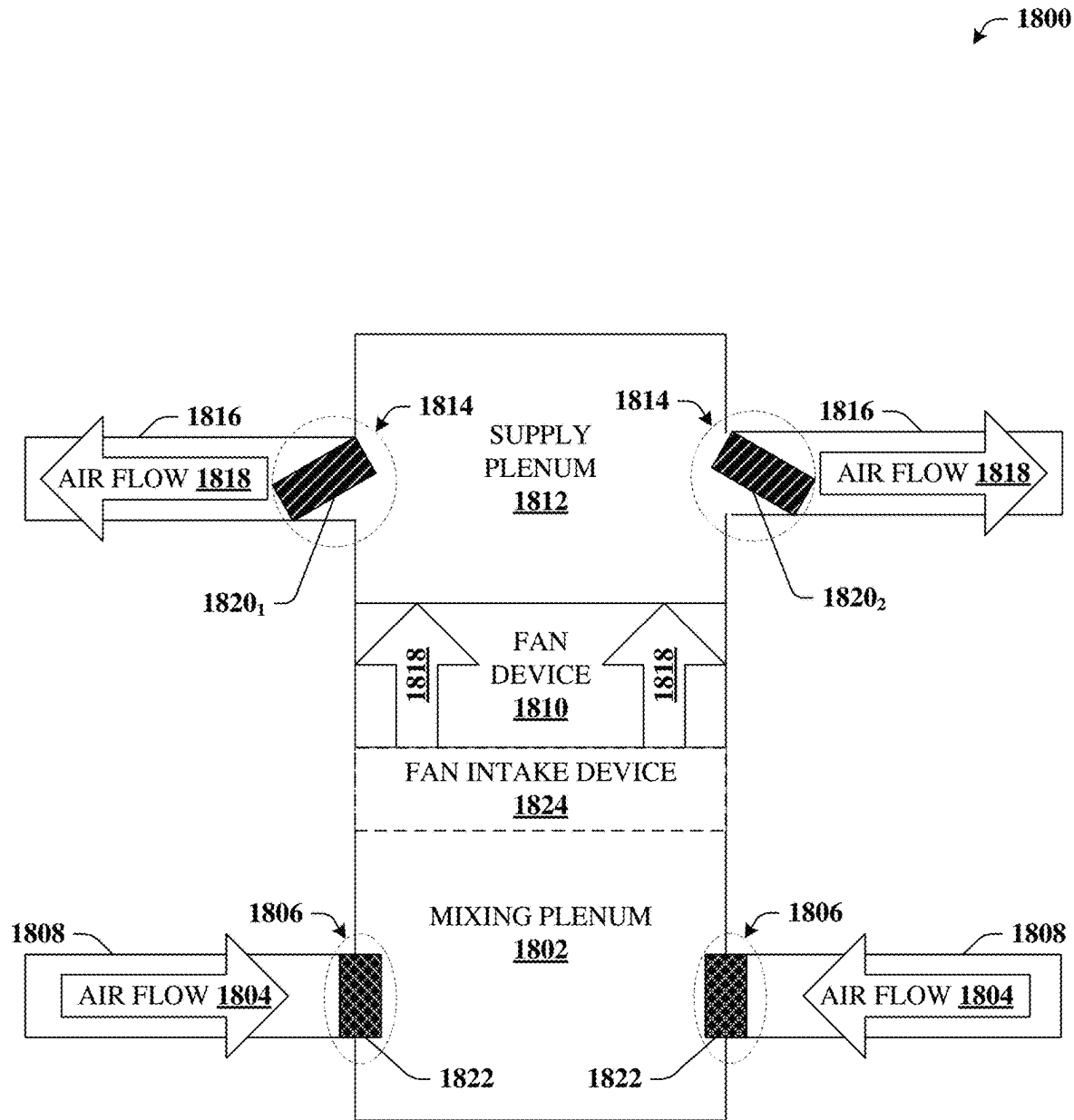


FIG. 18

1900

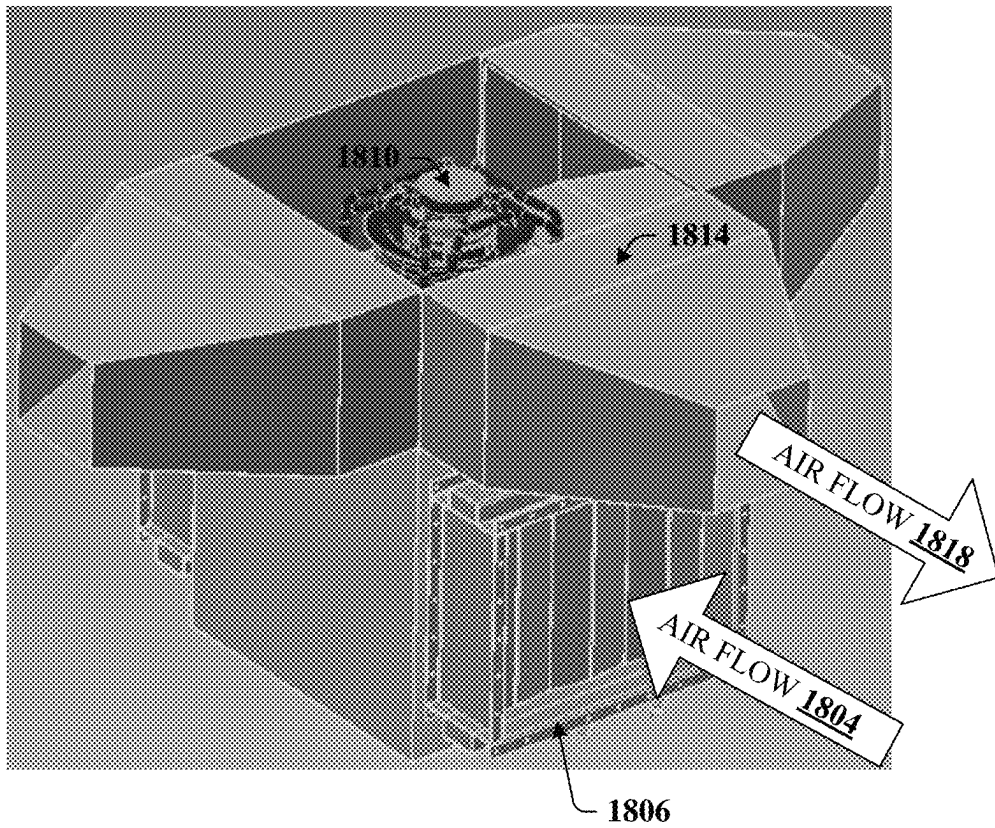


FIG. 19

2000

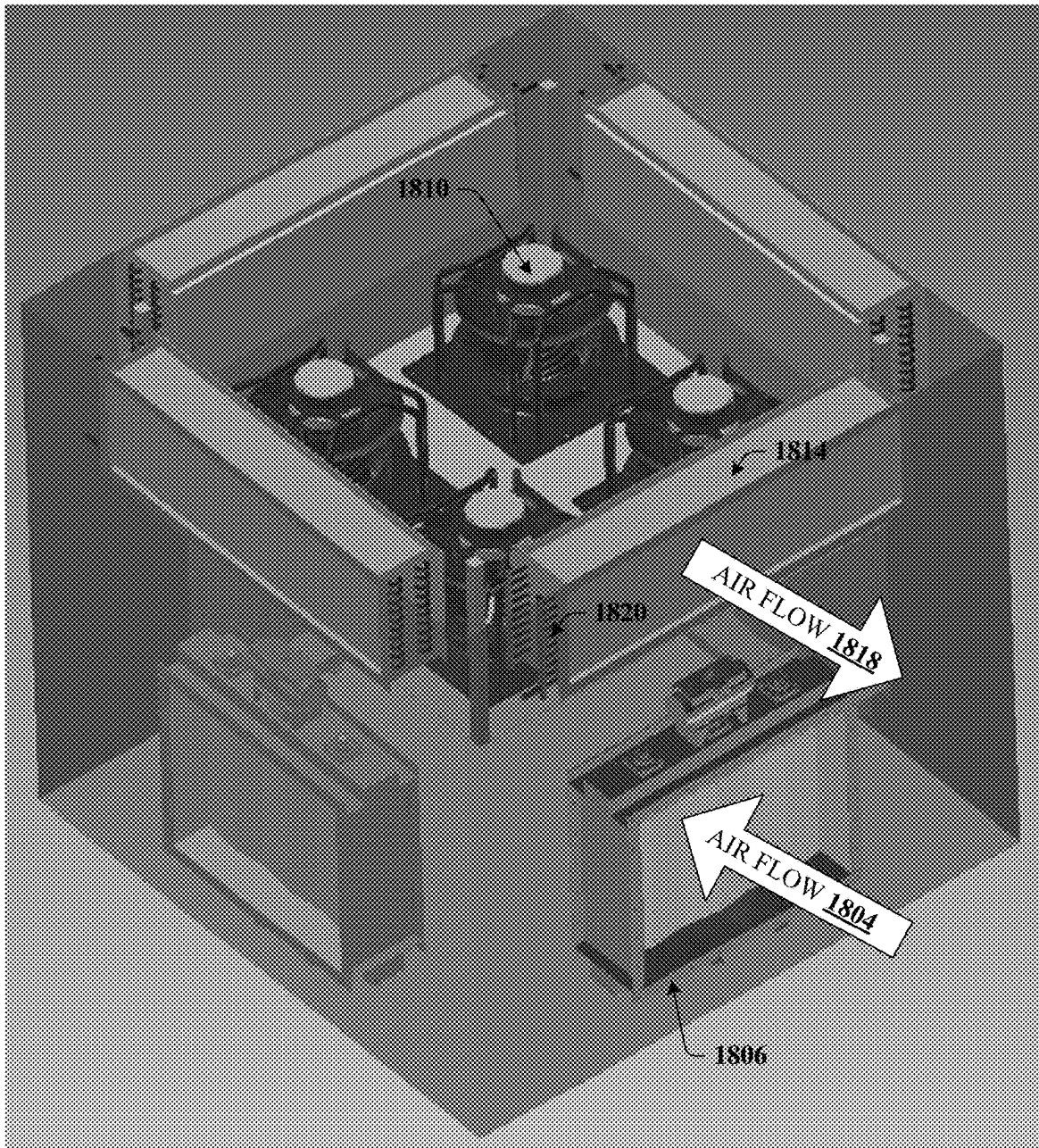


FIG. 20

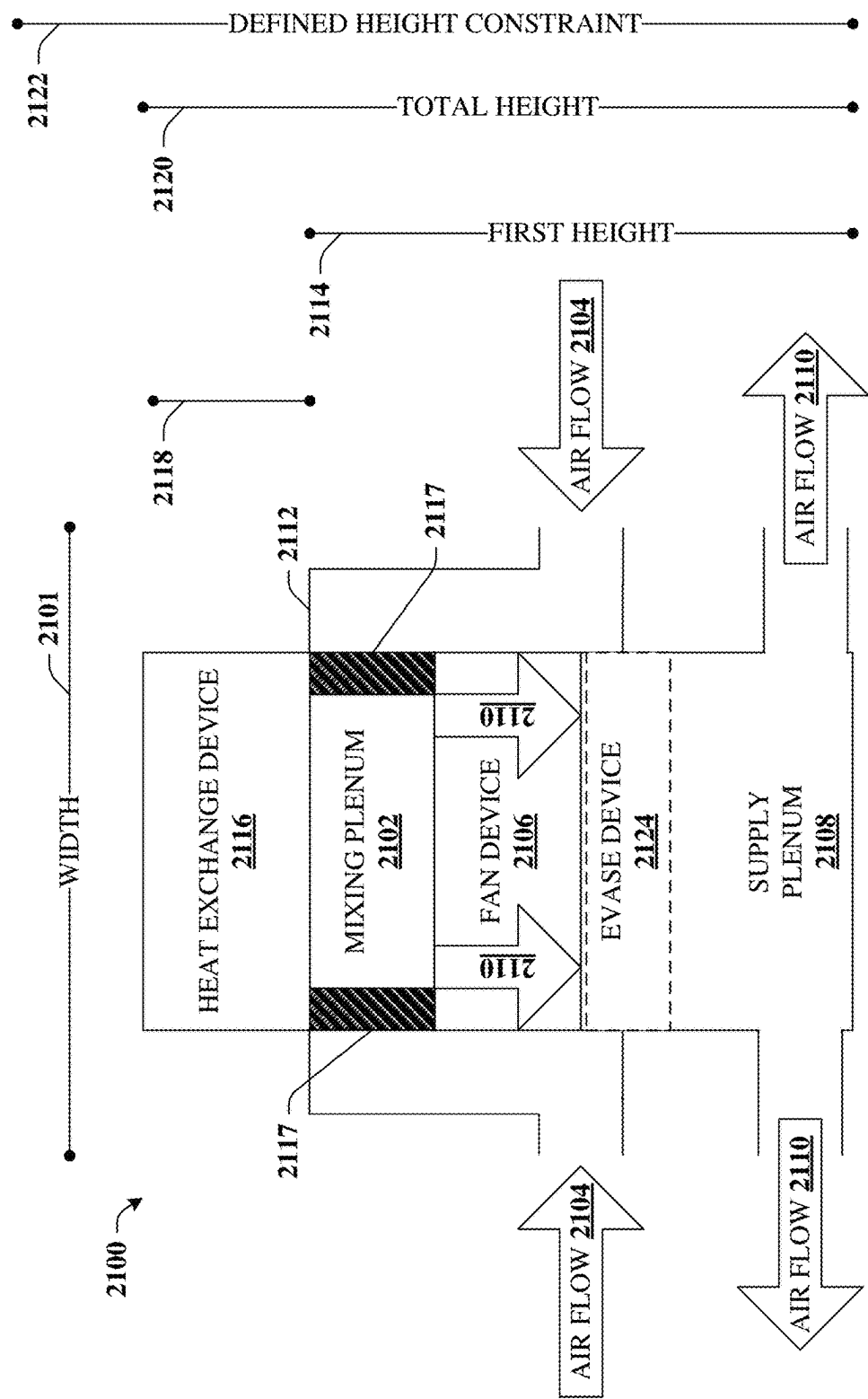


FIG. 21

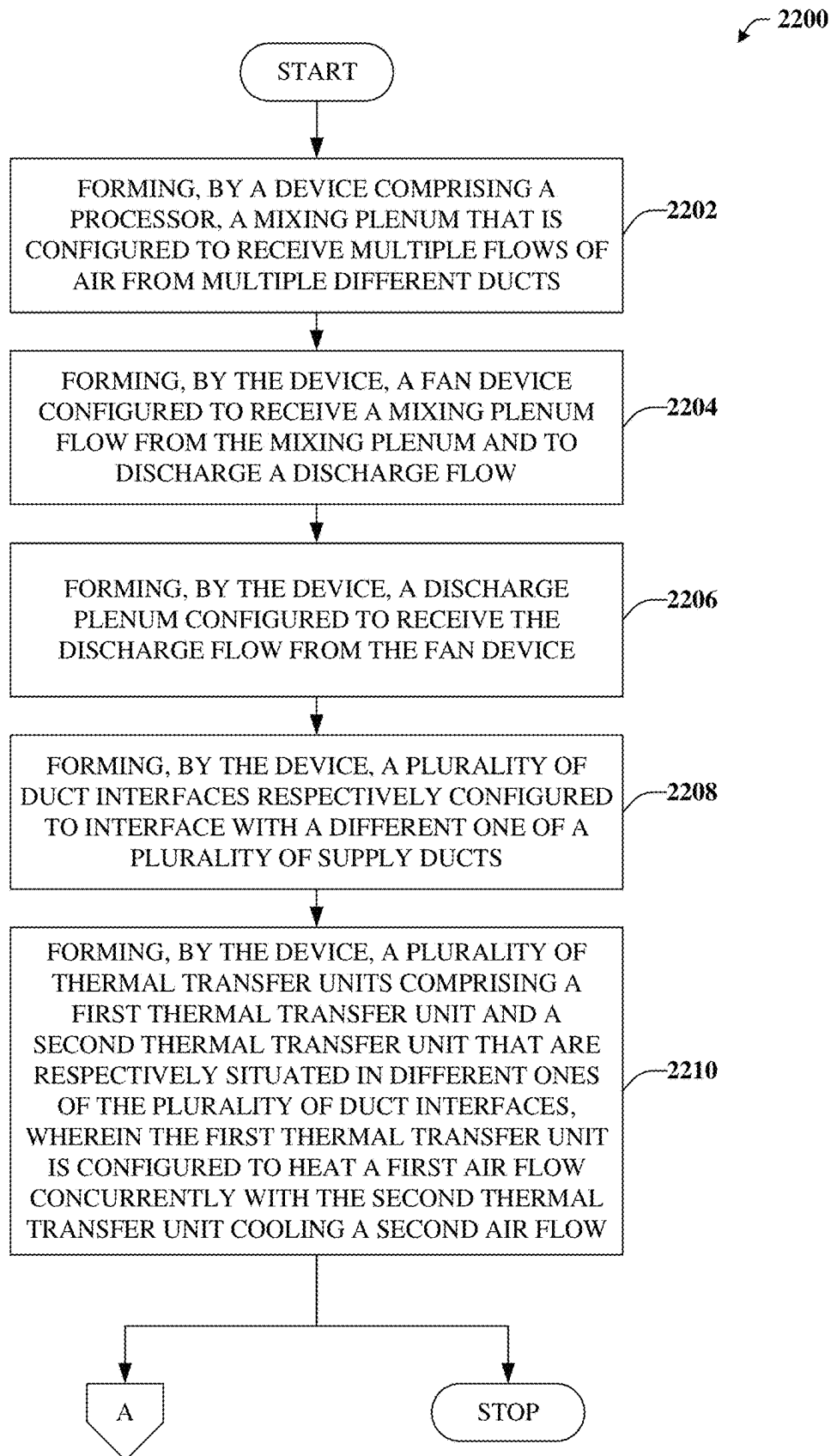
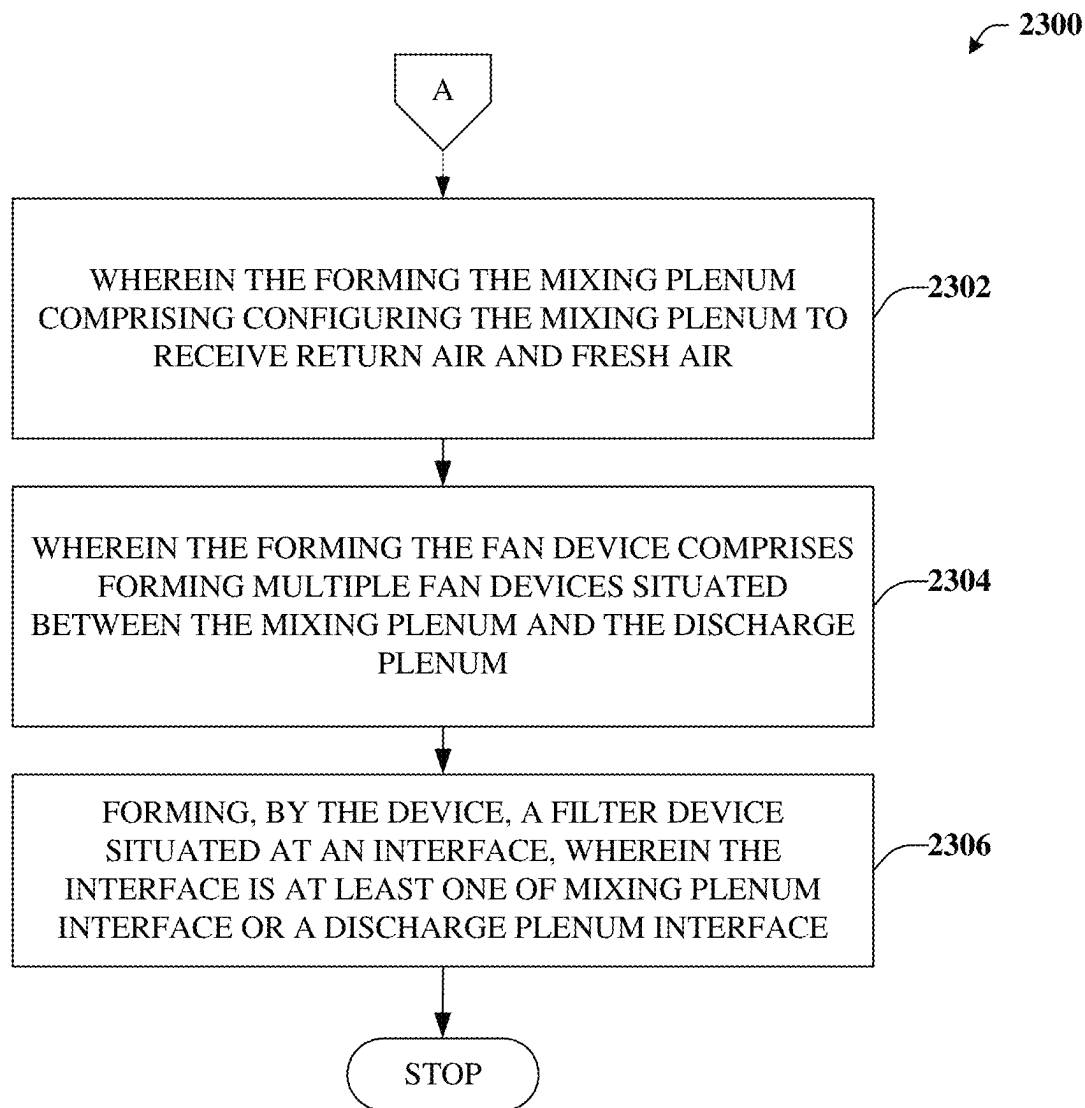


FIG. 22

**FIG. 23**

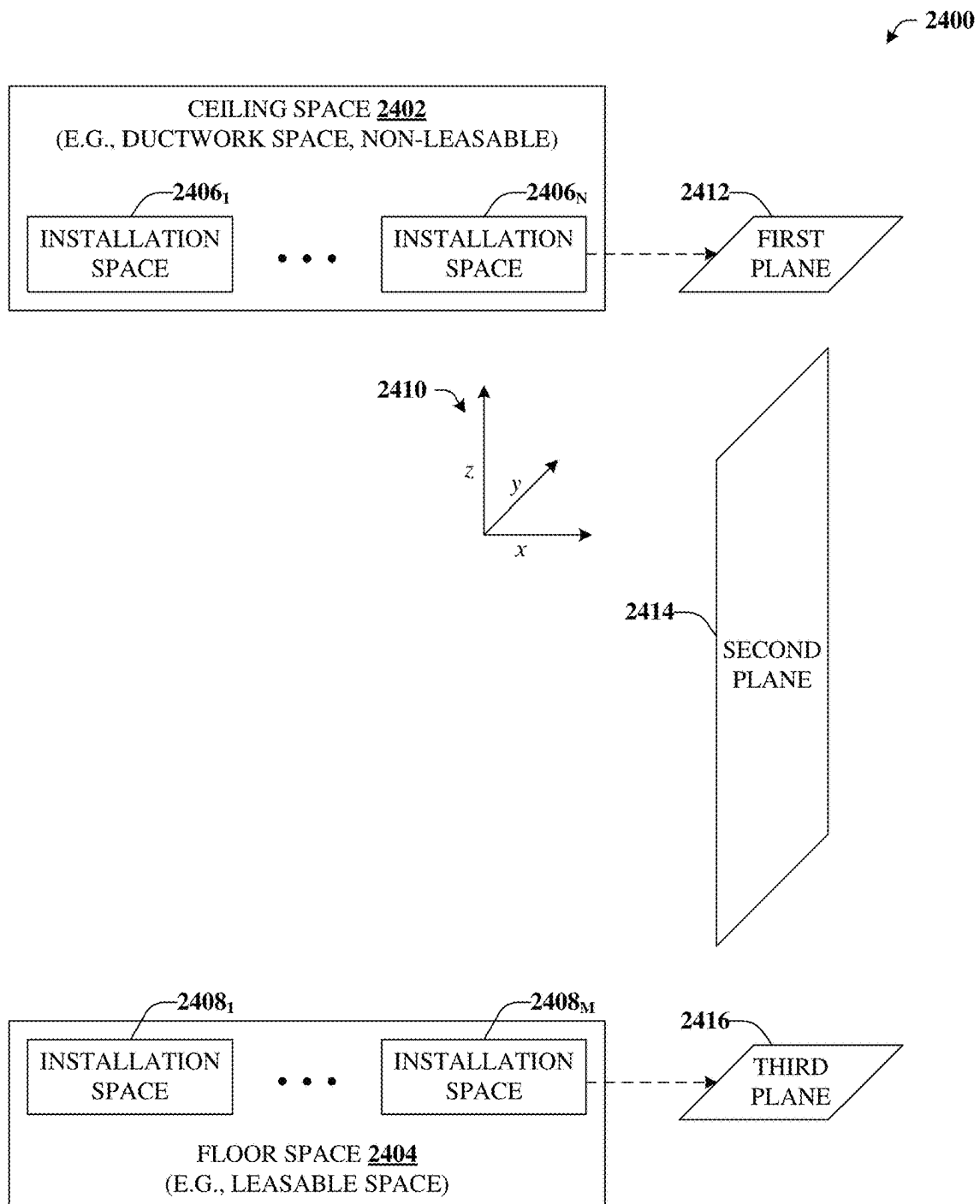


FIG. 24

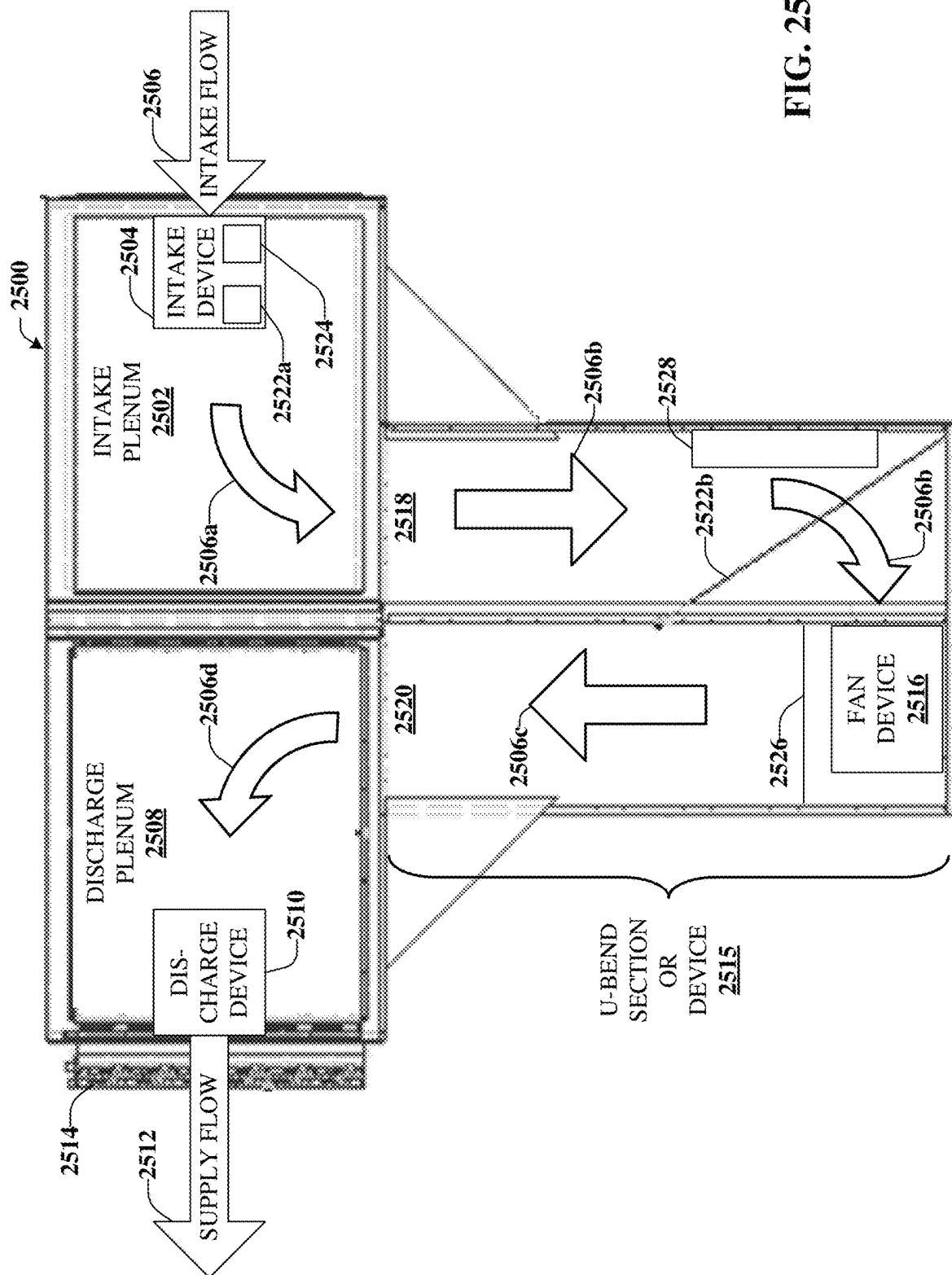
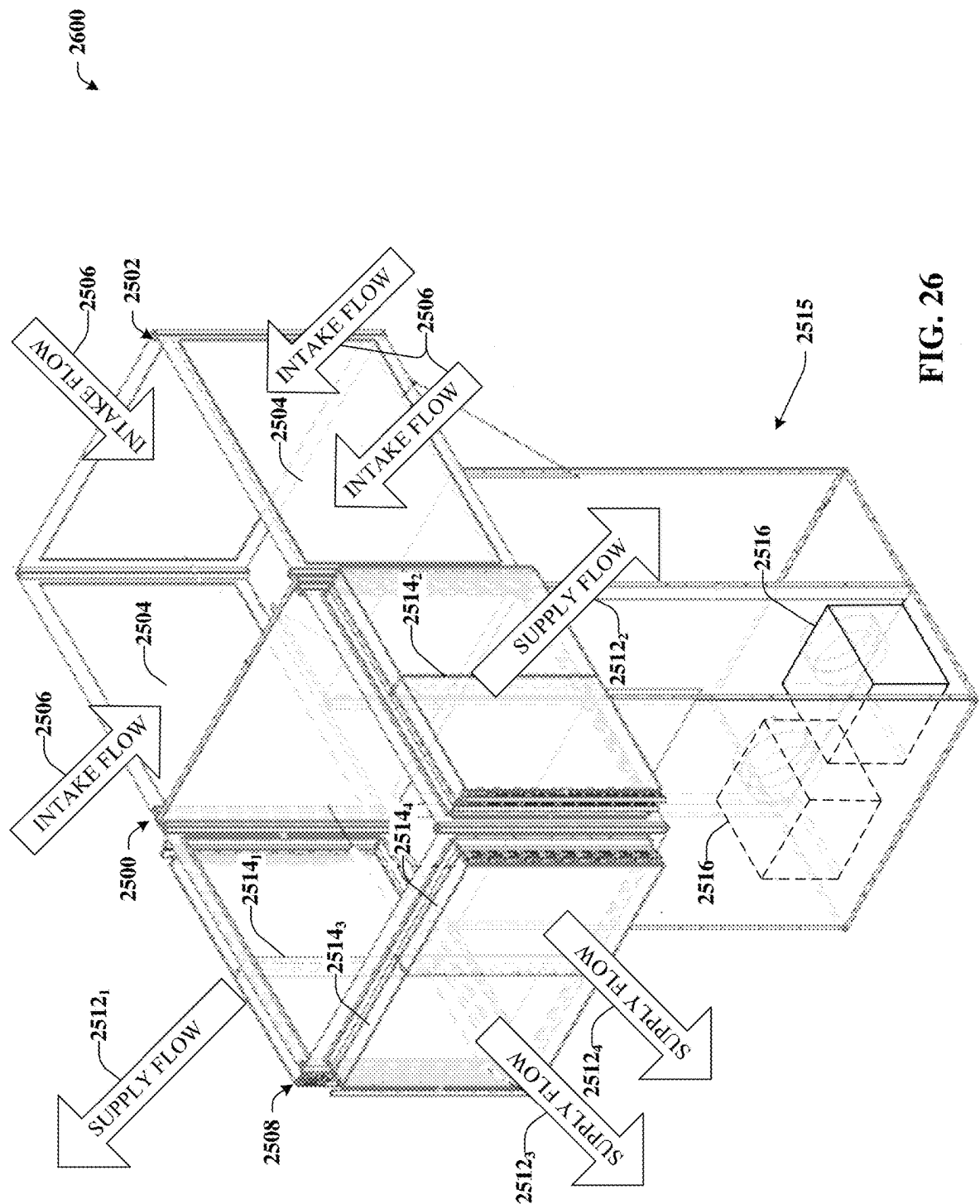


FIG. 25



2700

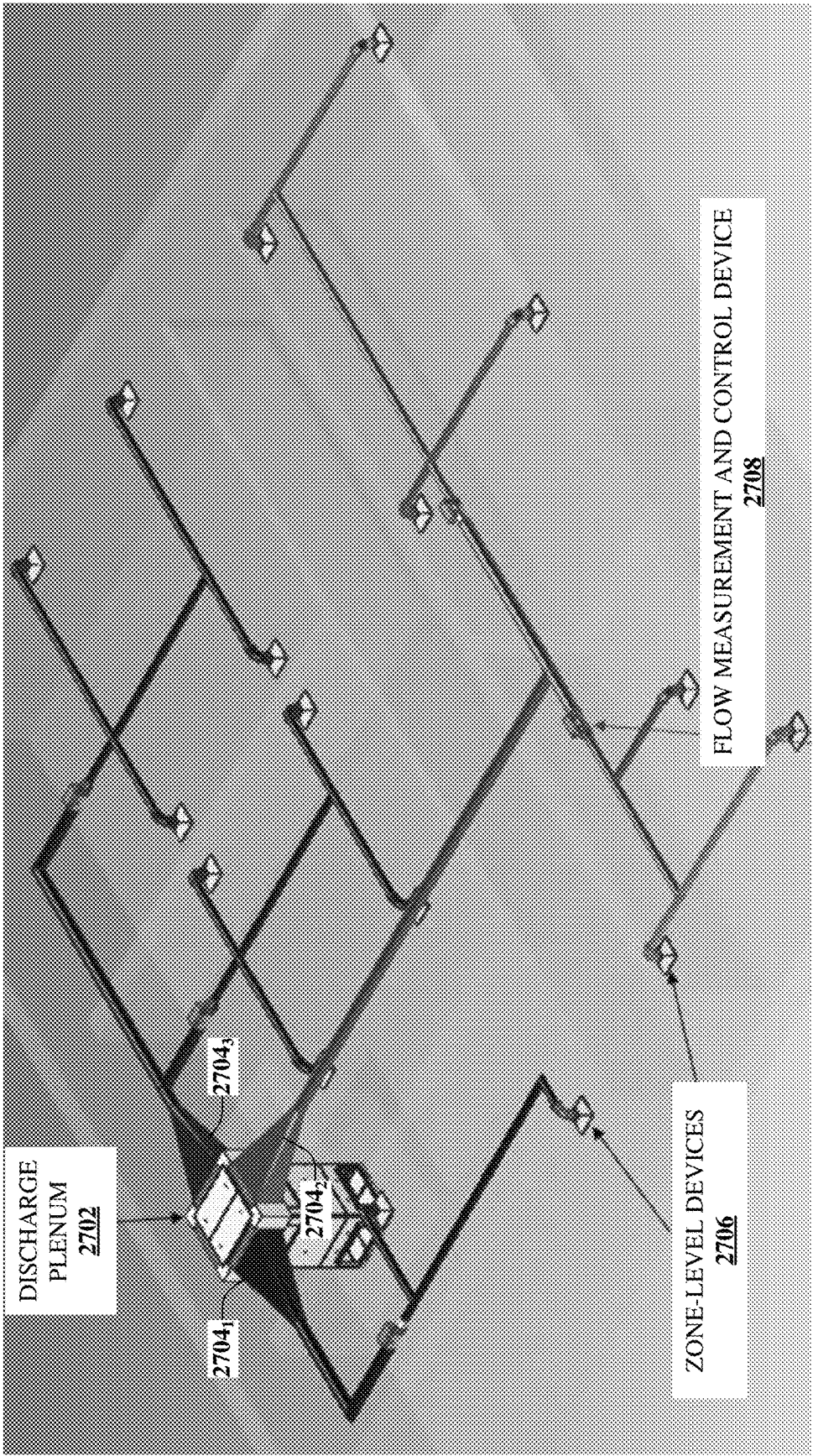


FIG. 27

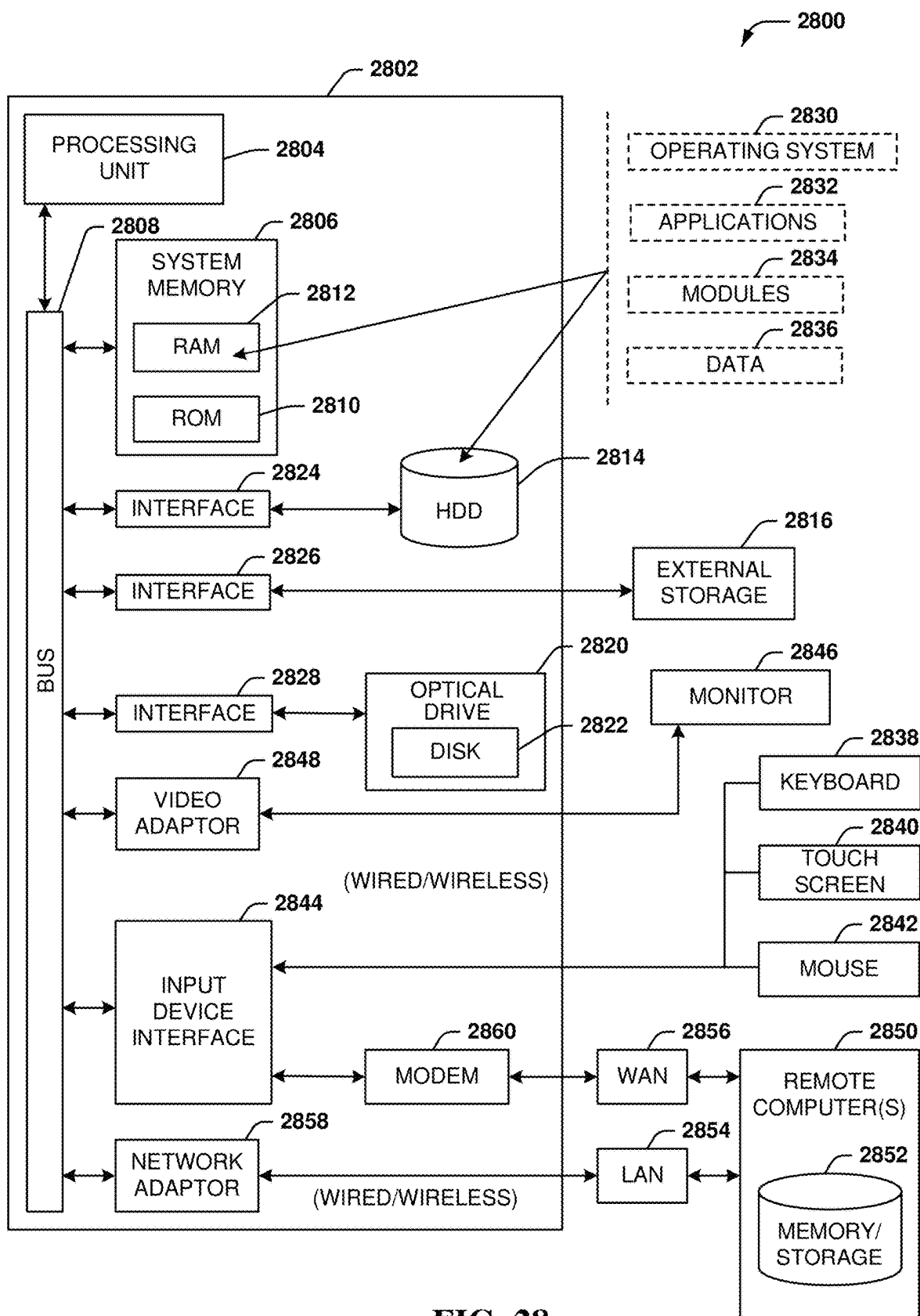


FIG. 28

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AIR HANDLER DEVICES WITH U-BEND DESIGN**RELATED APPLICATION**

The subject patent application is a continuation of, and claims priority to each of, U.S. patent application Ser. No. 18/314,420, filed May 9, 2023, and entitled "AIR HANDLER DEVICES WITH U-BEND DESIGN", which is a continuation-in-part of U.S. patent application Ser. No. 17/219,531 (Now U.S. Pat. No. 11,674,696), filed Mar. 31, 2021, and entitled "AIR HANDLER DEVICES WITH IMPROVED DESIGN AND FUNCTIONALITY", which is a continuation of U.S. patent application Ser. No. 16/930,635 (now U.S. Pat. No. 11,346,564), filed Jul. 16, 2020, and entitled "HVAC DEVICES WITH IMPROVED DESIGN AND FUNCTIONALITY", the entireties of which are hereby incorporated by reference herein.

TECHNICAL FIELD

The present disclosure is directed to improved designs for air handler devices, and more particularly to device designs comprising a U-bend design.

BACKGROUND

In several ways, modern air handler devices rely on structural designs or techniques that are many decades old without adequate improvement over that time. As such, improved designs, including those using advanced aero-acoustical physics, can provide much needed and long awaited benefits.

SUMMARY

The following presents a summary to provide a basic understanding of one or more embodiments of the disclosure. This summary is not intended to identify key or critical elements or delineate any scope of the particular embodiments or any scope of the claims. Its sole purpose is to present concepts in a simplified form as a prelude to the more detailed description that is presented later.

According to an embodiment of the present disclosure, an evase device is presented. The evase device can comprise a housing that encompasses a channel. The channel can extend in a longitudinal direction from a first side of the housing to a second side of the housing. The evase device can comprise a first opening that is situated at the first side of the housing. The first opening can be configured to receive a flow of a fluid discharged by a fan. The evase device can comprise a second opening that is situated at the second side of the housing. The second opening can be configured to discharge the flow into a duct. At the second side, the housing can have a rounded corner determined to mitigate a reverse flow of the fluid at corners of the duct.

According to an embodiment of the present disclosure, an intake device is presented. The intake device can be, e.g., intake air (or another fluid) for an HVAC system (or another system), and can operate with greatly reduced noise reduction. The intake device can comprise an intake duct. The intake duct can comprise a first opening by which a fluid enters the intake duct and a second opening by which the fluid exits the intake duct. The first opening and the second opening can be substantially circular about a longitudinal axis of the intake duct. A first circumference of the first opening can be larger than a second circumference of the

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second opening. The intake device can further comprise a top cover. The top cover can prevent the fluid from entering the intake duct in a direction along the longitudinal axis (e.g., vertical). However, the top cover can be situated a distance from the first opening, e.g., to permit the fluid to enter the intake duct in a radial direction that is radial about the longitudinal axis (e.g., horizontal). The intake device can further comprise an inner funnel that can be situated within the inner passageway of the intake duct. The inner funnel can comprise an upper portion that couples to the top cover and a lower portion that extends into the passageway. The inner funnel can comprise an outer surface that spans the upper portion and the lower portion. The outer surface can be sloped, causing the flow of the fluid entering the intake duct in the radial direction to change to the direction along the longitudinal axis.

According to an embodiment of this disclosure, an aero-acoustical fan intake device is presented. The fan intake device can be, e.g., represent an intake for air (or another fluid) for a fan of an HVAC system (or another system), and can operate with greatly reduced acoustical (e.g., noise) reduction without significant aerodynamic loss. The fan intake device can comprise an inlet face. The inlet face can comprise an inlet opening configured to receive a flow of a fluid. The fan intake device can further comprise a discharge face. The discharge face can comprise a discharge opening configured to discharge the flow of the fluid. Further still, the fan intake device can comprise a housing. The housing can encompass a flow channel that extends from the inlet opening to the discharge opening. Significantly, a cross-sectional area of the flow channel can vary between the inlet opening and the discharge opening in a manner that is determined to cause the flow of the fluid through the flow channel to continuously accelerate from a first location of the channel to the discharge opening.

According to a first embodiment of this disclosure, an air handler device is presented. The air handler device can comprise a mixing plenum. The mixing plenum can be configured to receive multiple flows of air from multiple different ducts that feed the mixing plenum. The air handler device can comprise a fan device. The fan device can be configured to receive a mixing plenum flow from the mixing plenum and to discharge a supply flow. The air handler device can further comprise a supply plenum. The supply plenum can be configured to receive the supply flow from the fan device. The supply plenum can comprise a plurality of duct interfaces. The duct interfaces can be respectively configured to interface with a different one of a plurality of supply ducts. The supply plenum can further comprise a plurality of thermal transfer units comprising a first thermal transfer unit and a second thermal transfer unit. The plurality of thermal transfer units can be respectively situated in different ones of the plurality of duct interfaces. Furthermore, the first thermal transfer unit can be configured to heat a first air flow concurrently with the second thermal transfer unit cooling a second air flow.

According to a second embodiment of this disclosure, another air handler device is presented. This air handler device (as well as the first air handler device) can be part of an HVAC product. The air handler device can be configured to circulate a flow of air within an HVAC system situated at a site the HVAC product is to be installed. The air handler device can comprise a top surface that is, relative to an installation at the site, on top of the air handler device. The air handler device can have a first height that is, relative to the installation, a height of the air handler device. The HVAC product can further comprise a heat exchange device

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that can be configured to exchange heat with the flow of air. The heat exchange device can have a second height that is, relative to the installation, a height of the heat exchange device. Further, the heat exchange device can be situated on the top surface of the air handler device, resulting in the HVAC product having a total height that is, relative to the installation, determined to be less than or equal to a defined height constraint

In some embodiments, elements described in connection with the systems and apparatuses above can be embodied in different forms such as a computer-implemented method of fabrication, or another form.

DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates a block diagram of two example views of an evase are depicted in accordance with certain embodiments of this disclosure;

FIG. 2 illustrates a block diagram of two example views of an improved evase design in accordance with certain embodiments of this disclosure;

FIG. 3 illustrates a three-dimensional graphical depiction of a first example improved evase device is illustrated in accordance with certain embodiments of this disclosure;

FIG. 4 illustrates a three-dimensional graphical depiction of a second example improved evase device is illustrated in accordance with certain embodiments of this disclosure;

FIG. 5 illustrates a graphical depiction of a first system that can be representative of an example exploded view of an example improved evase device in accordance with certain embodiments of this disclosure;

FIG. 6 illustrate a graphical depiction of a second system that can be representative of an example exploded view of an example improved evase device with an integrated fan in accordance with certain embodiments of this disclosure;

FIG. 7 illustrates a flow diagram of an example, non-limiting method for fabricating an evase device in accordance with one or more embodiments of the disclosed subject matter;

FIG. 8 illustrates a flow diagram of an example, non-limiting method that can provide additional aspects or elements in connection with fabricating an evase device in accordance with one or more embodiments of the disclosed subject matter;

FIG. 9 illustrates a three-dimensional example exploded view of an example improved intake device in accordance with certain embodiments of this disclosure;

FIG. 10 illustrates graphical depictions of an example three-dimensional view of the improved intake device and a corresponding two-dimensional cross-section view of the improved intake device in accordance with certain embodiments of this disclosure;

FIG. 11 illustrates a three-dimensional graphical depiction of an example improved intake device from a lower perspective showing a discharge of the intake device in accordance with certain embodiments of this disclosure;

FIG. 12 illustrates a flow diagram of an example, non-limiting method for fabricating an intake device in accordance with one or more embodiments of the disclosed subject matter;

FIG. 13 illustrates a flow diagram of an example, non-limiting method that can provide additional aspects or elements in connection with fabricating an intake device in accordance with one or more embodiments of the disclosed subject matter; and

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FIG. 14 illustrates a schematic diagram showing a cross-section of an a first example of a fan intake device in accordance with certain embodiments of this disclosure;

FIG. 15 illustrates a schematic diagram showing a cross-section of a second example of a fan intake device having a bulb-shaped inlet face in accordance with certain embodiments of this disclosure;

FIG. 16 illustrates a flow diagram of an example, non-limiting method for fabricating a fan intake device in accordance with one or more embodiments of the disclosed subject matter;

FIG. 17 illustrates a flow diagram of an example, non-limiting method that can provide additional aspects or elements in connection with fabricating a fan intake device in accordance with one or more embodiments of the disclosed subject matter; and

FIG. 18 illustrates a schematic diagram showing a cross-section of a first example air handler product in accordance with certain embodiments of this disclosure;

FIG. 19 illustrates a three-dimensional representation of a first example air handler product having three supply duct interfaces in accordance with certain embodiments of this disclosure;

FIG. 20 illustrates a three-dimensional representation of a second example air handler product having multiple fans and four supply duct interfaces in accordance with certain embodiments of this disclosure;

FIG. 21 illustrates a schematic diagram showing a cross-section of an a second example air handler product in accordance with certain embodiments of this disclosure; and

FIG. 22 illustrates a flow diagram of an example, non-limiting method for fabricating an air handler product in accordance with one or more embodiments of the disclosed subject matter;

FIG. 23 illustrates a flow diagram of an example, non-limiting method that can provide additional aspects or elements in connection with fabricating an air handler product in accordance with one or more embodiments of the disclosed subject matter;

FIG. 24 illustrates a schematic diagram 2400 depicting an example building layout for accommodating air handler devices and other HVAC components, devices, or elements in accordance with certain embodiments of this disclosure;

FIG. 25 illustrates a schematic block diagram of an example air handler device 2500 illustrated having a U-bend structure in accordance with certain embodiments of this disclosure;

FIG. 26 illustrates a schematic block diagram 2600 of the example air handler device 2500 having a U-bend structure 2515 illustrated in an isometric view in accordance with certain embodiments of this disclosure;

FIG. 27 illustrates a schematic diagram 2700 showing an example discharge plenum having multiple temperature controlled discharge devices integrated into a building duct layout in accordance with certain embodiments of this disclosure; and

FIG. 28 illustrates a block diagram of an example, non-limiting computing environment by which one or more embodiments described herein can be fabricated or otherwise facilitated.

DETAILED DESCRIPTION

The disclosed subject matter is now described with reference to the drawings, wherein like reference numerals are used to refer to like elements throughout. In the following description, for purposes of explanation, numerous specific

details are set forth in order to provide a thorough understanding of the disclosed subject matter. It may be evident, however, that the disclosed subject matter may be practiced without these specific details. In other instances, well-known structures and devices are shown in block diagram form in order to facilitate describing the disclosed subject matter.

Example Evase Apparatus

Referring now to the drawings, with initial reference to FIG. 1, a block diagram 100 of two example views of an evase are depicted in accordance with certain embodiments of this disclosure. In the HVAC domain, an evase can operate as a duct transition. For instance, the evase can connect a fan outlet, typically circular in shape to match fan impeller sweep, to a supply duct that is typically larger in size and rectangular in shape. This duct size and shape transition can lead to undesired consequences, some of which are discussed in connection with evase 102.

The left side of FIG. 1 illustrates a longitudinal axis perspective of evase 102, for instance a view as seen from the duct, with the longitudinal axis that extends into the page and intersects at point 103. Evase 102 can comprise inlet 104 that is circular in shape and can be configured to receive a flow of a fluid discharged by a fan (not shown). Evase 102 can further comprise outlet 108 that is rectangular in shape and can be configured to discharge the fluid into a supply duct (see duct 112).

The right side of FIG. 1 depicts evase 102 from the perspective of a cross-section along diagonal line 111 that runs from the top-right corner to the bottom-left corner, which can represent a projected diagonal view. Circular inlet 104 receives a flow of fluid from the fan, which is illustrated by fluid flow lines 106 (dashed lines). Because outlet 108 is larger in size, the fluid gradually expands through the interior chamber of evase 102. This gradual expansion continues well into duct 112.

As shown, a longest distance 110 between inlet 104 and outlet 108 is represented by some point on the circular ring of inlet 104 to a rectangular corner of the outlet. Distance 110 can represent a significant factor in the efficacy of evase 102 because it can approximately represent a potentially longest path for the flow of fluid through evase 102. Based on ordinary geometric principles, angle 114 is a function of and therefore constrained by evase length 116 and distance 110.

Further, due to the velocity of the fluid discharged by the fan, a common situation arises in other evase devices such as evase 102 in which angle 114 is too large to facilitate fluid flow to flow along longest path 110. As a result, significant reverse flow 118 arises. This reverse flow 118 leads to a number of disadvantages.

For example, in conventional systems, a decrease in kinetic energy between the fan discharge and larger downstream duct is entirely lost, being converted into heat carried by the flow. The effective fan efficiency is greatly reduced, in some cases by nearly 50%. To account for this loss, a larger fan motor than would otherwise be required is generally utilized and/or the fan is operated at a higher revolutions per minute (RPM) than needed otherwise. Generally, higher operating RPM's mean a noisier equipment room and reduced motor lifetime.

Further, because HVAC systems are generally configured to supply cool air to the building, the heating of the flow outlined above requires either increasing the total flow to obtain the same cooling effect from the warmer air or lowering heat rejection temperature to compensate for that

extra heat. In any case the extra heat places an additional burden on the thermal rejection system, which must also extract heat equal to the heating caused by the evase energy loss. Poor evase efficiency is paid for by increased operating cost for the fan and heat rejection sections.

Further still, practical HVAC systems rarely have sufficient space (e.g., 5 to 10 duct diameters of duct length) required for the flow to straighten out downstream of the ineffective evase. In practice the flow is often turned and/or divided almost immediately following the evase. The non-uniform flow increases losses in turns and will not follow the geometry of a split unless downstream dampers are feathered to limit flow to the favored channel, contributing to additional losses to the system together with additional noise from dampers up in the ceiling space, which can adversely affect occupants below the dampers.

Referring now to FIG. 2, a block diagram 200 of two example views of an improved evase design are depicted in accordance with certain embodiments of this disclosure. In that regard, a longitudinal axis perspective of evase 202 is illustrated on the left side of FIG. 2, while the right side of FIG. 2 depicts a cross-section along diagonal line 211 that runs from the top-right corner to the bottom-left corner, which can represent a projected diagonal view. Evase 202 can comprise housing 203 shown in dashed lines. Housing 203 can encompass a channel that extends in a longitudinal direction from a first side (e.g., inlet side) of housing 203 to a second side (e.g., outlet side) of housing 203. A length of this channel is illustrated by reference numeral 216.

Evase 202 can further comprise first opening 204 situated at the first side of housing 203 (e.g., inlet side). First opening 204 can be configured to receive a flow 206 (e.g., indicated by dashed lines) of a fluid discharge by a fan. As depicted, first opening can have a circular shape that can match or scale to the fan or impeller blades of the fan, however first opening 204 can be any suitable shape.

Evase 202 can further comprise second opening 208. Second opening 208 can be situated at the second side of housing 203 (e.g., outlet side). Second opening 208 can be configured to discharge flow 206 into duct 212. Advantageously, at second opening 208, housing 203 can have rounded corners 205. Rounded corners 205 can be configured to or determined to mitigate a reverse flow (e.g., see reverse flow 118 of FIG. 1) at corners of duct 212. In some implementations, reverse flow 118 can be entirely prevented, while in other cases reverse flow 118 can be significantly reduced, resulting in much smaller effective reverse flow shown here at reference numeral 218.

In more detail, duct 212 can have a rectangular shape and corners of duct 212 can be squared corners. As evase 202 can be coupled to duct 212 and/or serve as an interface to duct 212, corners of an exterior portion of housing 203 can be rectangular shaped that can be variably sized to correspond to or match a size and shape of duct 212. However, an interior portion of housing 203 can exhibit rounded corners 205.

By way of comparison with evase 102 of FIG. 1, due to rounded corners 205, length 210 is shorter than the corresponding length 110 of FIG. 1, the latter of which extends to the corner of duct 112. Assuming channel length 216 is approximately the same as length 116 (which is often a physical constraint of a given system or customer site), one result of length 210 being shorter is that angle 214 is less than angle 114. As such, flow 206 can readily flow through a larger volume of both the evase channel and duct 212 instead of being more inclined flow in regions not much

larger than opening **104** until much farther downstream of duct **112**, as shown in FIG. 1.

In some embodiments, a shape of rounded corners **205** is determined or designed based on a Reynolds number calculation. It is appreciated that the fluid discharged by the fan can have a velocity pressure that is converted to static pressure less an impact loss. In some embodiments, the shape of rounded corners **205** can be determined to reduce this impact loss and therefore cause a net positive change in static pressure.

It is appreciated that the shape of rounded corners **205** in this example is representative of a square shaped housing **203** with a suitably sized duct **212**. In other embodiments, housing **203** and/or ducts **212** might be different shapes, for example, rectangular in shape. In those cases, and further based on a difference between sizes or shapes of housing **203** and duct **212**, the prominence of rounded corners **205** can differ from what is depicted in this example. For instance, consider the case of a more rectangular shape in which a width of the longitudinal axis perspective is greater than the height. In that case, rounded corners **205** can have a similar height to what is depicted, but with a greater length. At some threshold, the rounded corners **205** may meet one of the two neighboring rounded corners **205**. For example, both of the rounded corners **205** at the top of the figure can intersect with those at the bottom of the figure, causing the shape of the opening to resemble a flattened oval. In other embodiments, such as when a given rounded corner **205** intersects with both neighboring rounded corners **205**, the shape of the opening can resemble a circle. These different shapes, as well as other suitable shapes are considered to be within the scope of the disclosed subject matter.

To continue the above description, when comparing evase **102** (e.g., comprising squared corners) to evase **202** (e.g., comprising rounded corners **205**), a change in static pressure (ASP) is expected to be zero. In contrast, ASP for evase **202** can be a function of a difference between a velocity pressure (VP) at first opening **204** (e.g., VP_1) and a VP within the duct **212** at some defined distance downstream of evase **202** (e.g., VP_0). As one example, ASP can equal $8 \cdot (VP_1 - VP_0)$. This can reduce utilized fan horsepower by 20-30%, sometimes allowing selection of the next smaller motor size, which can significantly reduce costs and overhead.

Furthermore, certain disadvantages listed above with respect to other systems (e.g., evase **102**) are reversed for improved evase **202**. For instance, evase **202** can result in reduced fan RPM's and installed horsepower, quieter equipment rooms, longer motor life, and more even discharge flow so that elbows and splitters work more efficiently. In addition, heat rejection load can be reduced. Both fan and heat rejection operating costs are reduced.

While not shown here, in some embodiments, evase **202** can further comprise an intermediate baffle that can further enhance advantages discussed herein, which is further detailed in connection with FIGS. 4 and 5. Further, in some embodiments, some portions of housing **203** or another housing or container can be filled with a material that absorbs sound, which is also discussed in more detail in connection with FIG. 4.

As previously noted first opening **204** can have a circular or annular shape. In some embodiments, this circular or annular shape can have a diameter that corresponds to or matches an impeller hub diameter of the fan. In some embodiments, the fan can be mounted to or embedded in housing **203**, which is further detailed in connection with FIG. 6

Turning now to FIG. 3, a three-dimensional graphical depiction of a first example improved evase device **300** is illustrated in accordance with certain embodiments of this disclosure. As illustrated, evase device **300** can comprise housing **302** that encompasses a channel (e.g., in which fluid flows) that extends in a longitudinal direction. This longitudinal direction can be represented by longitudinal axis **304** and the channel can extend from first side **306** of housing **302** (e.g., right hand side) to second side **308** of housing **302** (e.g., left hand side).

Evase device **300** can comprise first opening **310** situated at first side **306** of housing **302**. First opening **310** can be configured to receive flow **312** of a fluid discharged by a fan. Evase device **300** can further comprise second opening **314** situated at second side **308** of housing **302**. Second opening **314** can be configured to discharge flow **312** into a duct. Beneficially, at second side **308**, housing **302** has one or more rounded corners **316**. Rounded corners **316** can be determined to mitigate a reverse flow of the fluid that might otherwise occur at corners of the duct.

Referring now to FIG. 4, a three-dimensional graphical depiction of a second example improved evase device **400** is illustrated in accordance with certain embodiments of this disclosure. As illustrated, evase device **400** comprises all or a portion of example evase device **300**. In this view rounded corners **316** can be seen. In addition, evase device **400** comprises an exterior housing **402** that encloses evase device **300** and other elements. Housing **402** can further include a material that absorbs or mitigate sound.

In addition, evase device **400** can further include an intermediate baffle **404**. Intermediate baffle **404** can further improve functional advantages such as improving mitigation of reverse flow **116**. Intermediate baffle **404** can operate reduce necessary length (e.g., evase channel length **216**) of the evase by about half. For example, by including intermediate baffle **404**, evase channel length **216** can be about half the size as what might otherwise be needed in order to effectuate proper flow with mitigated reverse flow. Such can be a significant advantage, particularly in implementations where there is not a lot of space at the installation site for an evase device

Intermediate baffle **404** can operate to guide the outer portion of the flow to expand at nearly twice the angle (e.g., angle **214**) otherwise possible without engendering complete flow separation from the rapidly expanding outer boundaries. Intermediate baffle **404** can also provides superior sound attenuation by placing additional absorption material in the middle of the flow where the outer and inner sound absorbing materials are least effective. As illustrated, intermediate baffle **404** can also exhibit or comprise rounded corners **406**. Rounded corners **406** of intermediate baffle **404** can exhibit the same or a different gradient as rounded corners **316** of evase device **300**, either of which can be based on a Reynolds number calculation.

Turning now to FIG. 5, a graphical depiction illustrates system **500** that can be representative of an example exploded view of evase device **400** in accordance with certain embodiments of this disclosure. In this example, additional elements of evase device **400** can be identified. It is appreciated that evase device **400** can contain all or only a portion of elements described in connection with system **500**, which are intended to be exemplary or representative, but also non-limiting. For instance, other elements may be present and certain elements discussed here may be optional or excluded.

System **500** can include evase **502**, which can be substantially similar to evase **300**. At opposing sides of evase

502, the device can be coupled to interface elements such as annular fan interface element **504** and rectangular duct interface element **506**. Elements **504** and **506** can essentially line opposing openings (e.g., first opening **306** and second opening **308**). Hence, rectangular duct interface element **506** can exhibit rounded corners that match or correspond to rounded corners **316**.

System **500** can include intermediate baffle **508**, which can be substantially similar to intermediate baffle **404**. Likewise intermediate baffle **508** can be coupled to interface elements **510** and **512** that are situated on opposing sides of intermediate baffle **508**. When assembled, intermediate baffle **508** can fit inside evase device **502** and a central axis (e.g., longitudinal axis **304**) can include central pod **514**. Sizing for central pod **514** can match the impeller hub, eliminating the impact loss that otherwise occurs at the impeller hub region, and which can be built into the fan curves according to testing. A fan tested at, e.g., 78% efficient may become, e.g., 83% efficient, representing a 5-10% increase in efficiency. Central pod **514** may be conical in shape, resulting in a smaller area at the discharge, further reducing impact losses. The net effect of central pod **514** can translate to an 80% to 90% recovery of the impact loss behind the impeller hub.

Support for the assembled elements at the intake side can be provided by support elements **516**, while similar support at the opposing side can be provided by support elements **522**. Rectangular frame **518** and intake side face plate **520** can further be assembled.

On the opposing side (e.g., discharge side), L-shaped support elements **524** and support rod elements **526** can be assembled. These support elements (e.g., **522**, **524**, and **526**) can provide support, such as support for elements fitted inside housing **528**, which can include evase **502**, intermediate baffle **508**, and central pod **514**. System **500** can further include discharge side rectangular frame **530**, discharge side face plate **532** and top frame **534**.

With reference now to FIG. 6, a graphical depiction illustrates system **600** that can be representative of an example exploded view of evase device **400** with an integrated fan in accordance with certain embodiments of this disclosure. System **600** can include all or a portion of elements detailed in connection with system **500**, including all or some portion of elements **502-534**. In addition, system **600** can further include an integrated fan.

For example, system **600** can include fan hub **602** that can couple to all or a portion of central pod **514**, interface elements **504**, **510**, intermediate baffle **508**, and/or evase **502**. As illustrated, impeller housing **602** can include straightening vanes and a sleeve having an impeller hub diameter to contain the motor. System **600** can further include motor **604** and impeller **606**. Hence, in some embodiments, housing **528** of evase system **500**, or elements therein such as intermediate baffle **508** or evase **502**, can operate as a housing for certain elements of the fan, such as motor **604**.

As can be observed, in some embodiments, motor **604** can be situated within the interior channel of evase device **300** and/or within an interior channel of intermediate baffle **508**, which itself can be situated within the interior channel of evase device **300**. In some embodiments, central pod **514** can have dimensions that match or correspond to dimensions of motor **604**. In some embodiments, central pod **514** can contain all or portions of motor **604** such that central pod **514** can match up right behind the impeller hub.

Advantageously, situating fan elements (e.g., motor **604**, etc.) inside the interior channel of evase device **300** can

result in significant space savings, which can further increase the efficacy of evase devices detailed herein. For example, turning back FIG. 2, flow **206** can be considered to begin just behind location of impeller **606**. In other systems, where the fan motor is farther upstream, the length of the motor reduces the available length for evase **202** because an evase channel length **216** can be constrained by the locations of the fan and duct **212**. However, by placing motor **604** within the channel of evase **202** (or other evase devices detailed herein), evase channel length **216** can be increased by a similar amount. As such, angle **214** can be decreased, which can further prevent or mitigate reverse flow **218** as well as further other advantages detailed herein.

Example Methods of Fabricating an Evase Device

FIGS. 7 and 8 illustrate various methodologies in accordance with the disclosed subject matter. While, for purposes of simplicity of explanation, the methodologies are shown and described as a series of acts, it is to be understood and appreciated that the disclosed subject matter is not limited by the order of acts, as some acts can occur in different orders and/or concurrently with other acts from that shown and described herein. For example, those skilled in the art will understand and appreciate that a methodology could alternatively be represented as a series of interrelated states or events, such as in a state diagram. Moreover, not all illustrated acts can be required to implement a methodology in accordance with the disclosed subject matter. Additionally, it should be further appreciated that the methodologies disclosed hereinafter and throughout this specification are capable of being stored on an article of manufacture to facilitate transporting and transferring such methodologies to computers.

FIG. 7 illustrates a flow diagram **700** of an example, non-limiting method for fabricating an evase device in accordance with one or more embodiments of the disclosed subject matter. For example, a device comprising a processor can perform certain operations. Examples of said processor as well as other suitable computer or computing-based elements, can be found with reference to FIG. 24, and can be used in connection with implementing one or more of the devices or components shown and described in connection with figures disclosed herein.

At reference numeral **702**, the device comprising the processor can facilitate forming a housing that encompasses a channel. The channel can extend in a longitudinal direction from a first side of the housing to a second side of the housing. As used herein, the term 'forming' can comprise any suitable structural manipulation of a material or element including concepts directed to creating a material or element, structurally manipulating a material or element, or assembling a material or element.

At reference numeral **704**, the device can facilitate forming a first opening in the housing that is situated at the first side of the housing, wherein the first opening is configured to receive a flow of a fluid discharged by a fan. In some embodiments, the first opening can be sized to match or correspond to certain elements of a fan, such as an impeller of the fan. In some embodiments, the first side of the housing can be coupled to the fan.

At reference numeral **706**, the device can facilitate forming a second opening in the housing that is situated at the second side of the housing, wherein the second opening is configured to discharge the flow of a fluid into a duct. In some embodiments, the second opening can be sized to

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match or correspond to a duct. In some embodiments, the second side can be coupled to the duct.

At reference numeral **708**, the device can facilitate forming rounded corners at the second side of the housing, wherein the rounded corners are determined to mitigate a reverse flow of the fluid at corners of the duct. Method **700** can proceed to insert **A**, which is further detailed in connection with FIG. **8**, or terminate.

Turning now to FIG. **8**, illustrated is a flow diagram **800** of an example, non-limiting method that can provide additional aspects or elements in connection with fabricating an evase device in accordance with one or more embodiments of the disclosed subject matter.

At reference numeral **802**, the device can facilitate forming or assembling, by the device, an intermediate baffle situated in the channel. In some embodiments, the intermediate baffle can comprise rounded corners at an interface region to the duct. In some embodiments, the intermediate baffle can comprise a central pod situated within a baffle channel.

At reference numeral **804**, the device can facilitate assembling or forming the fan situated within the housing. As illustrated at reference numeral **806**, in some embodiments, a motor of the fan can be situated within the channel and/or within the baffle channel. In some embodiments, the motor can have dimensions that match or correspond to dimensions of the central pod.

Example Intake Apparatus (e.g., Radiax)

Turning now to FIG. **9**, a graphical depiction is illustrated of an example three-dimensional exploded view of an improved intake device **900** in accordance with certain embodiments of this disclosure. Intake device **900** can operate as an intake for a fluid, such as outside air, for an HVAC system. It is common practice for HVAC systems to mix outside air with tempered or conditioned air, typically mixed with return air from a controlled environment. However, conventional intake devices suffer from certain disadvantages.

For example, conventional intake device tend to be noisy, especially for large systems and/or large buildings. Standard inlet bell designs tend to be wide open at the inlet or mouth, both acoustically and aerodynamically. These designs can lead to significant acoustical noise issues and aerodynamic losses such that costs are increased. For instance, more energy is consumed and/or a larger system than might otherwise be needed is selected. In contrast, one significant advantage of certain embodiments detailed herein is that the intake flow is not wide open and is turned (e.g., about 90 degrees) and accelerated. This turning flow can mitigate direct acoustic radiation and can do so without introducing significant aerodynamic losses. Furthermore, due in part to the disclosed design elements, the disclosed intake apparatus can be made small enough so that an associated fan can be placed closer to a floor or wall than is possible using a standard inlet bell, which can mitigate potential building HVAC construction or upgrade issues and/or open up new possibilities in that regard.

FIG. **9** is intended to be referenced in conjunction with FIG. **10**, showing graphical depictions **1000** of an example three-dimensional view (left side of page) of an example assembled improved intake device **900** and a corresponding two-dimensional cross-section view (right side of page) of the improved intake device **900** in accordance with certain embodiments of this disclosure.

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It is appreciated that intake device **900**, which can be referred to herein as a “radiax”, “radiax device” or other similar variations, and can contain all or only a portion of elements described, which are intended to be exemplary or representative, but also non-limiting. For instance, other elements may be present and certain elements discussed here may be optional or excluded.

Intake device **900** can comprise intake duct **902**. Intake duct **902** can comprise first opening **904** by which a fluid enters the intake duct and second opening **906** by which the fluid exits intake duct **902**. First opening **904** and second opening **906** can be substantially circular or annular in shape about longitudinal axis **908** of intake duct **902**. It is appreciated that a first circumference of first opening **904** can be larger than a second circumference of second opening **906**, which is best observed with reference to the cross-section view illustrated by FIG. **10**. Intake duct **902** can further comprise interior surface **910**. Interior surface **910** can extend from first opening **904** to second opening **906**, providing a passageway for a fluid to flow. That second opening **906** is smaller than first opening **904** can be significant for reasons further detailed below such as, for instance, fluid flow through the passageway can undergo acceleration after entering intake duct **902**.

Intake device **900** can further comprise top cover **912**. Being situated on top, top cover **912** can prevent fluid from entering intake duct **902** in a direction along longitudinal axis **908**, which is illustrated by reference numeral **918**, showing fluid flow along longitudinal axis **908** being blocked by top cover **912**. As illustrated by reference numeral **919**, other non-radial flows can also be blocked by top cover **912**. On the other hand, because top cover **912** can be situated some distance **913** away from first opening **904**, such can permit the fluid to enter the intake duct **902** in a radial direction that is radial about the longitudinal axis, as illustrated by reference numeral **916**.

Intake device **900** can further comprise inner funnel **914**. Inner funnel **914** can comprise upper portion **920** that can couple to top cover **912**. Inner funnel **914** can comprise lower portion **922** that can extend into the passageway of intake duct **902**. Inner funnel **914** can further comprise outer surface **924** that can span from upper portion **920** to lower portion **922**. This span of outer surface **924** can be sloped causing the flow entering intake device **900** in the radial direction (e.g., flow **916**) to change substantially to the direction along longitudinal axis **908**. As can be seen, the passageway of intake duct **902**, through which fluid flows, is bounded by the regions between interior surface **910** (of intake duct **902**) and outer surface **924** (of inner funnel **914**).

In some embodiments, interior surface **910** of intake duct **902** can provide a smoothly tapered surface that encompasses a substantially funnel-shaped passageway for the flow of the fluid. Such is best illustrated by the white regions of the two-dimensional cross-section view illustrated by FIG. **10**. As noted, such can create a gradual change in the angle of the fluid flow. In some embodiments, the angular difference of the change in direction of the flow, e.g., representing a difference between the radial direction and the direction along longitudinal axis **908** can be in a range of about 80 degrees to about 100 degrees.

As can be observed in this embodiment, a cross-sectional area of the passageway (e.g., taking slices along longitudinal axis **908**), can decrease when moving from first opening **904** to second opening **906**. In other words, the passageway narrows as fluid flows farther into intake device **900**. In some embodiments, this narrowing can be determined to cause the flow of the fluid in the passageway to increase in

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velocity and/or to accelerate when flowing toward second opening 906, where the cross-sectional area can be the smallest. This increase in velocity and/or acceleration can be determined to have a damping effect on turbulence flow, which can, inter alia, significantly decrease noise of intake device 900 relative to other intake devices known in the marketplace.

In some embodiments, geometries of outer surface 924 of inner funnel 914 and interior surface 910 of intake duct 902 can be determined to cause the flow to be laminar. A laminar flow can be one that has high momentum diffusion while maintaining low momentum convection. Typically, a laminar flow occurs when the fluid flows in parallel layers with no disruption between them (e.g., no eddies or swirls). In some embodiments, these geometries of outer surface 924 and interior surface 910 can be determined to mitigate losses due to flow separation along bounding surfaces of a turning flow (e.g., the flow that is turning within intake device 900). The turning flow can represent the flow entering in the radial direction 916 and turning toward the longitudinal direction 908. In some embodiments, these geometries are determined to cause at least a portion of the flow entering intake device 900 to follow an elliptical path when changing from radial direction 916 to the direction along longitudinal axis 908.

In addition to elements detailed above, in some embodiments, intake device 900 can optionally include several other elements that are now described. For example, intake device 900 can include angled cover support 926 that can couple to top cover 912 and can include bottom funnel cover 928 that can attach to lower portion 922 of inner funnel 914. Intake device 900 can further include center support structure 930 that can couple to one or both inner funnel 914 and intake duct 902. For instance center support structure 930 can support the positioning or orientation of inner funnel 914 within the passageway of intake duct 902. Further, intake device 900 can include bottom duct cover 932 that can couple to a bottom side of intake duct 902. Support rods 934 and angled ring 936 can also be included in intake device 900.

It is appreciated that in some embodiments, interior portions of inner funnel 914 and interior portions of intake duct 902 can be filled with a material that absorbs or mitigates noise or sound. For example, one or both inner funnel 914 and intake duct 902 can be filled with fiberglass or another material having sound absorption properties.

Turning now to FIG. 11, a three-dimensional graphical depiction of an example assembled intake device 1100 is illustrated from a lower perspective showing a discharge of the intake device in accordance with certain embodiments of this disclosure. In this example, intake duct 902 is prominent and shown from the lower perspective. Inner funnel 914 is apparent both at the intake region (e.g., upper portion 920) and the discharge region (e.g., lower portion 922). Center support structures 930 that can support inner funnel 914 within the passageway of intake duct 902 can also be observed from this perspective, as well as bottom duct cover 932 and bottom funnel cover 928.

Example Methods of Fabricating an Intake Device

FIGS. 12 and 13 illustrate various methodologies in accordance with the disclosed subject matter. While, for purposes of simplicity of explanation, the methodologies are shown and described as a series of acts, it is to be understood and appreciated that the disclosed subject matter is not limited by the order of acts, as some acts can occur in different orders and/or concurrently with other acts from that

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shown and described herein. For example, those skilled in the art will understand and appreciate that a methodology could alternatively be represented as a series of interrelated states or events, such as in a state diagram. Moreover, not all illustrated acts can be required to implement a methodology in accordance with the disclosed subject matter. Additionally, it should be further appreciated that the methodologies disclosed hereinafter and throughout this specification are capable of being stored on an article of manufacture to facilitate transporting and transferring such methodologies to computers.

FIG. 12 illustrates a flow diagram 1200 of an example, non-limiting method for fabricating an intake device in accordance with one or more embodiments of the disclosed subject matter. For example, a device comprising a processor can perform certain operations. Examples of said processor as well as other suitable computer or computing-based elements, can be found with reference to FIG. 24, and can be used in connection with implementing one or more of the devices or components shown and described in connection with figures disclosed herein.

At reference numeral 1202, the device comprising the processor can facilitate forming an intake duct. The intake duct can have a cover plate that is configured to prevent a fluid from entering the intake duct in a longitudinal direction. Further, the intake duct can be configured to receive, at a first end, the fluid in a radial direction and to discharge, at a second end, the fluid substantially in the longitudinal direction.

At reference numeral 1204, the device can facilitate forming an inner funnel. The inner funnel can be situated between the cover plate and the second end. In some embodiments, the inner funnel can be coupled to the cover plate, e.g., to a bottom side of the cover plate. Advantageously, the inner funnel can have a funnel geometry that causes the fluid to follow an elliptical path after entering the intake device from substantially the radial direction. In other words, the flow of the fluid, when changing from the radial direction to the longitudinal direction within the intake device is determined to follow the elliptical path. Method 1200 can proceed to insert A, which is further detailed in connection with FIG. 13, or terminate.

Turning now to FIG. 13, illustrated is a flow diagram 1300 of an example, non-limiting method that can provide additional aspects or elements in connection with fabricating an intake device in accordance with one or more embodiments of the disclosed subject matter.

At reference numeral 1302, the forming the intake duct and the forming the inner funnel further comprises determining, by the device, that geometries of the intake duct and the inner funnel cause a flow of the fluid through the intake device to be laminar.

At reference numeral 1304, the forming the intake duct and the forming the inner funnel further comprises determining, by the device, that geometries of the intake duct and the inner funnel result in a continuously decreasing cross-sectional area when moving along the longitudinal axis toward the second end.

At reference numeral 1306, the forming the intake duct and the forming the inner funnel further comprises determining, by the device, that geometries of the intake duct and the inner funnel cause a flow of the fluid through the intake device to accelerate when moving toward the second end.

Example Fan Intake Apparatus (e.g., Uniax)

Turning now to FIG. 14, a schematic diagram is illustrated showing a cross-section of an example improved fan intake

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device **1400** in accordance with certain embodiments of this disclosure. Fan intake device **1400** can operate as an intake for a fluid, such as air, for a fan of an HVAC system. For instance a discharge of the fan intake device can feed into an HVAC fan or other suitable device. Conventional fan intake devices can lead to significant noise. Attempts by conventional fan intake devices to mitigate noise tend to result in pressure loss and flow intake irregularities, which can lead to a less efficient system. Designs and techniques disclosed herein can provide a fan intake device that can significantly reduce noise without resulting in pressure losses and/or flow intake irregularities common to previous systems or devices.

It is appreciated that fan intake device **1400**, which can be referred to herein as an “aero-acoustical fan intake device”, a “uniax”, a “uniax device” or other similar variations, can contain all or only a portion of elements described, which are intended to be exemplary or representative, but also non-limiting. For instance, other elements may be present and certain elements discussed here may be optional or excluded, one example of which is duct/plenum **1420**.

As illustrated, aero-acoustical fan intake device **1400** can comprise an inlet face that can broadly represent a side or face of device **1400** that receives a fluid. This inlet face is illustrated in FIG. **14** by element **1402** which encompasses the inlet face. Hereinafter, this inlet face is referred to as inlet face **1402**. Inlet face **1402** can comprise an inlet opening(s), illustrated by elements **1404** that encompass the opening(s). Hereinafter the inlet openings are referred to as inlet opening (s) **1404**, which can be configured to receive a flow of a fluid **1406**. As illustrated, fluid **1406** on the left side of FIG. **14** flows toward inlet openings **1404**. It is understood, fluid **1406** can flow toward inlet opening (2) **1404** from any suitable direction and/or point of origin, which can vary based a size and shape of (optional) duct/plenum **1420** as well as based on whether duct/plenum **1420** is present.

Fan intake device **1400** can further comprise a discharge face. Moving toward the right side of FIG. **14**, element **1408** encompasses the discharge face, which is hereinafter referred to as discharge face **1408**. Likewise, discharge face **1408** can comprise a discharge opening(s) **1410** that can be configured to discharge the flow of the fluid **1406** to a fan device. As illustrated fluid **1406** ultimately gets discharged toward a fan device (not shown), such as toward impellers of the fan device. It is appreciated that the fan device can be a centrifugal fan, a plenum fan, an axial fan or another suitable type of fan, which can be selected based upon implementation.

Fan intake device **1400** can further comprise housing **1412**. Housing **1412** can encompass flow channel **1414** that can extend from inlet opening **1404** to discharge opening **1410**. In other words, flow channel **1414** represents a constrained path through which fluid **1406** must flow in order to reach the fan device. Based on the geometry and/or design of fan intake device **1400**, and specifically flow channel **1414**, the flow of fluid **1406** can be manipulated to provide certain advantages detailed above and herein. It is appreciated that although this view illustrates a cross-section of fan intake device **1400**, it can be readily visualized that inlet opening **1404** and discharge opening **1410** can have an annulus shape (e.g., ring-shaped).

Flow channel **1414** can be designed such that a cross-sectional area of flow channel **1414** (e.g., a cross-sectional area of the annulus or ring-shaped inlet opening **1404**) can vary between inlet opening **1404** and discharge opening **1410** in a manner that is determined to cause the flow of fluid **1406** through flow channel **1414** to continuously accelerate. For instance, the cross-sectional area at inlet opening **1404**

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can be larger than the cross-sectional area of discharge opening **1410**, which can cause acceleration in general.

More particularly, the variance in cross-sectional area can be determined to cause the flow of fluid **1406** through flow channel **1414** to continuously accelerate from some identified point (e.g., first location **1416**) to discharge opening **1410**. The portions of flow channel **1414** where it is determined that the flow continuously accelerates can depend on a particularly implementation, and three representative examples are discussed herein.

For instance, in some embodiments, first location **1416** (e.g., **1416i**) can be at inlet opening **1404**. As such, in this embodiment, flow channel **1414** is designed such that continuous acceleration of fluid **1406** occurs throughout the entire length of flow channel **1414**. In other embodiments, first location **1416** can be at other locations along flow channel **1414**, such as about one third the distance to discharge opening **1410** (e.g., illustrated by first location **1416₂**) or such as about one half the distance to discharge opening **1410** (e.g., illustrated by first location **1416₃**). Other potential locations are contemplated, but it is noted that at whatever point along flow channel **1414** that is selected to represent first location **1416**, flow of fluid **1406** is determined to continuously accelerate thereafter at least to discharge opening **1410**.

As noted, one technique to accomplish this continuous acceleration can be to ensure that the cross-sectional area of flow channel **1414** continuously decreases from at least first location **1416** to discharge opening **1410**. As one example, the design of fan intake device **1440** can be such that flow channel **1414** angles (e.g., see angles **1418₁** and **1418₂**) toward the center of the device when moving from inlet opening **1404** to discharge opening **1410**. It can be visualized that flow channel **1414** has an annulus or ring shape that decreases in size as fluid **1406** flows toward discharge opening **1410**. In other words, for each cross-sectional, ring-shaped, slice of flow channel **1414**, the size of the ring slices decrease, meaning their cross-sectional area decreases. This decrease in cross-sectional area can exist when angles **1418₁** (e.g., α_1) and **1418₂** (e.g., α_2) are the same, or even when those angles differ. For example, if α_1 is greater than α_2 , then it can be readily observed that the cross-sectional area will decrease both as a function of the decreasing ring size and as a function of the height of discharge opening **1410** (e.g., a distance from the inner surface and the outer surface of flow channel **1414**).

However, it is understood that, provided the difference is not too great between α_1 and α_2 , the decrease in cross-sectional area can exist even when α_1 is less than α_2 . In that case, the height of discharge opening **1410** can actually be greater than a height of inlet opening **1404**, even while the cross-sectional area of flow channel **1414** decreases (e.g., due to the shrinking ring size). As one representative example, α_1 can be approximately 73 degrees, while α_2 can be approximately 74 degrees, resulting in a greater opening height at discharge than inlet, yet still a smaller cross-sectional area, which can cause the continuous acceleration of fluid **1406** flowing through flow channel **1414**.

In some embodiments, the cross-sectional area of flow channel **1414** can monotonically decrease from inlet opening **1402** to discharge opening **1410** (or at least from first location **1416** to discharge opening **1410**). The terminal monotonically decreased cross-sectional area can be substantially at an area swept by impellers of a fan situated proximal to discharge opening **1410**.

In some embodiments, a geometry of flow channel **1414** that is determined to cause the flow of fluid **1406** to

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continuously accelerate is determined to result in a reduced energy loss across the aero-acoustical fan intake device **1400**. This can be contrasted with conventional fan intake devices that yield a significant energy loss and/or pressure loss, which is typically in the range of 0.2 in. wc. to 0.5 in. wc.

In some embodiments, this reduction in energy loss provided by the geometry of flow channel **1414** or other components of fan intake device **1400** can be representative of a decrease in total pressure through fan intake device **1400** that is less than about 10% of an impeller velocity pressure. In some embodiments, the reduction in energy loss provided by the geometry of flow channel **1414** or other components of fan intake device **1400** can be representative of a decrease in total pressure through aero-acoustical fan intake device **1400** that is less than about 5% of an impeller velocity pressure.

It is further appreciated that, in some embodiments, a cross-sectional area of flow channel **1414** at inlet opening **1404** can be less than one-half of a cross sectional area of duct **1402**. One advantage of such a design is that high frequency noise will tend to intersect inlet face **1402** at locations having solid or structural elements where that noise can be absorbed or constrained rather than entering flow channel **1414**, which is open to fluid **1406**. Thus, it can be advantageous for the diameter of flow-limiting structural elements to be greater than a fan impeller diameter, which is further detailed in connection with FIG. **15**. In some embodiments, fan intake device **1400** can further comprise a material determined to absorb noise, e.g., fiberglass or the like. This material can be distributed within housing **1412** and/or around flow channel **1414** and elsewhere. For example, regions marked with the text "FILL" can be suitable locations for the noise-absorbing material in certain embodiments.

With reference now to FIG. **15**, a schematic diagram showing a cross-section is illustrated of an example improved fan intake device **1500** having a bulb or hemisphere shaped inlet face in accordance with certain embodiments of this disclosure. For example, inlet face **1402** can be configured as a bulb **1502** (also referred to as hemisphere **1502**) and inlet opening **1404** surround bulb **1502**. Bulb **1502** can, relative to conventional fan intake devices, improve flow characteristics of fluid **1406** entering flow channel **1414**, which can represent an advantage of fan intake device **1500**. In this embodiment, fan intake device **1500** is illustrated without optional duct/plenum (e.g., see **1420**), however it is appreciated that a duct or plenum can exist and can be of any suitable shape or size.

However, bulb **1502** can increase manufacturing costs of a fan intake device, so a lower manufacturing cost can be yet another advantage of fan intake device **1400**, which is substantially similar to fan intake device **1500** in terms of having superior flow characteristics, but without bulb **1502**.

In some embodiments, bulb **1502** can have an inlet face diameter **1504** that is determined to less than an impeller diameter **1506** of a fan situated proximal to discharge opening **1410**. It is appreciated that greater inlet face diameter **1504** characteristic can apply to either inlet face, whether configured as a bulb **1502** (e.g., fan intake device **1500**) or otherwise (e.g., fan intake device **1400**). In some embodiments, a fan hub diameter **1508** can correspond to or be substantially similar to an inner diameter of flow channel.

Example Methods of Fabricating a Fan Intake Device

FIGS. **16** and **17** illustrate various methodologies in accordance with the disclosed subject matter. While, for

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purposes of simplicity of explanation, the methodologies are shown and described as a series of acts, it is to be understood and appreciated that the disclosed subject matter is not limited by the order of acts, as some acts can occur in different orders and/or concurrently with other acts from that shown and described herein. For example, those skilled in the art will understand and appreciate that a methodology could alternatively be represented as a series of interrelated states or events, such as in a state diagram. Moreover, not all illustrated acts can be required to implement a methodology in accordance with the disclosed subject matter. Additionally, it should be further appreciated that the methodologies disclosed hereinafter and throughout this specification are capable of being stored on an article of manufacture to facilitate transporting and transferring such methodologies to computers.

FIG. **16** illustrates a flow diagram **1600** of an example, non-limiting method for fabricating a fan intake device in accordance with one or more embodiments of the disclosed subject matter. For example, a device comprising a processor can perform certain operations. Examples of said processor as well as other suitable computer or computing-based elements, can be found with reference to FIG. **24**, and can be used in connection with implementing one or more of the devices or components shown and described in connection with figures disclosed herein.

At reference numeral **1602**, the device comprising the processor can facilitate forming an inlet face. The inlet face can be surrounded by an inlet opening. The inlet opening can be configured to receive a flow of a fluid. In some embodiments, the inlet opening can be representative of an annulus or ring about the inlet face. In some embodiments, the inlet face can have a shape characterized as a bulb or hemisphere.

At reference numeral **1604**, the device can facilitate forming a discharge face. The discharge face can be surrounded by a discharge opening. The discharge opening can be configured to discharge the flow of the fluid. The flow of the fluid can be discharged toward a proximally situated fan device and/or toward impellers of the fan device. In some embodiments, the discharge opening can be representative of an annulus or ring about the discharge face.

At reference numeral **1606**, the device can facilitate forming a housing. The housing can encompass a channel that extends from the inlet opening to the discharge opening. A cross-sectional area of the channel can vary between the inlet opening and the discharge opening in a manner that is determined to cause the flow of the fluid through the channel to continuously accelerate. Continuous acceleration for the fluid can occur from a first location of the channel to the discharge opening. Selection of the first location can be a function of a particular implementation. Method **1600** can proceed to insert A, which is further detailed in connection with FIG. **17**, or terminate.

Turning now to FIG. **17**, illustrated is a flow diagram **1700** of an example, non-limiting method that can provide additional aspects or elements in connection with fabricating a fan intake device in accordance with one or more embodiments of the disclosed subject matter.

At reference numeral **1702**, the forming the housing can further comprise determining, by the device, that the cross-sectional area of the channel at the first opening is less than one-half of a cross-sectional area of the inlet opening.

At reference numeral **1704**, the forming the housing can further comprise determining, by the device, that the cross-sectional area of the channel monotonically decreases from the inlet opening to the discharge opening at substantially an area swept by the fan impellers. In some embodiments, it can

be determined that the cross-sectional area of the channel monotonically decreases from the first location to the discharge opening at substantially an area swept by the fan impellers.

At reference numeral **1706**, the forming the housing can further comprise determining, by the device, that a geometry of the flow causes a reduced energy loss across the fan intake device.

Example Air Handler Apparatuses and/or Products (e.g., Aircube)

Turning now to FIG. **18**, a schematic diagram is illustrated showing a cross-section of a first example air handler product in accordance with certain embodiments of this disclosure. Air handler device **1800** (also referred to as air handler product **1800**) can operate to supply both heated and cooled air that can be independently selected based on the supply duct. Thus, for instance, cooled air can be provided to a first supply duct that serves one portion of a building (e.g., a south facing portion in direct sunlight) concurrently with heated air being provided to a second supply duct that serves a different portion or zone of the building (e.g. north facing portion). Such an advantage can be provided at a low cost, using only a single air handler device, which is distinct from conventional air handler devices that do not allow for concurrent deliver of both heated and cooled air. Another advantage can be observed in operational costs, since diverse heating or cooling needs can be satisfied during the same duty cycle rather than by multiple sequential duty cycles, which can reduce operational costs and increase equipment lifecycle.

It is appreciated that air handler device **1800**, which can be referred to herein as an “aircube”, an “aircube device/product” or other similar variations, can contain all or only a portion of elements described, which are intended to be exemplary or representative, but also non-limiting. Air handler device **1800** can comprise mixing plenum **1802**, which can also be referred to as a mixing chamber or central chamber. Mixing plenum **1802** can receive multiple air flows **1804** from multiple different ducts **1808** that feed mixing plenum **1802** as well as in some cases directly from the surrounding area (e.g., non-ducted intake). In conventional literature, the term ‘mixing’ usually refers to combining air flows of different temperatures such as outside air and return air. As used herein, mixing plenum **1802** is intended to refer to a plenum or other structure (upstream from a fan) that receives air from multiple flows, inclusive of cases where the multiple flows are not of substantially different temperatures.

Reference numeral **1806** illustrates an encircled area conceptually representing mixing plenum interfaces that couple mixing plenum **1802** to surrounding air or to ducts **1808**, referred to herein as mixing plenum interfaces **1806**. In other words, as used herein, ducts **1808** can represent structural ductwork, as depicted, or another exposure to air flow **1804** such as from outside air. In some embodiments, air handler device **1800** can be located against an outside wall with outside air louvers and dampers placed in that outside wall, such as at mixing plenum interface **1806**. In that case the mixing plenum interface **1806** can contain dampers enabling control of mixed air temperature, e.g., when outside air is cool and control of a minimum percentage of outside air, e.g., when outside air is hot. As further detailed below, mixing plenum interface **1806** can comprise air filters **1822** as well as dampers and louvers or other suitable elements.

Air handler device **1800** can comprise fan device **1810**. Fan device **1810** can be configured to receive a mixing plenum flow (e.g., flow **1804**) from mixing plenum **1802** and to discharge a supply flow **1818**. Fan device **1810** can be embodied as, for example, a centrifugal fan, a plenum fan, an axial fan, or any other suitable type of fan, whereas embodiments described herein with respect to air handler device **1800** generally assume a centrifugal fan embodiment. Air handler device **1800** can further comprise supply plenum **1812**. Supply plenum **1812** can be configured to receive supply flow **1818** from fan device **1810** and to discharge supply flow **1818** as explained below. It is appreciated that, as used herein, the term “supply plenum” can also refer to a region comprising vanes such as straightening vanes, which is typically more appropriate in cases where fan device **1810** is an axial fan, such as the generally presumed case with respect to FIG. **21**. In other words, “supply plenum” can refer to what is conventionally considered a supply plenum (e.g., in embodiments that employ a centrifugal fan) as well as a vane section (e.g., in embodiments that employ an axial fan).

For example, supply plenum **1812** can comprise a plurality of duct interfaces **1814**, which are conceptually illustrated by the encircled region where supply ducts **1816** intersect with supply plenum **1812**. Hence, duct interfaces **1814** can be configured to interface with a different one of a plurality of supply ducts **1816**. Supply plenum **1812** can further comprise a plurality of thermal transfer units (TTUs) **1820**. For instance, the plurality of TTUs **1820** can comprise first TTU **1820₁** and second TTU **1820₂**, that are respectively situated in different ones of the plurality of duct interfaces **1814**. Advantageously, first TTU **1820₁** affecting a first air flow **1818** can be configured to a first temperature concurrently with second TTU **1820₂** affecting a second air flow **1818** can be configured to a second temperature that differs from the first temperature.

In the present embodiment, two TTUs **1820** are depicted, but it is appreciated that any suitable number of TTUs **1820** can be employed. For instance, for each supply duct **1816** and/or duct interface **1814**, a different TTU **1820** can be employed, which can effectively allow individual (e.g., per-supply duct **1816**) of heating versus cooling versus neutral or matching (e.g., neither heating nor cooling) as well as individually controlling temperature gradients on a per-duct basis. For example, air handler device **1800** can be configured as a two-TTU design (e.g., FIG. **18**), a three-TTU design (e.g., FIG. **19**), a four-TTU design (e.g., FIG. **20**), or more. TTU **1820** can comprise coils that operate according to direct expansion, water-type, or any other suitable techniques for thermal transfer. The heat transfer medium of TTU **1820** can be any suitable fluid such as water, gas, refrigerant, CO₂, O₂, etc., that flows through pipe connecting an evaporative coil array to condensing coil arrays. Such can be used in any suitable configuration and in connection with heat pumps, air conditioners, compressors, or the like.

For example, in some embodiments, air handler device **1800** can be equipped with six-way valve water coils that can independently heat or cool supply flows **1818**. In some embodiments, valve packages can be factory installed such that air handling device **1800** can be fully assembled prior to delivery at an installation site, which can significantly reduce costs.

Further, either draw-through or blow-through configurations can be provided, or in some embodiments both concurrently. For example, mixing plenum interfaces **1806** as well as duct interfaces **1814** can comprise either or both TTUs **1820** or filters **1822**. As depicted, coils of a TTU **1820**

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can be situated in a slanted configuration, which can increase the thermal transfer between TTU **1820** and supply flow **1818**.

Although a single fan device **1810** is illustrated, it is appreciated that any suitable number of fan devices **1810** can be employed. For example, depending on size or implementation, some embodiments can provide for two, three, four, six or more fans situated between mixing plenum **1802** and supply plenum **1812** (for instance see FIG. **20**, showing four fan devices).

In some embodiments, fan device **1810** can comprise or be operatively coupled to fan intake device **1824** at an upstream location. Fan intake device **1824** can operate to straighten or improve air flow **1804** and/or to significantly reduce noise without significant pressure loss. In that regard, fan intake device **1824** employ designs or techniques detailed herein in connection with fan intake device **1400** or fan intake device **1500**, of which advantages described herein with respect to those devices can be incorporated into air handler device **1800** (as well as embodiments of air handler product **2100** detailed in connection with FIG. **21**).

In some embodiments, fan device **1810** can comprise or be operatively coupled to an evase device (not shown, but see evase device **2124** of FIG. **21**) at a downstream location (e.g., toward supply plenum **1812**). Such an evase device can be substantially similar to any of the evase devices detailed herein (e.g., evase device **400**, evase device(s) **202**, **302**). As explained, the evase device can therefore operate to efficiently convert velocity pressure to static pressure. Hence, the evase device can be particularly advantageous in cases where fan device **1820** is an axial fan (e.g., see FIG. **21**), which tends to generate significantly more velocity pressure than centrifugal or plenum fans. In some embodiments, either or both of the evase device or the fan intake device **1824** can be built into fan device **1810** and/or can share a common housing.

Further, fan device(s) **1810** can be configured to discharge supply flow **1818** in a vertical direction, a horizontal direction, or some angle in between. Likewise, air handler devices or products disclosed herein can be configured to blow air upward (e.g., a floor unit such as air handler device **1800**) or blow air downward (e.g., a rooftop unit an example of which is provided in connection with FIG. **21**). It is to be further appreciated that an aspect ratio of various coils and/or TTUs **1820** can vary and/or be non-symmetrical. For instance, one side can be longer than other sides. In other words, coils of various TTUs **1820** can be configured to any suitable height, width, length specification. Such can provide better coil performance, such as, e.g., lower APD, additional face area, etc., and any TTU **1820** can be tailored specifically to a given duct or zone requirement.

In some embodiments, as depicted in FIG. **18**, supply flows **1818** can flow in different directions, that is one supply flow **1818** is flowing toward the right of the page, while the other supply flow is flowing toward the left side of the page. In other embodiments, at least two supply flows **1818** can flow into two of supply ducts **1816** in a same direction. For example, two adjacent supply ducts **1816** might carry air in a parallel direction (e.g., out of the page or into the page), while two other supply ducts **1816** can carry air in different directions such as to the right of the page and the left of the page, as depicted. In any case, each supply duct **1816** can have an individually controllable TTU **1820** that can independently heat or cool corresponding supply flows **1818**.

Turning now to FIG. **19**, illustrated is a three-dimensional representation of a first example air handler product **1900** having three supply duct interfaces in accordance with

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certain embodiments of this disclosure. Return air and/or a combination of return air and fresh air (e.g., air flow **1804**) can be received via mixing plenum interface **1806**, which can in some embodiments, include TTU **1820** and/or filter **1822**. Air flow **1804** can be controlled by dampers or another suitable mechanism or technique. Likewise, supply duct interface **1814** can also be configured to include TTU **1820** and/or filter **1822**. Supply flow **1818** can be received, via mixing plenum interface **1806**, by fan device **1810** (e.g., a centrifugal fan or another suitable type of fan) and discharged via supply duct interface **1814**. Supply duct interface **1814** can also be configured with dampers to control supply flow **1818** at any given supply duct interface **1814**. The illustrated embodiment represents a three-way supply duct design, but other suitable designs are contemplated.

Referring now to FIG. **20**, illustrated is a three-dimensional representation of a second example air handler product **2000** having multiple fans and four supply duct interfaces in accordance with certain embodiments of this disclosure. Return air and/or a combination of return air and fresh air can be received via mixing plenum interface **1806**, which can in some embodiments, include TTU **1820** and/or filter **1822**. Air flow **1804** can be controlled by dampers or another suitable mechanism or technique. Likewise, supply duct interface **1814** can also be configured to include TTU **1820** and/or filter **1822**. Supply flow **1818** can be received, via mixing plenum interface **1806**, by fan device **1810** (e.g., a centrifugal fan or another suitable type of fan) and discharged via supply duct interface **1814**. Supply duct interface **1814** can also be configured with dampers to control supply flow **1818** at any given supply duct interface **1814**. The illustrated embodiment represents a four-way supply duct design, but other suitable designs are contemplated.

Turning now to FIG. **21**, a schematic diagram is illustrated showing a cross-section of a second example air handler product in accordance with certain embodiments of this disclosure. Air handler device **2100** (also referred to as air handler product **2100** or HVAC product **2100**) is illustrated in the context of a rooftop unit, but it is appreciated that floor units are also contemplated. As such, air handler device **2100** is configured to blow air downward (as opposed to upward as illustrated in FIGS. **18-20**). Air handler device **2100** can be factory-assembled and/or shipped to an installation site fully assembled, in some cases including valve packages or the like.

One difficulty associated with factory-assembled air handler devices is that the size of the unit typically makes shipping and fabrication more expensive. For example, transportation codes, which can be based on the height of highway overpasses or the like typically have a height constraint. Likewise, building codes can also impose a height constraint for rooftop units. In order to meet these constraints, conventional HVAC devices are manufactured to be wide and short. That is, all the ordinary components (e.g., duct interfaces, mixing chambers and other plenums, fan, heat rejection and/or thermal transfer unit, filters, etc.) are not stacked on top of one another, but rather situated side-by-side. This conventional design allows the unit to meet building codes, but due to the large width (and potentially weight), can increase shipping costs as only one unit might fit on a single truck at a time. The large size of the unit is also more costly in terms of materials and fabrication.

Due in part to advantageous designs disclosed above, and herein, the inventors have discovered a way to stack air handler components on top of one another, which can greatly decrease width **2101** of the unit, while meeting height code constraints. In particular, a heat rejection section can be

placed on top, whereas conventional designs are unable to situate the heat rejection section on top and therefore place that component on the side of other air handler components. As a result, fewer units of conventional designs can be shipped per truck, which increases transportation costs as well as other costs, some noted herein.

HVAC product **2100** can comprise mixing plenum **2102**, fan device **2106**, and supply plenum **2108** (collectively referred to hereinafter as an air handler component). Collectively, these components can be configured to circulate flows of air within an HVAC system situated at a site HVAC product **2100** is to be installed. It is appreciated that air flow **2104** (e.g., return air and/or fresh air) follows a substantially “S” or “Z” shaped path to arrive at mixing plenum **2102** (also referred to as central chamber **2102**) before entering fan device **2106**. This design can significantly reduce noise and can also improve aerodynamic properties (e.g., reduce turbulence or shear flows, etc), which might otherwise damage fan device **2106**.

In some embodiments, fan device **2106** can be an axial fan, which typically generates a much greater velocity pressure than plenum or centrifugal fans. By utilizing an axial fan in this design, the rating of the unit can be much greater than conventional units of similar dimensions. However, several difficulties can arise with the use of axial fans. A first difficulty is that the fan impellers can break when confronted with shear flows, turbulence, or the like. This difficulty can be substantially mitigated by the “S” or “Z” shaped path of air flow **2104**, as detailed above as well as implementation of a fan intake device (not shown, but see fan intake device **1824** of FIG. **18**).

A second difficulty associated with axial fans is they produce a very large velocity pressure that tends to be inefficiently converted to static pressure in the remainder of the system. Thus, in conventional designs, the total pressure loss can be significant with axial fans. In order to mitigate this difficulty, evase device **2124** can be placed downstream of fan device **2106** in some embodiments. Evase device **2124** can be substantially similar to any of the devices detailed herein (e.g., evase device **400**, evase device(s) **202**, **302**). As explained previously, evase device **2124** can therefore operate to efficiently convert velocity pressure to static pressure. Hence, the second difficulty of using axial fans can be mitigated.

Because of the efficient, space-saving design, a top surface **2112** of the air handler component can have a significantly smaller height (e.g., first height **2114**) than other systems or products. As such, a heat exchange device **2116** can be situated on top surface **2112**. Heat exchange device **2116** can be configured to exchange heat with air flow **2104** and can include coils **2117** and/or filters, etc. situated in the path of air flow **2104**. In some embodiments, coils **2117** can be slanted as illustrated and discussed in connection with coils **1820**. Further, coils **2117** can be configured to heat, cool, or neither (e.g., provide neutral air) independently from other coils **2117** as discussed in connection with coils **1820**.

Heat exchange device **2116** can have a second height **2118** that, when combined with first height **2114**, represents a total height **2120** of HVAC product **2100**. Total height **2120**, which reflects heat exchange device **2116** being situated on top (rather than on the side), can be determined to be less than or equal to a defined height constraint **2122**. In some embodiments, the defined height constraint **2122** can be determined to satisfy a local building code of the installation site. In some embodiments, the defined height constraint **2122** can be determined to satisfy a transportation code

applicable to a transportation route between a manufacturing site of HVAC product **2100** and the installation site. By way of example, the defined height constraint **2122** can be, e.g., 14 feet, or 10 feet, or some other suitable value. Further, in some embodiments, a weight of HVAC product **2100** can be determined to satisfy a defined weight constraint.

Moreover, it is appreciated that the described HVAC product **2100** can be designed to discharge according to an overhead configuration or an under floor configuration. Hence, in some embodiments, heat exchange device **2116** can be situated on a bottom surface of the air handler component. For example, if HVAC product **2100** is rotated 180 degrees, for instance to accommodate a floor unit versus a rooftop unit, top surface **2112** would then be descriptive of a bottom surface, below which can be situated heat exchange device **2116**. In either embodiment, it can be seen that such is distinct from conventional designs in which heat rejection sections are situated side-by-side with other components.

Example Methods of Fabricating an Air Handler Product

FIGS. **22** and **23** illustrate various methodologies in accordance with the disclosed subject matter. While, for purposes of simplicity of explanation, the methodologies are shown and described as a series of acts, it is to be understood and appreciated that the disclosed subject matter is not limited by the order of acts, as some acts can occur in different orders and/or concurrently with other acts from that shown and described herein. For example, those skilled in the art will understand and appreciate that a methodology could alternatively be represented as a series of interrelated states or events, such as in a state diagram. Moreover, not all illustrated acts can be required to implement a methodology in accordance with the disclosed subject matter. Additionally, it should be further appreciated that the methodologies disclosed hereinafter and throughout this specification are capable of being stored on an article of manufacture to facilitate transporting and transferring such methodologies to computers.

FIG. **22** illustrates a flow diagram **2200** of an example, non-limiting method for fabricating an air handler product in accordance with one or more embodiments of the disclosed subject matter. For example, a device comprising a processor can perform certain operations. Examples of said processor as well as other suitable computer or computing-based elements, can be found with reference to FIG. **24**, and can be used in connection with implementing one or more of the devices or components shown and described in connection with figures disclosed herein.

At reference numeral **2202**, the device comprising the processor can facilitate forming a mixing plenum. The mixing plenum can be configured to receive multiple flows of air from multiple different ducts. In some embodiments, the multiple flows of air can be from multiple different directions.

At reference numeral **2204**, the device comprising the processor can facilitate forming a fan device. The fan device can be configured to receive a mixing plenum flow from the mixing plenum and to discharge a supply flow. At reference numeral **2206**, the device comprising the processor can facilitate forming a supply plenum. The supply plenum can be configured to receive the supply flow from the fan device.

At reference numeral **2208**, the device comprising the processor can facilitate forming a plurality of duct interfaces. The plurality of duct interfaces can be respectively configured to interface with a different one of a plurality of

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supply ducts. In some embodiments, the plurality of supply ducts can be configured to transport air in multiple different directions. In some embodiments, at least two of the plurality of supply ducts can be configured to transport air in a same direction.

At reference numeral **2210**, the device comprising the processor can facilitate forming a plurality of thermal transfer units. The plurality of thermal transfer units can comprise a first thermal transfer unit and a second thermal transfer unit that can be, respectively, situated in different ones of the plurality of duct interfaces. The first thermal transfer unit can be configured to heat a first air flow concurrently with the second thermal transfer unit cooling a second air flow. Method **2200** can proceed to insert A, which is further detailed in connection with FIG. **23**, or terminate.

Turning now to FIG. **23**, illustrated is a flow diagram **2300** of an example, non-limiting method that can provide additional aspects or elements in connection with fabricating an air handler device in accordance with one or more embodiments of the disclosed subject matter.

At reference numeral **2302**, the device can facilitate configuring the mixing plenum to receive return air and fresh air from the multiple different ducts. Said configuring can be accomplished in connection with the forming the mixing plenum that is detailed above in connection with reference numeral **2202** of FIG. **22**.

At reference numeral **2304**, and potentially in connection with the forming the fan device discussed at reference numeral **2204** of FIG. **22**, the device can facilitate forming multiple fan devices situated between the mixing plenum and the supply plenum.

At reference numeral **2306**, the device can facilitate forming a filter device situated at an interface. The interface can be at least one of a mixing plenum interface or a supply plenum interface. In other words, the filter can be situated at either one of or both of the mixing plenum interface or the supply plenum interface.

Example Air Handler with U-Bend Structure

Certain air handler devices detailed herein (e.g., those associated with FIGS. **18-20**, **25**, and **26**) can be implemented according to designs or techniques typically have small footprints that take up very little floor space, allowing for additional leasable space. Moreover, improved or superior acoustics can also lead to additional space savings such as by reducing or eliminating mechanical rooms with acoustical abatement. Embodiments with floor-level fans and/or filters can save on maintenance by providing simplified access to fans, filters, and/or controls. Further, when coupling to downstream devices having accurate air flow devices, air handler devices with multi-sided coil configurations can reduce or eliminate zone level coils or other fluid distribution infrastructure, resulting in a sustainability and economic advantage over previous HVAC designs or systems.

It is understood that an air handler device, as used herein, can be a centralized unit that provides supply air to multiple downstream zones or zone-level devices and can receive return air from the multiple downstream zones or zone-level devices as well as outside air or fresh air that can be mixed with the return air. In situations in which outside air is added to the return air, it can be advantageous to thoroughly mix the outside air with the return air. Such can produce adequately mixed supply air that is discharged by the air handler device to the downstream zones and/or zone-level devices.

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Air handler devices and other HVAC components are installed in buildings or other structures. Generally, these buildings allocate specific spaces for HVAC equipment, such as ceiling spaces, floor spaces, and may potentially have equipment rooms devoted to HVAC equipment.

With reference now to FIG. **24**, schematic diagram **2400** depicts an example building layout for accommodating air handler devices and other HVAC components, devices, or elements in accordance with certain embodiments of this disclosure. In this example, two primary spaces are illustrated, namely, ceiling space **2402** and floor space **2404**.

Ceiling space **2402** can be considered at or above a ceiling structure and/or an elevated space, and is typically where much of a building's ductwork exists or is foreseen to be installed. In addition to ductwork, ceiling space **2402** can comprise one or more (ceiling) installation spaces **2406**, denoted here as installation spaces **2406_{1-2406_N}**, where N can be any whole number. Installation space(s) **2406** can be representative of a ceiling space that is suitable for installation of all or a portion of the disclosed air handler devices and/or other suitable HVAC equipment detailed herein.

Floor space **2404** can comprise one or more (floor) installation spaces **2408**, denoted here as installation spaces **2408_{1-2408_M}**, where M can be any whole number. Installation space(s) **2408** can be representative of a floor space that is suitable for installation of all or a portion of the disclosed air handler devices and/or other suitable HVAC equipment detailed herein. While ceiling space **2402** can relate to elevated spaces, floor space **2404** can relate to floor-level space which is often leasable space and therefore commands a much greater premium than certain ceiling space **2402** that is non-leasable space. Thus, in some embodiments, it can be advantageous to reduce HVAC equipment footprint on floor space **2404**, e.g., by shifting HVAC equipment typically installed in installation space **2408** (e.g., at floor level) to installation space **2406** (e.g., at ceiling level).

In order to better describe the disclosed air handler unit with a U-bend structure in the context of building layouts, consider coordinate system **2410**, illustrating a commonly referenced Cartesian coordinate system, having three dimensional axes, x, y, and z. In coordinate system **2410**, first plane **2412** can exist that can comprise the dimensions indicated by the x-y axes. First plane **2412** can intersect installation space(s) **2406** in some embodiments. In some embodiments, first plane **2412** can intersect spaces within ceiling space **2402** in which ductwork exists or is allocated for ductwork installation.

Consider as well, second plane **2414**, which can be substantially perpendicular to first plane **2412**. Second plane **2414** can therefore comprise dimensions indicated by the y-z axes as illustrated here or the x-z axes, or any angle of rotation about the z-axis, as all examples are substantially perpendicular to first plane **2412**.

Third plane **2416** can be substantially parallel with first plane **2412** and can therefore also comprise dimensions of the x-y axes. Third plane **2416** can intersect installation space(s) **2408** in some embodiments. In some embodiments, third plane **2412** can intersect a floor or leasable floor space within a building. It is noted that in conventional buildings, a typical vertical distance between (ceiling) installation space **2406** and (floor) installation space **2408** is about 12 feet. Hence, although such can depend on the implementation or embodiment, about 12 feet can represent a common distance between first plane **2412** and third plane **2416**. As a common range, this distance may be between 8 and 15 feet between the two planes **2412**, **2416**, which can also represent the regions of interest of second plane **2414**, namely a

distance between the floor and ceiling and/or the distance between the intersection of second plane **2414** with first plane **2412** and third plane **2416**.

While still referring to FIG. **24**, but turning as well to FIG. **25**, a schematic block diagram of an example air handler device **2500** is illustrated having a U-bend structure in accordance with certain embodiments of this disclosure. This U-bend structure can be indicative of an air flow path that changes directions about 180 degrees. This directional change can coincide with an intake and discharge of a fan device that is situated within the U-bend structure, and can be within a single plane (e.g., both paths can be on second plane **2414**).

Air handler device **2500** can comprise intake plenum **2502**. Intake plenum **2502** can comprise intake device **2504** that is configured to receive intake flow **2506**. Intake flow **2506** can comprise intake air. This intake air can comprise one or both of return air or outside air that is to be mixed together by air handler device **2500**. In some embodiments, intake device **2504** can comprise filter **2522a**. Filter **2522a** can be configured to filter the intake air. In some embodiments, intake device **2504** can comprise economizer device **2524**. Economizer device **2524** can comprise at least one of a damper or a fan, e.g., to provide additional control over intake flow **2506**.

Air handler device **2500** can further comprise discharge plenum **2508**. Discharge plenum **2508** can comprise discharge device **2510** that is configured to discharge supply flow **2512**. Supply flow **2512** can comprise supply air that is composed of the intake air that was mixed by air handler device **2500**. In some embodiments, discharge device **2510** can comprise or be coupled to a thermal transfer unit (TTU) **2514**. TTU **2514** can be configured to control a temperature of the supply air of supply flow **2512**.

As illustrated, discharge plenum **2508** can be situated, or configured to be installed, adjacent to intake plenum **2500**. As one example, intake plenum **2502** and discharge plenum **2508** can be adjacently situated in the same (ceiling) installation space **2406** or respectively installed in adjacent or contiguous installation spaces **2406₁**, **2406_N**.

In this example, both intake plenum **2502** and discharge plenum **2508** can be adjacently situated along and/or intersecting first plane **2412**. Likewise, intake flow **2506** and discharge flow **2512** can be situated on (or substantially parallel to) first plane **2412**. Even though intake flow **2506** and discharge flow **2512** can be in the same plane along with intake plenum **2502** and discharge plenum **2508** that are adjacent to one another, the entire air flow path through air handler device **2500** does not remain on that plane, but rather extends in other dimensions, due at least in part to U-bend device **2515**, which may also be referred to herein as U-bend section **2515**.

Hence, air handler device **2500** can further comprise U-bend device **2515** that can provide an air flow path between (adjacently situated) intake plenum **2502** and discharge plenum **2508**. In more detail, U-bend device **2515** can comprise fan device **2516**. Fan device **2516** can be configured to draw air (e.g., the intake air that is being mixed) from intake plenum **2502** and provide the air to discharge plenum **2508**. Fan device **2516** can be a plenum fan, an axial fan, a centrifugal fan, or any other suitable type of fan. Fan device **2516** can comprise one or more fan devices **2516**.

U-bend device **2515** can further comprise first channel **2518** and second channel **2520**, one of which can comprise or encompass fan device **2516**. First channel **2518** can extend from intake plenum **2502** to some region or location

that is adjacent to fan device **2516**. In some embodiments, first channel **2518** can comprise filter **2522b**. Filter **2522b** can be configured to filter the intake air.

Second channel can extend from discharge plenum **2508** to a second location adjacent to fan device **2516**. As intake plenum **2502** and discharge plenum **2508** are situated adjacently, first channel **2518** and second channel **2520** can be similarly adjacent.

In this example, all portions of fan device **2516** are situated in second channel **2520**, but all or portions of fan device **2516** might otherwise be situated in first channel **2518** in other implementations or embodiments. Regardless, it is appreciated that fan device **2516** can be installed in (floor) installation space **2408**. Thus, fan device **2516** can intersect third plane **2416**, whereas first channel **2518** and second channel **2520** can intersect or extend along second plane **2414**.

As illustrated, after being received by intake device **2504**, intake flow **2506** changes direction within intake plenum **2502**, as illustrated at flow **2506a**. Hence, flow **2506a** potentially leaves first plane **2412** while potentially following a direction that is on second plane **2414**. Upon entering first channel **2518**, flow **2506b** can be drawn into fan device **2516** and discharged by fan device **2516** into second channel **2520** as flow **2506c** that can also be on second plane **2414**. Once entering discharge plenum **2508**, flow **2506d** can again change direction, potentially rejoining first plane **2412** as supply flow **2512**.

Turning now to FIG. **26**, a schematic block diagram **2600** of the example air handler device **2500** having a U-bend structure is illustrated in an isometric view in accordance with certain embodiments of this disclosure. As shown, U-bend section **2515** comprises multiple fan devices **2516**.

Intake plenum **2502** can comprise multiple intake devices **2504** that can be respectively configured to receive respective intake flows **2506**. In some embodiments, different intake flows **2506** can be fed into intake plenum **2502** from separate ducts or other equipment. As shown, any two intake flows **2506** may be received in the same direction as one another or may instead be received in different directions. In some embodiments, all or a portion of the various intake flows **2506** can be received in directions that are on first plane **2412**, even though, within first plane **2412**, the individual directions of respective intake flows **2506** can be in different directions. As noted, a first intake flow **2506** can be comprised of return air, while a second intake flow **2506** can be comprised of outside air to be mixed, and any or all of the associated intake device(s) **2504** can comprise an economizer device **2524** with a damper or a fan attached, as well as filter **2522a** or other elements.

Further, discharge plenum **2508** can comprise multiple discharge devices **2510** that can be respectively configured to discharge respective supply flows **2512**, denoted here as supply flows **2512₁**, **2512₂**, for clarity. In some embodiments, different supply flows **2512** can be generated for separate ducts or other equipment. As shown, any two supply flows **2512** may be discharged in the same direction as one another (e.g., supply flows **2512₃** and **2512₄**) or may instead be discharged in different directions (e.g., supply flows **2512₁** vs. **2512₂** or **2512₃**). In some embodiments, all or a portion of the various supply flows **2512** can be discharged in directions that are on first plane **2412**, even though, within first plane **2412**, the individual directions of respective supply flows **2512** can be in different directions. In some embodiments, discharge device **2510** can comprise a damper to control an associated supply flow **2512**.

As noted, potentially each discharge device **2510** can comprise a respective TTU **2514**, which are labeled here as TTU **2514₁-2514₄** for the sake of clarity, that can be configured to control an associated temperature of an associated supply flow **2512**. In some embodiments, TTUs **2514** can be configured to independently control the associated temperature such that a first temperature, e.g., controlled by TTU **2514₁** that affects a first supply flow **2512₁** can concurrently differ from a second temperature, e.g., controlled by TTU **2514₂** that affects a second supply flow **2512₂**.

In some embodiments, TTU **2514** can be independently controllable to perform one or more of a group of thermal transfer operations. For instance, these thermal transfer operations can comprise a heating operation that heats an associated supply flow **2512**, a cooling operation that cools the associated supply flow **2512**, or a neutral operation that does not affect the temperature of the associated supply flow **2512**. In some embodiments, TTU **2514₁** can perform one of the thermal transfer operations (e.g., the cooling operation) concurrently with TTU **2514₂** performing a different thermal transfer operation (e.g., the heating operations or the neutral operation).

These designs and techniques, suitably illustrated in FIGS. **25** and **26**, can be advantageous for a number of reasons. For example, it is understood that each supply flow **2512** can supply air to different downstream zones or zone-level devices. Because each associated TTU **2514** can be independently controlled to meet the individual (and potentially different) downstream demands from a single location (e.g., the location of air handler device **2500**), there is no need to install thermal transfer coils or other equipment at downstream zones as is done conventionally. Such can result in significant cost savings in terms of piping costs and mitigate the potential for damage due to leaks in piping infrastructure that, in conventional systems, extend throughout the majority of most buildings.

Another example advantage is that, even though intake plenum **2502** and discharge plenum **2508** can be adjacent to one another in ceiling space **2402**, the total flow path can be substantial. For instance, U-bend section **2516** can provide a flow path that in many implementations is well over 20 feet (e.g., extending from ceiling space **2402** to floor space **2404** and back). Such can provide for superior mixing of the intake air over conventional designs. It is observed that mixing can occur for all flows **2506a-2506d** between intake device **2504** to discharge device **2510**.

As another example advantage, intake plenum **2502** and discharge plenum **2508**, collectively representing the largest elements of air handler device **2500**, can be situated in ceiling space **2402**, thereby significantly reducing the footprint on potential leasable space. As one example, an equipment room can be much smaller than conventional expectations, both because intake plenum **2502** and discharge plenum **2508** can be adjacently situated and because both intake plenum **2502** and discharge plenum **2508** can be situated in ceiling space **2402**.

However, fan device **2516**, which is smaller by comparison and typically represents the portion of air handler device **2500** that is more frequently serviced, can be situated in floor space **2404**, which can dramatically simplify replacement, repair, or maintenance of fan device **2516**. In some embodiments, U-bend section **2516** can comprise service door **2526**. Service door **2526** can be a full-service door, not merely a hatch that allows access to fan device **2516**. In that regard, service door **2526** can have dimensions that are

greater than dimensions of fan device **2516** and/or that are sufficient to remove or install fan device **2516** via service door **2526**.

Likewise, U-bend device **2516** can comprise filter door or hatch **2528** that can facilitate access to filter **2522b**. As illustrated here, service door **2526** can, depending on the implementation, be in either one of first channel **2518** or second channel **2520**, which can depend on the location of fan device **2516**. As illustrated here, service door **2526** is situated in second channel **2520** and filter door or hatch **2528** is in first channel **2518**, but other implementations are contemplated.

Furthermore, other installation orientations or locations can vary based on the customer constraints. For example, while the present illustration illustrates intake plenum **2502** and discharge plenum **2508** being in ceiling space **2402** and fan device **2516** being in floor space **2404**, this orientation can be reversed. Conceptually, such can be imaged by rotating the illustrated design 180 degrees and potentially making suitable other modifications to certain elements. In that case, intake plenum **2502** and discharge plenum **2508** may be situated in floor space **2404** and fan device **2516** may be situated in ceiling space **2402**. Such an orientation may address specific ductwork layout, or other constraints of a customer.

In a different embodiment, substantially all portions of air handler device **2500** (e.g., intake plenum **2502**, discharge plenum **2508**, and fan device **2516**) can be in either ceiling space **2402** or floor space **2404**. Conceptually, such can be imaged by rotating the illustrated design 90 degrees. In ceiling-only embodiments, such can have the advantage of minimizing the leasable space footprint. In floor-only embodiments, such can have the advantage of simplifying all service operations for any portion of air handler device **2500**.

In some embodiments, intake plenum **2502** and discharge plenum **2508** may not both be situated at an intersection of first plane **2412** and second plane **2414**. For example, intake plenum **2502** and discharge plenum **2508** can be situated diagonally or catty-corner to one another (albeit, sill adjacent). In that case, it can be imaged that the shape of U-bend device **2516** would more closely resemble a "V" shape, however, such variations are considered to be within the spirit and scope of this disclosure. In other embodiments, there may be multiple instances of U-bend device **2516** and/or additional channels, such as a third channel (e.g., similar in flow direction to first channel **2518**) and a fourth channel (e.g., similar in flow direction to second channel **2520**). Such can provide additional potential flow path length, which can further improve mixing or provide other advantages.

Still another advantage of the illustrated designs or techniques is a dramatic improvement in aero-acoustics. U-bend section **2516** can function to significantly reduce noise levels over conventional air handler devices. For example, the long flow paths can reduce noise-producing turbulence and acoustical lining can be applied to the first channel **2518** and/or the second channel **2520**. Acoustical lining can also be applied to intake plenum **2502** and discharge plenum **2508**.

Yet another advantage is that many of the individual elements of air handler device **2500** can be fabricated, shipped, and/or installed in a modular manner. For example, each of intake plenum **2502**, discharge plenum **2508**, and U-bend device **2512** can individually be sized to fit in a standard elevator, dramatically simplifying installation or retrofit installation.

As used herein, the term ‘adjacent’ is intended to mean being near to one another with no like element between. Thus, in the context of intake plenum **2502** and discharge plenum **2508** being adjacent, such can mean that these two elements (or associated housings or structures) are in contact or can mean that these two elements are near to one another with no other plenum device in between.

FIG. **27** illustrates a schematic diagram **2700** showing an example discharge plenum **2702** having multiple temperature controlled discharge devices **2704** integrated into a building duct layout in accordance with certain embodiments of this disclosure. Discharge plenum **2702** can in some embodiments be substantially similar to discharge plenum **2508** of FIGS. **25** and **26**, supply plenum **1812**, and/or other suitable plenums disclosed herein.

Hence, discharge plenum **2702** can have multiple discharge devices **2704**₁-**2704**₃ that respectively serve different portions of a building or a floor of a building. Thus, different supply flows (e.g., supply flows **2512**₁-**2512**₃) can be delivered to different discharge device **2704** and associated downstream devices such as zone-level devices **2706** (e.g., diffusers or other suitable devices) and flow measurement and control devices **2708**, either of which may comprise a downstream damper device.

Because discharge devices **2704** can comprise independently controllable TTU (e.g., TTUs **2514**₁-**2514**₃), a zone or zones served by discharge device **2704**₁ can be provided supply flows having a different temperature than downstream zones served by discharge devices **2704**₂ or **2704**₃. For example, a first downstream zone (e.g., on a south side of the building) served by discharge device **2704**₁ can be served cool air, while a second downstream zone (e.g., on an east side of the building) can be served neutral or ambient temperature air by discharge device **2704**₂, and while a third downstream zone (e.g., on a north side of the building) can be served heated air by discharge device **2704**₃. Associated flow measurement and control device can modulate the supply air according to the demands of individual zones. In other words, discharge plenum **2702** and/or an associated air handler device can control the temperatures of all downstream zones directly (e.g., as requested by respective zone thermostats), by concurrently supplying hot air, cold air, or any temperature in between.

Individual zone demands can be met by controlling an exact amount of air (e.g., cubic feet per minute) commensurate to temperature using flow measurement and control device(s) **2708**, which can have turndowns exceeding **300-1**. Such can also facilitate self-balancing, which can minimize or reduce zone-level thermal transfer devices and hydronic infrastructure, adding sustainability and potentially resulting in a higher return on investment.

Zone-level device(s) **2706** can be configured for proper room mixing over a wide dynamic range of flows, including low flows. Such can also facilitate hybrid flex space that can reduce costly mechanical retrofits, which can enable healthy space and occupancy management to meet personal and organizational needs.

Example Operating Environments

In order to provide additional context for various embodiments described herein, FIG. **28** and the following discussion are intended to provide a brief, general description of a suitable computing environment **2800** in which the various embodiments of the embodiment described herein can be implemented, for example, a device or product fabrication environment.

Generally, program modules include routines, programs, components, data structures, etc., that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the inventive methods can be practiced with other computer system configurations, including single-processor or multiprocessor computer systems, minicomputers, mainframe computers, Internet of Things (IoT) devices, distributed computing systems, as well as personal computers, hand-held computing devices, microprocessor-based or programmable consumer electronics, and the like, each of which can be operatively coupled to one or more associated devices.

The illustrated embodiments of the embodiments herein can be also practiced in distributed computing environments where certain tasks are performed by remote processing devices that are linked through a communications network. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

Computing devices typically include a variety of media, which can include computer-readable storage media, machine-readable storage media, and/or communications media, which two terms are used herein differently from one another as follows. Computer-readable storage media or machine-readable storage media can be any available storage media that can be accessed by the computer and includes both volatile and nonvolatile media, removable and non-removable media. By way of example, and not limitation, computer-readable storage media or machine-readable storage media can be implemented in connection with any method or technology for storage of information such as computer-readable or machine-readable instructions, program modules, structured data or unstructured data.

Computer-readable storage media can include, but are not limited to, random access memory (RAM), read only memory (ROM), electrically erasable programmable read only memory (EEPROM), flash memory or other memory technology, compact disk read only memory (CD-ROM), digital versatile disk (DVD), Blu-ray disc (BD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, solid state drives or other solid state storage devices, or other tangible and/or non-transitory media which can be used to store desired information. In this regard, the terms “tangible” or “non-transitory” herein as applied to storage, memory or computer-readable media, are to be understood to exclude only propagating transitory signals per se as modifiers and do not relinquish rights to all standard storage, memory or computer-readable media that are not only propagating transitory signals per se.

Computer-readable storage media can be accessed by one or more local or remote computing devices, e.g., via access requests, queries or other data retrieval protocols, for a variety of operations with respect to the information stored by the medium.

Communications media typically embody computer-readable instructions, data structures, program modules or other structured or unstructured data in a data signal such as a modulated data signal, e.g., a carrier wave or other transport mechanism, and includes any information delivery or transport media. The term “modulated data signal” or signals refers to a signal that has one or more of its characteristics set or changed in such a manner as to encode information in one or more signals. By way of example, and not limitation, communication media include wired media, such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media.

With reference again to FIG. 28, the example environment 2800 for implementing various embodiments of the aspects described herein includes a computer 2802, the computer 2802 including a processing unit 2804, a system memory 2806 and a system bus 2808. The system bus 2808 couples system components including, but not limited to, the system memory 2806 to the processing unit 2804. The processing unit 2804 can be any of various commercially available processors. Dual microprocessors and other multi-processor architectures can also be employed as the processing unit 2804.

The system bus 2808 can be any of several types of bus structure that can further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and a local bus using any of a variety of commercially available bus architectures. The system memory 2806 includes ROM 2810 and RAM 2812. A basic input/output system (BIOS) can be stored in a non-volatile memory such as ROM, erasable programmable read only memory (EPROM), EEPROM, which BIOS contains the basic routines that help to transfer information between elements within the computer 2802, such as during startup. The RAM 2812 can also include a high-speed RAM such as static RAM for caching data.

The computer 2802 further includes an internal hard disk drive (HDD) 2814 (e.g., EIDE, SATA), one or more external storage devices 2816 (e.g., a magnetic floppy disk drive (FDD) 2816, a memory stick or flash drive reader, a memory card reader, etc.) and an optical disk drive 2820 (e.g., which can read or write from a CD-ROM disc, a DVD, a BD, etc.). While the internal HDD 2814 is illustrated as located within the computer 2802, the internal HDD 2814 can also be configured for external use in a suitable chassis (not shown). Additionally, while not shown in environment 2800, a solid state drive (SSD) could be used in addition to, or in place of, an HDD 2814. The HDD 2814, external storage device(s) 2816 and optical disk drive 2820 can be connected to the system bus 2808 by an HDD interface 2824, an external storage interface 2826 and an optical drive interface 2828, respectively. The interface 2824 for external drive implementations can include at least one or both of Universal Serial Bus (USB) and Institute of Electrical and Electronics Engineers (IEEE) 2894 interface technologies. Other external drive connection technologies are within contemplation of the embodiments described herein.

The drives and their associated computer-readable storage media provide nonvolatile storage of data, data structures, computer-executable instructions, and so forth. For the computer 2802, the drives and storage media accommodate the storage of any data in a suitable digital format. Although the description of computer-readable storage media above refers to respective types of storage devices, it should be appreciated by those skilled in the art that other types of storage media which are readable by a computer, whether presently existing or developed in the future, could also be used in the example operating environment, and further, that any such storage media can contain computer-executable instructions for performing the methods described herein.

A number of program modules can be stored in the drives and RAM 2812, including an operating system 2830, one or more application programs 2832, other program modules 2834 and program data 2836. All or portions of the operating system, applications, modules, and/or data can also be cached in the RAM 2812. The systems and methods described herein can be implemented utilizing various commercially available operating systems or combinations of operating systems.

Computer 2802 can optionally comprise emulation technologies. For example, a hypervisor (not shown) or other intermediary can emulate a hardware environment for operating system 2830, and the emulated hardware can optionally be different from the hardware illustrated in FIG. 28. In such an embodiment, operating system 2830 can comprise one virtual machine (VM) of multiple VMs hosted at computer 2802. Furthermore, operating system 2830 can provide runtime environments, such as the Java runtime environment or the .NET framework, for applications 2832. Runtime environments are consistent execution environments that allow applications 2832 to run on any operating system that includes the runtime environment. Similarly, operating system 2830 can support containers, and applications 2832 can be in the form of containers, which are lightweight, standalone, executable packages of software that include, e.g., code, runtime, system tools, system libraries and settings for an application.

Further, computer 2802 can be enable with a security module, such as a trusted processing module (TPM). For instance with a TPM, boot components hash next in time boot components, and wait for a match of results to secured values, before loading a next boot component. This process can take place at any layer in the code execution stack of computer 2802, e.g., applied at the application execution level or at the operating system (OS) kernel level, thereby enabling security at any level of code execution.

A user can enter commands and information into the computer 2802 through one or more wired/wireless input devices, e.g., a keyboard 2838, a touch screen 2840, and a pointing device, such as a mouse 2842. Other input devices (not shown) can include a microphone, an infrared (IR) remote control, a radio frequency (RF) remote control, or other remote control, a joystick, a virtual reality controller and/or virtual reality headset, a game pad, a stylus pen, an image input device, e.g., camera(s), a gesture sensor input device, a vision movement sensor input device, an emotion or facial detection device, a biometric input device, e.g., fingerprint or iris scanner, or the like. These and other input devices are often connected to the processing unit 2804 through an input device interface 2844 that can be coupled to the system bus 2808, but can be connected by other interfaces, such as a parallel port, an IEEE 1394 serial port, a game port, a USB port, an IR interface, a BLUETOOTH® interface, etc.

A monitor 2846 or other type of display device can be also connected to the system bus 2808 via an interface, such as a video adapter 2848. In addition to the monitor 2846, a computer typically includes other peripheral output devices (not shown), such as speakers, printers, etc.

The computer 2802 can operate in a networked environment using logical connections via wired and/or wireless communications to one or more remote computers, such as a remote computer(s) 2850. The remote computer(s) 2850 can be a workstation, a server computer, a router, a personal computer, portable computer, microprocessor-based entertainment appliance, a peer device or other common network node, and typically includes many or all of the elements described relative to the computer 2802, although, for purposes of brevity, only a memory/storage device 2852 is illustrated. The logical connections depicted include wired/wireless connectivity to a local area network (LAN) 2854 and/or larger networks, e.g., a wide area network (WAN) 2856. Such LAN and WAN networking environments are commonplace in offices and companies, and facilitate enter-

prise-wide computer networks, such as intranets, all of which can connect to a global communications network, e.g., the Internet.

When used in a LAN networking environment, the computer **2802** can be connected to the local network **2854** through a wired and/or wireless communication network interface or adapter **2858**. The adapter **2858** can facilitate wired or wireless communication to the LAN **2854**, which can also include a wireless access point (AP) disposed thereon for communicating with the adapter **2858** in a wireless mode.

When used in a WAN networking environment, the computer **2802** can include a modem **2860** or can be connected to a communications server on the WAN **2856** via other means for establishing communications over the WAN **2856**, such as by way of the Internet. The modem **2860**, which can be internal or external and a wired or wireless device, can be connected to the system bus **2808** via the input device interface **2844**. In a networked environment, program modules depicted relative to the computer **2802** or portions thereof, can be stored in the remote memory/storage device **2852**. It will be appreciated that the network connections shown are example and other means of establishing a communications link between the computers can be used.

When used in either a LAN or WAN networking environment, the computer **2802** can access cloud storage systems or other network-based storage systems in addition to, or in place of, external storage devices **2816** as described above. Generally, a connection between the computer **2802** and a cloud storage system can be established over a LAN **2854** or WAN **2856** e.g., by the adapter **2858** or modem **2860**, respectively. Upon connecting the computer **2802** to an associated cloud storage system, the external storage interface **2826** can, with the aid of the adapter **2858** and/or modem **2860**, manage storage provided by the cloud storage system as it would other types of external storage. For instance, the external storage interface **2826** can be configured to provide access to cloud storage sources as if those sources were physically connected to the computer **2802**.

The computer **2802** can be operable to communicate with any wireless devices or entities operatively disposed in wireless communication, e.g., a printer, scanner, desktop and/or portable computer, portable data assistant, communications satellite, any piece of equipment or location associated with a wirelessly detectable tag (e.g., a kiosk, news stand, store shelf, etc.), and telephone. This can include Wireless Fidelity (Wi-Fi) and BLUETOOTH® wireless technologies. Thus, the communication can be a predefined structure as with a conventional network or simply an ad hoc communication between at least two devices.

As used in this application, the terms “component,” “system,” “platform,” “interface,” and the like, can refer to and/or can include a computer-related entity or an entity related to an operational machine with one or more specific functionalities. The entities disclosed herein can be either hardware, a combination of hardware and software, software, or software in execution. For example, a component can be, but is not limited to being, a process running on a processor, a processor, an object, an executable, a thread of execution, a program, and/or a computer. By way of illustration, both an application running on a server and the server can be a component. One or more components can reside within a process and/or thread of execution and a component can be localized on one computer and/or distributed between two or more computers. In another example, respective components can execute from various

computer readable media having various data structures stored thereon. The components can communicate via local and/or remote processes such as in accordance with a signal having one or more data packets (e.g., data from one component interacting with another component in a local system, distributed system, and/or across a network such as the Internet with other systems via the signal). As another example, a component can be an apparatus with specific functionality provided by mechanical parts operated by electric or electronic circuitry, which is operated by a software or firmware application executed by a processor. In such a case, the processor can be internal or external to the apparatus and can execute at least a part of the software or firmware application. As yet another example, a component can be an apparatus that provides specific functionality through electronic components without mechanical parts, wherein the electronic components can include a processor or other means to execute software or firmware that confers at least in part the functionality of the electronic components. In an aspect, a component can emulate an electronic component via a virtual machine, e.g., within a cloud computing system.

In addition, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or.” That is, unless specified otherwise, or clear from context, “X employs A or B” is intended to mean any of the natural inclusive permutations. That is, if X employs A; X employs B; or X employs both A and B, then “X employs A or B” is satisfied under any of the foregoing instances. Moreover, articles “a” and “an” as used in the subject specification and annexed drawings should generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form. As used herein, the terms “example” and/or “exemplary” are utilized to mean serving as an example, instance, or illustration and are intended to be non-limiting. For the avoidance of doubt, the subject matter disclosed herein is not limited by such examples. In addition, any aspect or design described herein as an “example” and/or “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects or designs, nor is it meant to preclude equivalent exemplary structures and techniques known to those of ordinary skill in the art.

As it is employed in the subject specification, the term “processor” can refer to substantially any computing processing unit or device comprising, but not limited to, single-core processors; single-processors with software multithread execution capability; multi-core processors; multi-core processors with software multithread execution capability; multi-core processors with hardware multithread technology; parallel platforms; and parallel platforms with distributed shared memory. Additionally, a processor can refer to an integrated circuit, an application specific integrated circuit (ASIC), a digital signal processor (DSP), a field programmable gate array (FPGA), a programmable logic controller (PLC), a complex programmable logic device (CPLD), a discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. Further, processors can exploit nano-scale architectures such as, but not limited to, molecular and quantum-dot based transistors, switches and gates, in order to optimize space usage or enhance performance of user equipment. A processor can also be implemented as a combination of computing processing units. In this disclosure, terms such as “store,” “storage,” “data store,” “data storage,” “database,” and substantially any other information storage component relevant to operation and functionality of a component are utilized to refer to

“memory components,” entities embodied in a “memory,” or components comprising a memory. It is to be appreciated that memory and/or memory components described herein can be either volatile memory or nonvolatile memory or can include both volatile and nonvolatile memory. By way of illustration, and not limitation, nonvolatile memory can include read only memory (ROM), programmable ROM (PROM), electrically programmable ROM (EPROM), electrically erasable ROM (EEPROM), flash memory, or non-volatile random-access memory (RAM) (e.g., ferroelectric RAM (FeRAM)). Volatile memory can include RAM, which can act as external cache memory, for example. By way of illustration and not limitation, RAM is available in many forms such as synchronous RAM (SRAM), dynamic RAM (DRAM), synchronous DRAM (SDRAM), double data rate SDRAM (DDR SDRAM), enhanced SDRAM (ESDRAM), Synchlink DRAM (SLDRAM), direct Rambus RAM (DR- RAM), direct Rambus dynamic RAM (DRDRAM), and Rambus dynamic RAM (RDRAM). Additionally, the disclosed memory components of systems or computer-implemented methods herein are intended to include, without being limited to including, these and any other suitable types of memory.

What has been described above include mere examples of systems and computer-implemented methods. It is, of course, not possible to describe every conceivable combination of components or computer-implemented methods for purposes of describing this disclosure, but one of ordinary skill in the art can recognize that many further combinations and permutations of this disclosure are possible. Furthermore, to the extent that the terms “includes,” “has,” “possesses,” and the like are used in the detailed description, claims, appendices and drawings such terms are intended to be inclusive in a manner similar to the term “comprising” as “comprising” is interpreted when employed as a transitional word in a claim. The descriptions of the various embodiments have been presented for purposes of illustration but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

Example Aspects

Aspects denoted with the letter “A” generally relate to an evase device, aspects denoted with the letter “B” generally relate to a fluid intake device, aspects denoted with the letter “C” generally relate to a fan intake device, and aspects denoted with the letter “D” generally relate to an air handler device. It is appreciated that aspects denoted with a same letter can generally be combined with together in any suitable combination. In some cases, if relevant, any aspect noted below can be combined with any other aspect. In some cases aspects of different letters can be combined to produce a combined device or product such as, for example, an aspect having the letter D (e.g., an air handler device) can be combined with suitable any combination of aspects having letters A-C (e.g., an air handler device further improved by an evase device, a fluid intake device, and/or a fan intake device).

Aspect A1. An evase device, comprising: a housing that encompasses a channel that extends in a longitudinal direc-

tion from a first side of the housing to a second side of the housing; a first opening, situated at the first side of the housing, configured to receive a flow of a fluid discharged by a fan; and a second opening, situated at the second side of the housing, configured to discharge the flow into a duct, wherein, at the second side, the housing has a rounded corner determined to mitigate a reverse flow of the fluid at corners of the duct.

Aspect A2. The system or device in accordance with aspect A1, wherein a shape of the rounded corners is designed based on a Reynolds number calculation.

Aspect A3. The system or device in accordance with aspect A1 or any suitable previous aspect, wherein the fan is an axial fan.

Aspect A4. The system or device in accordance with aspect A1 or any suitable previous aspect, wherein the fan is an axial fan.

Aspect A5. The system or device in accordance with aspect A1 or any suitable previous aspect, wherein the corners of the duct are squared corners.

Aspect A6. The system or device in accordance with aspect A1 or any suitable previous aspect, wherein the fluid discharged by the fan flows in the longitudinal direction from the first opening to the second opening.

Aspect A7. The system or device in accordance with aspect A1 or any suitable previous aspect, wherein the fluid discharged by the fan has a velocity pressure that is converted to static pressure less an impact loss.

Aspect A8. The system or device in accordance with aspect A7 or any suitable previous aspect, wherein the fluid discharged by the fan has a velocity pressure that is converted to static pressure less an impact loss.

Aspect A9. The system or device in accordance with aspect A1 or any suitable previous aspect, wherein the first opening has an annular shape having a diameter that matches an impeller hub diameter of the fan.

Aspect A10. The system or device in accordance with aspect A1 or any suitable previous aspect, further comprising the fan, wherein the fan is mounted to the housing at the first opening.

Aspect A11. The system or device in accordance with aspect A10 or any suitable previous aspect, wherein the housing operates as a fan housing for the fan.

Aspect A12. The system or device in accordance with aspect A10 or any suitable previous aspect, wherein a motor of the fan is situated outside the channel.

Aspect A13. The system or device in accordance with aspect A10 or any suitable previous aspect, wherein a motor of the fan is situated within the channel.

Aspect A14. The system or device in accordance with aspect A1 or any suitable previous aspect, further comprising an intermediate baffle situated in the channel.

Aspect A15. The system or device in accordance with aspect A14 or any suitable previous aspect, wherein the intermediate baffle has rounded corners at an end that discharges the fluid into the channel.

Aspect A16. The system or device in accordance with aspect A1 or any suitable previous aspect, further comprising a container for the evase device that is filled with a material that absorbs sound.

Aspect A17. A method of fabricating an evase device, comprising: forming, by a device comprising a processor, a housing that encompasses a channel that extends in a longitudinal direction from a first side of the housing to a second side of the housing; forming, by the device, a first opening in the housing that is situated at the first side of the housing, wherein the first opening is configured to receive a

flow of a fluid discharged by a fan; forming, by the device, a second opening in the housing that is situated at the second side of the housing, wherein the second opening is configured to discharge the flow of a fluid into a duct; and forming, by the device, rounded corners at the second side of the housing, wherein the rounded corners are determined to mitigate a reverse flow of the fluid at corners of the duct.

Aspect A18. The method in accordance with aspect A17 or any suitable previous aspect, further comprising forming or assembling, by the device, an intermediate baffle situated in the channel.

Aspect A19. The method in accordance with aspect A17 or any suitable previous aspect, further comprising forming or assembling, by the device, the fan situated within the housing.

Aspect A20. The method in accordance with aspect A19 or any suitable previous aspect, further comprising forming or assembling, by the device, a motor of the fan that is situated within the channel.

Aspect B 1. An intake device, comprising: an intake duct comprising: a first opening by which a fluid enters the intake duct and a second opening by which the fluid exits the intake duct, wherein the first opening and the second opening are substantially circular about a longitudinal axis of the intake duct, and wherein a first circumference of the first opening is larger than a second circumference of the second opening; and an interior surface that extends from the first opening to the second opening, providing a passageway for a flow of the fluid; the intake device further comprising: a top cover, situated a distance from the first opening, that prevents the fluid from entering the intake duct in a direction along the longitudinal axis, and that permits the fluid to enter the intake duct in a radial direction that is radial about the longitudinal axis; and an inner funnel, comprising: an upper portion that couples to the top cover; a lower portion extends into the passageway; and an outer surface, spanning the upper portion and the lower portion, that is sloped, causing the flow of the fluid entering the intake device in the radial direction to change to the direction along the longitudinal axis.

Aspect B2. The system or device in accordance with aspect B1 or any suitable previous aspect, wherein the interior surface of the intake duct provides a smoothly tapered surface that encompasses a substantially funnel-shaped passageway for the flow of the fluid.

Aspect B3. The system or device in accordance with aspect B1 or any suitable previous aspect, wherein an angular difference of the change in direction of the flow, representing a difference between the radial direction and the direction along the longitudinal axis, is between about 80 degrees and 100 degrees.

Aspect B4. The system or device in accordance with aspect B1 or any suitable previous aspect, wherein an angular difference of the change in direction of the flow, representing a difference between the radial direction and the direction along the longitudinal axis, is approximately 90 degrees.

Aspect B5. The system or device in accordance with aspect B1 or any suitable previous aspect, wherein a cross-section of the passageway of the intake duct has an area that is a difference between a first area of the interior surface of the intake duct at the cross-section and a second area of the outer surface of the inner funnel at the cross-section.

Aspect B6. The system or device in accordance with aspect B5 or any suitable previous aspect, wherein the area

of the cross-section of the passageway decreases when moving along the longitudinal axis from the first opening to the second opening.

Aspect B7. The system or device in accordance with aspect B6 or any suitable previous aspect, wherein the area that decreases when moving from the first opening to the second opening is determined to cause the flow of the fluid in the passageway to increase in velocity while flowing toward the second opening.

Aspect B8. The system or device in accordance with aspect B7 or any suitable previous aspect, wherein the increase in velocity is determined to have a damping effect on turbulence of the flow.

Aspect B9. The system or device in accordance with aspect B1 or any suitable previous aspect, wherein geometries of the outer surface of the inner funnel and the interior surface of the intake duct are determined to cause the flow to be laminar.

Aspect B10. The system or device in accordance with aspect B9 or any suitable previous aspect, wherein the geometries are determined to mitigate losses due to flow separation along bounding surfaces of a turning flow, and wherein the turning flow represents the flow entering in the radial direction and turning toward the longitudinal direction.

Aspect B11. The system or device in accordance with aspect B9 or any suitable previous aspect, wherein the geometries are determined to cause at least a portion of the flow entering the intake device to follow an elliptical path when changing from the radial direction to the direction along the longitudinal axis.

Aspect B12. An intake device, comprising: an intake duct, having a cover plate that is configured to prevent a fluid from entering the intake duct in a longitudinal direction, wherein the intake duct is configured to receive, at a first end, the fluid in a radial direction and to discharge, at a second end, the fluid substantially in the longitudinal direction; and an inner funnel situated between the cover plate and the second end, wherein the inner funnel has a funnel geometry that causes the fluid to follow an elliptical path when changing from the radial direction to the longitudinal direction.

Aspect B13. The system or device in accordance with aspect B12 or any suitable previous aspect, wherein an area of a cross-section of an inner chamber of the intake duct, through which the fluid flows, decreases when moving along the longitudinal axis from the first end to the second end.

Aspect B14. The system or device in accordance with aspect B12 or any suitable previous aspect, wherein the intake duct has a duct geometry configured to reduce a surface area normal to a flow of the fluid as the fluid flows from the first end to the second end.

Aspect B15. The system or device in accordance with aspect B14 or any suitable previous aspect, wherein the duct geometry is determined to cause the flow to be laminar.

Aspect B16. The system or device in accordance with aspect B14 or any suitable previous aspect, wherein the duct geometry is determined to mitigate losses due to flow separation along bounding surfaces of a turning flow, and wherein the turning flow represents the flow entering in the radial direction and turning toward the longitudinal direction.

Aspect B17. A method of fabricating an intake device, comprising: forming, by a device comprising a processor, an intake duct, having a cover plate that is configured to prevent a fluid from entering the intake device in a longitudinal direction, wherein the intake device is configured to receive, at a first end, the fluid in a radial direction and to discharge,

at a second end, the fluid substantially in the longitudinal direction; and forming, by the device, an inner funnel situated between the cover plate and the second end, wherein the inner funnel has a funnel geometry that causes the fluid to follow an elliptical path when changing from the radial direction to the longitudinal direction.

Aspect B18. The method in accordance with aspect B17 or any suitable previous aspect, wherein the forming the intake duct and the forming the inner funnel further comprises, determining, by the device, that geometries of the intake duct and the inner funnel cause a flow of the fluid through the intake device to be laminar.

Aspect B19. The method in accordance with aspect B17 or any suitable previous aspect, wherein the forming the intake duct and the forming the inner funnel further comprises, determining, by the device, that geometries of the intake duct and the inner funnel result in a continuously decreasing cross-sectional area when moving along the longitudinal axis toward the second end.

Aspect B20. The method in accordance with aspect B17 or any suitable previous aspect, wherein the forming the intake duct and the forming the inner funnel further comprises, determining, by the device, that geometries of the intake duct and the inner funnel cause a flow of the fluid through the intake device to accelerate when moving toward the second end.

Aspect C1. An aero-acoustical fan intake device, comprising: an inlet face comprising an inlet opening configured to receive a flow of a fluid; a discharge face comprising a discharge opening configured to discharge the flow of the fluid; and a housing that encompasses a flow channel that extends from the inlet opening to the discharge opening, wherein a cross-sectional area of the flow channel varies between the inlet opening and the discharge opening in a manner that is determined to cause the flow of the fluid through the flow channel to continuously accelerate from a first location of the channel to the discharge opening.

Aspect C2. The system or device in accordance with aspect C1 or any suitable previous aspect, of claim 1, wherein the inlet opening has an annulus shape.

Aspect C3. The system or device in accordance with aspect C1 or any suitable previous aspect, of claim 1, wherein the discharge opening has an annulus shape.

Aspect C4. The system or device in accordance with aspect C1 or any suitable previous aspect, of claim 1, wherein the first location is at the inlet opening.

Aspect C5. The system or device in accordance with aspect C1 or any suitable previous aspect, of claim 1, wherein the first location is about midway between the inlet opening and the discharge opening.

Aspect C6. The system or device in accordance with aspect C1 or any suitable previous aspect, of claim 1, wherein the first location is about one third of a distance between the inlet opening and the discharge opening.

Aspect C7. The system or device in accordance with aspect C1 or any suitable previous aspect, of claim 1, wherein the inlet opening receives the flow of the fluid from an inlet duct or plenum.

Aspect C8. The system or device in accordance with aspect C7 or any suitable previous aspect, of claim 1, wherein the cross-sectional area of the flow channel at the first opening is less than one-half of a cross-sectional area of the inlet face.

Aspect C9. The system or device in accordance with aspect C1 or any suitable previous aspect, of claim 1, further comprising a material determined to absorb noise that is distributed within the housing around the flow channel.

Aspect C10. The system or device in accordance with aspect C1 or any suitable previous aspect, of claim 1, wherein the cross-sectional area of the flow channel monotonically decreases from the inlet opening to the discharge opening at substantially an area swept by impellers of a fan situated proximal to the discharge opening.

Aspect C11. The system or device in accordance with aspect C1 or any suitable previous aspect, of claim 1, wherein the inlet face is shaped as a bulb and the inlet opening surrounds the bulb.

Aspect C12. The system or device in accordance with aspect C11 or any suitable previous aspect, of claim 1, wherein the bulb has a bulb diameter that is determined to be greater than an impeller diameter of a fan.

Aspect C13. The system or device in accordance with aspect C1 or any suitable previous aspect, of claim 1, wherein a geometry of the flow channel that is determined to cause the flow of the fluid to continuously accelerate is determined to result in a reduced energy loss across the aero-acoustical fan intake device.

Aspect C14. The system or device in accordance with aspect C13 or any suitable previous aspect, of claim 1, wherein the reduced energy loss across the aero-acoustical fan intake device is representative of a decrease in total pressure through the aero-acoustical fan intake device that is less than about 10% of an impeller velocity pressure.

Aspect C15. The system or device in accordance with aspect C13 or any suitable previous aspect, of claim 1, wherein the reduced energy loss across the aero-acoustical fan intake device is representative of a decrease in total pressure through the aero-acoustical fan intake device that is less than about 50% of an impeller velocity pressure.

Aspect C16. A method of fabricating a fan intake device, comprising: forming, by a device comprising a processor, an inlet face surrounded by an inlet opening configured to receive a flow of a fluid; forming, by the device, a discharge face surrounded by a discharge opening configured to discharge the flow of the fluid; and forming, by the device, a housing that encompasses a channel that extends from the inlet opening to the discharge opening, wherein a cross-sectional area of the channel varies between the inlet opening and the discharge opening in a manner that is determined to cause the flow of the fluid through the channel to continuously accelerate from a first location of the channel to the discharge opening.

Aspect C17. The method in accordance with aspect C16 or any suitable previous aspect, wherein the forming the housing comprises determining that the cross-sectional area of the channel at the first opening is less than one-half of a cross-sectional area of the inlet opening.

Aspect C18. The method in accordance with aspect C16 or any suitable previous aspect, wherein the forming the housing comprises determining that the cross-sectional area of the channel monotonically decreases from the inlet opening to the discharge opening at substantially an area swept by the fan impellers.

Aspect C19. The method in accordance with aspect C16 or any suitable previous aspect, wherein the forming the housing comprises determining that a geometry of the flow causes a reduced energy loss across the fan intake device.

Aspect D1. An air handler device, comprising: a mixing plenum configured to receive multiple flows of air from multiple different ducts or intakes that feed the mixing plenum; a fan device configured to receive a mixing plenum flow from the mixing plenum and to discharge a supply flow; and a supply plenum configured to receive the supply flow from the fan device, wherein the supply plenum comprises:

a plurality of duct interfaces respectively configured to interface with a different one of a plurality of supply ducts; and a plurality of thermal transfer units comprising a first thermal transfer unit and a second thermal transfer unit that are respectively situated in different ones of the plurality of duct interfaces, wherein the first thermal transfer unit affecting a first flow is configured to a first temperature concurrently with the second thermal transfer affecting a second air flow being configured to a second temperature that differs from the first temperature.

Aspect D2. The system or device in accordance with aspect D1 or any suitable previous aspect, of claim 1, wherein a first flow of the multiple flows comprises return air of a heating, ventilation, and air conditioning (HVAC) system.

Aspect D3. The system or device in accordance with aspect D1 or any suitable previous aspect, of claim 1, wherein a second flow of the multiple flows comprises fresh air.

Aspect D4. The system or device in accordance with aspect D1 or any suitable previous aspect, of claim 1, wherein a duct of the multiple different ducts that feed the mixing plenum comprises at least one of a group comprising: a thermal transfer device configured to exchange heat with a corresponding flow through the duct and a filter device configured to filter the corresponding flow.

Aspect D5. The system or device in accordance with aspect D1 or any suitable previous aspect, of claim 1, wherein the fan is a centrifugal fan.

Aspect D6. The system or device in accordance with aspect D1 or any suitable previous aspect, of claim 1, further comprising multiple fan devices situated between the mixing plenum and the supply plenum.

Aspect D7. The system or device in accordance with aspect D1 or any suitable previous aspect, of claim 1, wherein the plurality of thermal transfer units are individually configured to heat, cool, or match in temperature a flow of air independently of other members of the plurality of thermal transfer units.

Aspect D8. The system or device in accordance with aspect D1 or any suitable previous aspect, of claim 1, wherein the plurality of duct interfaces comprise four duct interfaces.

Aspect D9. The system or device in accordance with aspect D1 or any suitable previous aspect, of claim 1, wherein the plurality of duct interfaces comprise three duct interfaces.

Aspect D10. The system or device in accordance with aspect D1 or any suitable previous aspect, of claim 1, wherein a plurality of supply air flows that flow into the plurality of duct interfaces flow in different directions.

Aspect D11. The system or device in accordance with aspect D1 or any suitable previous aspect, of claim 1, wherein at least two of a plurality of supply air flows that flow into two of the plurality of duct interfaces flow in a same direction.

Aspect D12. A heating, ventilation, and air conditioning (HVAC) product, comprising: an air handler component configured to circulate a flow of air within an HVAC system situated at a site the HVAC product is to be installed, wherein the air handler device comprises a top surface that is, relative to an installation at the site, on top of the air handler component and has a first height that is, relative to the installation, a height of the air handler component; and a heat exchange device configured to exchange heat with the flow of air, wherein the heat exchange device has a second height that is, relative to the installation, a height of the heat

exchange device, and wherein the heat exchange device is situated on the top surface of the air handler component, resulting in the HVAC product having a total height that is, relative to the installation, determined to be less than or equal to a defined height constraint.

Aspect D13. The system or device in accordance with aspect D12 or any suitable previous aspect, of claim 1, wherein the defined height constraint is determined to satisfy a local building code of the installation site.

Aspect D14. The system or device in accordance with aspect D12 or any suitable previous aspect, of claim 1, wherein the defined height constraint is determined to satisfy a transportation code applicable to a transportation route between a manufacturing site of the HVAC product and the installation site.

Aspect D15. The system or device in accordance with aspect D12 or any suitable previous aspect, of claim 1, wherein the defined height constraint is 14 feet.

Aspect D16. The system or device in accordance with aspect D12 or any suitable previous aspect, of claim 1, wherein the defined height constraint is 10 feet.

Aspect D17. The system or device in accordance with aspect D12 or any suitable previous aspect, of claim 1, wherein the air handler component comprises an evase device, wherein the evase device comprises a housing configured to couple, at an interface, to a duct or plenum at the site, and wherein the housing of the evase has rounded corners at the interface that are determined to mitigate a reverse flow of the flow of air at corners of the duct.

Aspect D18. The system or device in accordance with aspect D17 or any suitable previous aspect, of claim 1, wherein the rounded corners have a shape that is determined based on a Reynolds number calculation.

Aspect D19. The system or device in accordance with aspect D18 or any suitable previous aspect, of claim 1, wherein a height of the evase device is determined to facilitate the total height satisfying the defined height constraint based on the shape of the rounded corners that, by mitigating the reverse flow, reduce turbulence in the flow of air over a shorter distance represented by the height of the evase device.

Aspect D20. The system or device in accordance with aspect D17 or any suitable previous aspect, of claim 1, further comprising a fan that is integrated into the housing of the evase.

Aspect D21. The system or device in accordance with aspect D20 or any suitable previous aspect, of claim 1, wherein a height of the evase device is determined to be reduced in response to situating a motor of the fan on a downstream side of an impeller of the fan.

Aspect D22. The system or device in accordance with aspect D17 or any suitable previous aspect, of claim 1, further comprising a fan that is integrated into the housing of the evase.

Aspect D23. The system or device in accordance with aspect D12 or any suitable previous aspect, of claim 1, wherein the air handler component comprises a mixing plenum that receives the flow of air, wherein the mixing plenum comprises multiple intake openings, comprising: a first opening that receives into the mixing plenum a first portion of the flow of air from a first direction; and a second opening that receives into the mixing plenum a second portion of the flow of air from a second direction that differs from the first direction.

Aspect D24. The system or device in accordance with aspect D23 or any suitable previous aspect, of claim 1,

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wherein the heat exchange device comprises a separate coil array unit for each of the multiple intake openings.

Aspect D25. The system or device in accordance with aspect D24 or any suitable previous aspect, of claim 1, wherein the heat exchange device comprises: a first coil array unit that exchanges heat with the first portion of the flow prior to entering the mixing plenum from the first direction; and a second coil array unit that exchanges heat with the second portion of the flow prior to entering the mixing plenum from the second direction.

Aspect D26. The system or device in accordance with aspect D24 or any suitable previous aspect, of claim 1, wherein the separate coil array unit further comprises a filter that filters contaminants from the flow of air.

Aspect D27. The system or device in accordance with aspect D12 or any suitable previous aspect, of claim 1, wherein a total weight of the HVAC product is determined to satisfy a defined weight constraint.

Aspect D28. The system or device in accordance with aspect D12 or any suitable previous aspect, of claim 1, wherein the mixing plenum comprises a single, axial fan that feed the supply flow to a vane.

Aspect D29. The system or device in accordance with aspect D12 or any suitable previous aspect, of claim 1, wherein the HVAC product is shipped to the site fully assembled as a single unit.

Aspect D30. A method of fabricating an air handler device, comprising: forming, by a device comprising a processor, a mixing plenum that is configured to receive multiple flows of air from multiple different ducts; forming, by the device, a fan device configured to receive a mixing plenum flow from the mixing plenum and to discharge a supply flow; forming, by the device, a supply plenum configured to receive the supply flow from the fan device; forming, by the device, a plurality of duct interfaces respectively configured to interface with a different one of a plurality of supply ducts; and forming, by the device, a plurality of thermal transfer units comprising a first thermal transfer unit and a second thermal transfer unit that are respectively situated in different ones of the plurality of duct interfaces, wherein the first thermal transfer unit is configured to a first temperature concurrently with the second thermal transfer unit being configured to a second temperature that differs from the first temperature.

Aspect D31. The method in accordance with aspect D30 or any suitable previous aspect, wherein the forming the mixing plenum comprising configuring the mixing plenum to receive return air and fresh air.

Aspect D32. The method in accordance with aspect D30 or any suitable previous aspect, wherein the forming the fan device comprises forming multiple fan devices situated between the mixing plenum and the supply plenum.

Aspect D33. The method in accordance with aspect D30 or any suitable previous aspect, further comprising forming, by the device, a filter device situated at an interface, wherein the interface is at least one of a mixing plenum interface or a supply plenum interface.

Aspect D34. The system or device in accordance with aspect D1 or any suitable previous aspect, of claim 1, that is configured according to a blowthrough configuration or a drawthrough configuration.

Aspect D35. The system or device in accordance with aspect D1 or any suitable previous aspect, of claim 1, that is configured according to an overhead discharge configuration or an under floor configuration.

Aspect D36. A heating, ventilation, and air conditioning (HVAC) product, comprising: an air handler component

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configured to circulate a flow of air within an HVAC system situated at a site the HVAC product is to be installed, wherein the air handler device comprises a bottom surface that is, relative to an installation at the site, on bottom of the air handler component and has a first height that is, relative to the installation, a height of the air handler component; and an air handler component configured to circulate a flow of air within an HVAC system situated at a site the HVAC product is to be installed, wherein the air handler device comprises a bottom surface that is, relative to an installation at the site, on bottom of the air handler component and has a first height that is, relative to the installation, a height of the air handler component.

What is claimed is:

1. An air handler device, comprising:

an intake plenum comprising an intake device configured to receive an intake flow comprising intake air;

a discharge plenum that is situated adjacent to the intake plenum, the discharge plenum comprising a discharge device configured to discharge a supply flow comprising supply air composed of the intake air mixed in transit from the intake plenum to the discharge plenum, wherein the discharge device comprises a thermal transfer unit that is configured to control a temperature of the supply air;

a fan device configured to facilitate intake of the intake air via the intake plenum, and to facilitate discharge of the supply air via the discharge plenum; and

a U-bend device that provides an air flow path that changes direction between the intake plenum and the discharge plenum, the U-bend device comprising:

a first channel that extends in a first direction from the intake plenum; and

a second channel, situated adjacent to the first channel, that extends in a second direction, different than the first direction, to the discharge plenum.

2. The air handler device of claim 1, wherein the intake air comprises at least one of: outside air or return air.

3. The air handler device of claim 1, wherein the intake plenum comprises multiple intake devices configured to receive respective portions of the intake air from respective different directions.

4. The air handler device of claim 1, further comprising an acoustical lining that lines at least one element of the air handler device.

5. The air handler device of claim 1, further comprising a thermal lining that lines at least one element of the air handler device.

6. The air handler device of claim 1, wherein the intake device comprises at least one of:

an economizer device comprising at least one of: a damper or a fan;

a filter device;

a heat recovery device;

a pre-heat device; or

a humidifier device.

7. The air handler device of claim 1, wherein the discharge plenum comprises multiple discharge devices configured to discharge respective portions of the supply air in respective different directions, comprising a first supply flow that is discharged in a first supply flow direction and a second supply flow that is discharged in a second supply flow direction.

8. The air handler device of claim 7, wherein the multiple discharge devices comprise respective thermal transfer units comprising a first thermal transfer unit configured to control a first temperature of the first supply flow and a second

thermal transfer unit configured to control a second temperature of the second supply flow.

9. The air handler device of claim 8, wherein the first thermal transfer unit and the second thermal transfer unit are independently controllable such that the first temperature concurrently differs from the second temperature.

10. The air handler device of claim 8, wherein the respective thermal transfer units are independently controllable to perform one of a group of thermal transfer operations, comprising a heating operation that heats an associated supply flow, a cooling operation that cools the associated supply flow, or a neutral operation that does not affect the temperature of the associated supply flow.

11. The air handler device of claim 10, wherein the first thermal transfer unit performs a first one of the group of thermal transfer operations concurrently with the second thermal transfer unit performing a second one, different than the first one, of the group of thermal transfer units.

12. The air handler device of claim 1, wherein:

the intake plenum and the discharge plenum are configured to be installed adjacent to one another within a first installation space of a building; and
the fan device is configured to be installed in a second installation space of the building that differs from the first installation space.

13. The air handler device of claim 12, wherein the first installation space is in a ceiling space of the building or an elevated space that is above a floor space of the building and the second installation space is within the floor space of the building.

14. The air handler device of claim 12, wherein the first installation space is in a floor space or under the floor space and the air handler device is compatible with an underfloor air distribution device or system.

15. The air handler device of claim 1, wherein a setting for the fan device is adjusted in response to feedback received from at least one downstream damper device from among a group of downstream damper devices.

16. The air handler device of claim 15, wherein the setting for the fan is a speed setting and the speed setting is decreased, which causes members of the group of downstream damper devices to further open in order to satisfy an associated demand, until one of the members is entirely open, thereby indicating a minimum speed setting for the fan that is still sufficient to satisfy the associated demands of the members.

17. An air handler device, comprising:

an intake plenum comprising an intake device configured to receive an intake flow comprising intake air;

a discharge plenum, situated adjacent to the intake plenum, comprising a discharge device configured to discharge a supply flow comprising supply air composed of the intake air that was mixed, wherein the discharge device comprises a thermal transfer unit configured to control a temperature of the supply air;
a fan device configured to facilitate intake of the intake air via the intake plenum, and to facilitate discharge of the supply air via the discharge plenum; and

a U-bend device that provides an air flow path that changes direction between the intake plenum and the discharge plenum, the U-bend device comprising:
a first channel that extends in a first direction from the intake plenum; and

a second channel, situated adjacent to the first channel, that extends in a second direction, different than the first direction, to the discharge plenum.

18. The air handler device of claim 17, wherein discharge plenum comprises multiple discharge device, comprising a first discharge device that discharges a first supply flow in a first direction and a second discharge device that discharges a second supply flow in a second direction that differs from the first direction.

19. The air handler device of claim 18, wherein a first thermal transfer unit if the first discharge device and a second thermal transfer unit of the second discharge device are independently controllable such that the first temperature concurrently differs from the second temperature.

20. The air handler device of claim 19, wherein the first thermal transfer unit and the second thermal transfer unit are independently controllable to perform one of a group of thermal transfer operations, comprising a heating operation that heats an associated supply flow, a cooling operation that cools the associated supply flow, or a neutral operation that does not affect the temperature of the associated supply flow.

21. The air handler device of claim 20, wherein the first thermal transfer unit performs a first one of the group of thermal transfer operations concurrently with the second thermal transfer unit performing a second one, different than the first one, of the group of thermal transfer units.

22. The air handler device of claim 17, wherein the U-bend device comprises a service door configured to facilitate access to a filter device.

23. An air handler device, comprising:

an intake plenum configured to be installed within a first installation space of a building that intersects a first coordinate plane and a second coordinate plane that differs from the first coordinate plane, the intake plenum comprising an intake device configured to receive an intake flow, comprising intake air, in a first intake direction of the first coordinate plane;

a discharge plenum configured to be installed within the first installation space adjacent to the intake plenum, the discharge plenum comprising a discharge device configured to discharge a supply flow, comprising supply air composed of the intake air that was mixed, in a first discharge direction of the first coordinate plane;

a fan device configured to facilitate intake of the intake air via the intake plenum, and to facilitate discharge of the supply air via the discharge plenum; and

a U-bend device that provides an air flow path along the second coordinate plane between the intake plenum and the discharge plenum, the U-bend device comprising:
a first channel that extends along the second coordinate plane from the intake plenum to a second installation space; and
a second channel, situated adjacent to the first channel, that extends along the second coordinate plane from the second installation space to the discharge plenum.

24. The air handler device of claim 23, wherein the intake plenum comprises multiple intake devices comprising a first intake device configured to receive a first intake flow in the first intake direction of the first coordinate plane and a second intake device configured to receive a second intake flow in a second intake direction, different than the first intake direction, of the first coordinate plane.

25. The air handler device of claim 23, wherein the discharge plenum comprises multiple discharge devices comprising a first discharge device configured to discharge a first supply flow in the first discharge direction of the first coordinate plane and a second discharge device configured

to discharge a second supply flow in a second discharge direction, different than the first discharge direction, of the first coordinate plane.

26. The air handler device of claim 25, wherein discharge plenum comprises multiple discharge devices, comprising a first discharge device comprises a first thermal transfer unit configured to control a first temperature of a first supply flow, and a second discharge device comprises a second thermal transfer unit configured to control a second temperature of a second supply flow independently of control of the first temperature by the first thermal transfer unit.

27. The air handler device of claim 26, wherein the first thermal transfer unit performs a first operation concurrently with a second operation performed by the second thermal transfer unit, wherein the first operation and the second operation are selected from a group comprising a heating operation that heats an associated supply flow, a cooling operation that cools the associated supply flow, or a neutral operation that does not affect the temperature of the associated supply flow.

28. The air handler device of claim 23, wherein the first installation space, which intersects the first coordinate plane and the second coordinate plane, is indicative of a ceiling space of the building or an elevated space that is above a floor space of the building.

29. The air handler device of claim 23, wherein the second installation space, which intersects the second coordinate plane and the third coordinate plane, is indicative of a floor space of the building.

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